ACE MANUFACTURING SYSTEMS LTD

INDEX - MACHINE HISTORY

AMS
In Pursuit of Excellence

MACHINE NUMBER : M11-772

SL NO	DESCRIPTION	FORMAT NUMBER	ENCLOSED
1	MACHINE DISPATCH CLEARANCE	F-QA-FA -16	✓
2	PACKING LIST / DISPATCH DETAILS	F-TSG-18 & F-PAC-01	✓
3	MINUTES OF MEETING	F-TSG-17	—
4	WORK ORDER	F-TSG-15	✓
5	MACHINE SEQUENCE RUNNING REPORT	F-QA-FA-12	✓
6	SPINDLE STABILISATION REPORT	F-QA-FA-11	✓
7	MACHINE HEALTH CHECK POINTS	F-QA-FA-09	✓
8	CHECK LIST - MECHANICAL	F-QA-FA-07	✓
9	MACHINE HISTORY - MECHANICAL	F-QA-FA-10	✓
10	CHECK LIST - ELECTRICAL	F-QA-FA-08	✓
11	MACHINE HISTORY - ELECTRICAL	F-QA-FA-14	✓
12	FINAL ASSEMBLY REPORT - MECHANICAL	F-QA-FA-05/15	✓
13	GEOMETRICAL TEST REPORT	DS-DSN-PROD-TC 01~ 06	✓
14	PRACTICAL TEST REPORT	F-TSG -34	✓
15	SPINDLE BALANCING REPORT	F-QA-FA-04	✓
16	SPINDLE ASSEMBLY	F-QA-IPR-SPL- 01	✓
17	HYDRAULIC ASSEMBLY REPORT (OPTIONAL)	F-QA-IPR-HYD-01	— ✓
18	BALL BAR REPORT	F-QA-FA-02	✓
19	AXIS CALIBRATION REPORT	F-QA-FA-01	✓
20	GEAR BOX / ATC ASSEMBLY	F-QA-IPR-GBA- 01 / SUPPLIER REPORT	✓
21	BALL SCREW ASSEMBLY	F-QA-IPR-BSA- 01	✓
22	BASE BUILDING	F-QA-IPR-BBA- 01/ 02	✓

Date :

CHECKED BY

APPROVED BY

ACE MANUFACTURING SYSTEMS LTD

MACHINE DISPATCH CLEARANCE

AMS
In Pursuit of Excellence

M/c No.: M11-772	CUSTOMER : SNY AUTOTECH
MODEL : MCV-400	DATE : 16/02/12

DEVIATION DETAILS

ASSEMBLY IN CHARGE | QUALITY IN CHARGE | AUTHORIZED SIGNATURE

CORRECTIVE ACTION (BY CSG)

RESPONSIBLE PERSON : SIGNATURE OF CSG HOD

Note : Customer Support Group (CSG) will confirm the closure of deviation and Return to QA.

9:00

ACE MANUFACTURING SYSTEMS LTD.

MACHINE DISPATCH CHECK POINTS

AMS — In Pursuit of Excellence

CUSTOMER'S NAME: SNY Autotech Pvt. Ltd. Kanchipuram

M/C No.: M11-778 APC MCV-408 DATE: 14-09-2012 9:00PM

S.No	DESCRIPTION	YES/ NO/NA	S.No.	DESCRIPTION	YES/ NO/NA
1	Packing list available	Yes	21	Chip Trolley cleaned	NA
2	Hydraulic / Pneumatic Counter Balance Cylinder removed	NA	22	Standard Spares & m/c manual available	Yes
3	Z- Motor Cover Removed & Packed	NO	23	Coolant tank assly. packed	Yes
4	Side door handles removed & Packed	NO	24	Index Table Cover packed properly	NA
5	Base guard removed & Packed	Yes	25	Oil skimmer bin cleaned	NA
6	Coolant Gun tied properly	Yes	26	Voltage Stabilizer packed	NA
7	Air gun tied properly	NA	27	Machine Platform cleaned	NA
8	Leveling Screws, Nuts removed & Packed	Yes	28	Wanner Pump & cover cleaned & put back	NA
9	Leveling blocks 100/180/220/300 Ht. Qty: - 4/6/7/8/11/20	Yes	29	Customer components packed properly	NA
10	Automatic Tool Changer	Yes	30	Index Table packed properly	NA
11	APC Assembly packed Properly	Yes	31	Chiller unit cleaned	NA
12	Pallet removed & Packed	NA	32	Rotary Table packed properly	NA
13	APC leveling screws - 1set	Yes	33	Toolings packed properly	NA
14	Power pack stand	Yes	34	Tail Stock packed properly	NA
15	Lub Tank oil drained - retain 2 ltrs.	Yes	35	Practical Test piece packed	NA
16	Machine oil Bin cleaned	Yes	36	Fixture/s packed properly	NA
17	Coolant tank mesh cleaned & put back	Yes	37	Air removed from cover with Vaccum cleaner	NA
18	Chip Bin cleaned & put back	Yes	38	Desiccant Bags Hung	NA
19	Chip Bin mesh cleaned & put back	Yes	39	Fumigation Certificate available	NA
20	TPM points stickers fixed properly	NA	40		

REMARKS / SHORTAGES IF ANY:

1) Electrical documination Manuals & backup cd-Oiser

ASSEMBLY	STORES	FIXTURE	ELECTRICAL	DISPATCH	AUTHORISED SIGNATORY

F-PAC-01-RO

ACE MANUFACTURING SYSTEMS LTD.

AMS
In Pursuit of Excellence

DISPATCH CHECK POINTS

CUSTOMER'S NAME : SNY Autotech Pvt. Ltd. Kanchipuram

MACHINE NUMBER : M11-772 APC MCV-900 **DATE :** 14-02-2012

S.No	CHECK POINTS	STATUS - YES / NO	REMARKS
1	PACKING CASE IS IN GOOD CONDITION.	Yes	
2	CHIP CONVEYOR IS FREE FROM CHIPS & UNWANTED MATERIALS.	Yes	
3	PAINT SHOULD NOT BE PEELED OFF IN ANY PART OF THE MACHINE.	Yes	
4	ALL POCKETS IN ATC ARE EMPTY & CLEANED	Yes	
5	NO TOOL IN SPINDLE & CLEANED	Yes	
6	NO LOOSE PARTS KEPT ANYWHERE IN THE MACHINE.	Yes	
7	ENSURE RUST PREVENTIVE OIL IS APPLIED TO ALL MACHINED SURFACE.	Yes	
8	BOLT & NUT ARE FITTED AT ALL RELAVENT PLACES ON PACKING CASE.	Yes	
9	SPARE BOX KEPT INSIDE THE PACKING CASE.	Yes	
10	GEOMETRICAL, AXIS CALIBRATION, CUTTING TEST REPORTS ARE SENT WITH THE MACHINE Inside the cabinet	Yes	
11	ELECTRICAL CABINET CLEANED.	Yes	
12	ENSURE MACHINE IS CLEANED & NO DUST ON ANY WHERE ON M/c	Yes	
13	SUPRABHA COVER IS SEALED PROPERLY.	Yes	
14	ENSURE PACKING BOX SHOULD HAVE MINIMUM 3" GAP ALL OVER	Yes	
15	COOLANT WATER FROM ALL PIPES REMOVED BY AIR JET.	Yes	
16	NO RUST OBSERVED ANYWHERE IN THE MACHINE & RUST PREVENTIVE SPRAYED TO ALL FASTENERS & BLACKEDNED PARTS	Yes	
17	THERE IS NO TOUCHUP PAINT DEPOSITED ON MATCHING PARTS.	NO	
18	MACHINE LIFTING PROCEDURE CHART FIXED ON THE PACKING CASE	Yes	
19	CABINET KEY SENT WITH MACHINE TIED TO THE AC UNIT EYE BOLT	Yes	
20	PANEL COOLER FILTER IS CLEANED	Yes	
21	CHECK ITEMS AS PER PACKING LIST	Yes	
22	WOODEN PACKING PROVIDED BELOW THE APC STATION.	Yes	
23	WOODEN support given below the grundfoss pump mounting bracket	———	This proposal is dropped as per A 23/4/2011
24.	grundfoss pump Removed and packed seperately	Yes/NA	
25	MACHINE LIFTING PROCEDURE SKTCH PASTED ON THE GOURD	yes	

ATTENDED BY QUALITY ASSURANCE DISPATCH - INCHARGE

F-PAC-02-R0

CONTROLLED COPY

Sign: M Date: 1/8

ACE MANUFACTURING SYSTEMS LTD

PACKING CASE IDENTIFICATION LIST

AMS

MACHINE NUMBER : MH-772 DATE: 14-02-2013

SL.NO	CASE-1 CONTENTS	QUANTITY
1	Machine assembly	01NO
2	Std spare and Mic Manuals	01Box
3	Leveling screws and Lock Nut	4+4Nos
4	leveling Block 180MM	04Nos
5	Spin-Rods	01&2Nos
6	Coolant gun	01NO
7	Power Pack	01NO
8	Power Pack stand	01NO
9	APC pad	01Set
10		
11		
12		

CASE-2 CONTENTS

1	Coolant tank assembly	01NO
2	Coolant Pump	04Nos
3	Chipbin	01NO
4	chipbin Mesh	08No
5	Chucker plate	01NO
6	Orbin	01NO
7	base guard	01NO
8		
9		
10		
11		
12		

CASE-3 CONTENTS

1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		

CHECKED BY : S

ACE MANUFACTURING SYSTEMS LTD.

AMS
In Pursuit of Excellence

848

M/C.Sl. No. M111-772

PACKING LIST ANNEXTURE-II

M/s, SNY Autotech Pvt Ltd. Kancheepuram

Sl.No.	DESCRIPTION	QTY.
1	O'Ring Dia 19.1X1.6	2
2	O' Ring Dia 8.1X1.6	6
3	O' RingDia 11.1X1.6	4
4	O'Ring Dia 57.6X2.4	6
5	O'Ring Dia 24.5x3.0	6
6	O'Ring Dia 51.6x2.4	2
7	Cord Of O'Ring Dia 3 X 2 Mtrs	1
8	Spanner 10 x 11	1
9	Spanner 12 x 13	1
10	Spanner 18 x 19	1
11	Spanner 30 x 32	1
12	Spanner 36 x 41	1
13	Allen key set Metric.	1
14	Extension Allen Key **8mm**	1
15	Split Collet With Stud Push	2
16	Fuse Set (Fanuc Make)	1
17	Glass Fuse Set (Kept in Electrical Cabinet)	1
18	IML-Process Oil - 12 (Kept in Machine) **Servo Spin-12**	1
19	Pull Stud BT-40 (**DIN**,~~MAS 403~~)	24
20	Fanuc Series Oi Mate Model D / Oi-Model D **USER`S MANUAL**	1
21	Fanuc Series Oi-Model D / Oi Mate Model D **MAINTENANCE MANUAL**	1
22	Fanuc AC SERVO & SPINDLE MOTOR **Bis series /~~alpha series~~** MAINTENANCE MANUAL	1
23	Operating Instructions And Maintainance Manual	1
24	Spare Parts Manual	1
25	Air Cooler Manual	1
26		
27		
28		
29		
30		
31		
32		
33		
34		

Ace Micromatic
Group Company

Sign.....

DATE. 14-02-12

F-STR-27-R0

ACE MANUFACTURING SYSTEMS LIMITED.	AMS In Pursuit of Excellence
DISPATCH ADVICE	

Sl. No.	Description Of Dispatch details	
1	Customer Name	SNY AUTOTECH PVT LTD [Party Id - SNYAU001]
2	Telephone No.	044 - 27107578 / 9840949815
3	Fax No.	
4	E-Mail	snyautotech@yahoo.co.in
5	Customers ECC No.	AAHCS2789JXM002
6	Customers Sales Tax No. & Date	CST: 797370 21/07/2003 TIN: 33221662405
7	Sales Order no.	MO10002171
8	Machine Model	MCV-400
9	Machine No.	M11-772
10	Invoice Address	SNY AUTOTECH PVT. LTD., SURVEY NO. 297, 300, 301 & 302, SUGAMTHARUMPEDU VILLAGE ROAD,IRUNGATTUKOTTAI, SRIPERUMBUDUR TALUK, DIST., KANCHEEPURAM - 602 105
11	Place of Delivery	Default Address
12	Purchase Order No. & Date	CON/11-12/PUR/112089 Date:13/02/2012
13	P.O Value (Excluding Taxes & Duties)	Rs.28,90,000.00
14	Transportation	Door delivery through any reliable carriers and freight to-pay basis
15	Insurance	To be arranged and borne by AMS. Dispatch details to be faxed to the customer.
16	Scheduled Dispatch	14/02/2012
17	Payment	Refer payment collection detail sheet
18	Special Instructions	Keep the machine ready for dispatch. Contact HOD before dispatch Shortage list to be made while preparing machine Invoice All Pending Points to be attended before machine dispatch

Copy: Production/Electrical/Design/Purchase/Accounts/Dispatch/TSG

F-TSG-19-R1

For ACE MANUFACTURING SYSTEMS LTD.

Authorised Signatory

ACE MANUFACTURING SYSTEMS LIMITED

PACKING LIST

AMS
In Pursuit of Excellence

PACKING LIST NO.	DNo#006759	Date: 14/02/2012
SALES ORDER NO.	MO10002171	Date: 13/02/2012
CUSTOMER	SNY AUTOTECH PVT. LTD., KANCHEEPURAM	
P.O. NO.	CON/11-12/PUR/112089 Date : 13/02/2012	

Sl. No.	Item Id	Model	Qty
1	M11-772	MCV-400	1.000

Sl. No.	Description	Short Supply?
1	CNC Vertical Machining Center Model "MCV-400"	
	Spindle nose taper 7/24 No.40	
	AC Servo axes drive (Fanuc)	
	Twin Arm (Random) type High Speed Auto tool changer	
	Tool shank BT-40	
	LM Guideways for all the three axes	
	CNC System Fanuc 0IMATE-MD with 8.4" Colour LCD	
	Ball screw stretching for X & Y Axes	
	Panel cooler for electrical cabinet	
	Fail safe brake for Z-Axis	
	Standard work light	
	Coolant tank and chip tray	
	Three tier indication lamp	
	Flexible coupling for X, Y & Z Axes	
	Manual pulse generator	
	Rigid tapping	
	Total machine enclosure	
	Chip disposal to the rear side of the machine through two chutes on the sides	
	One set of manuals	
	Spindle speed of 60 - 6000 rpm (Indirect drive)	
	Rapid traverse of 30 / 30 / 30 m/min. for X, Y & Z Axes	
	Tool life management	
	Air drier unit (Small)	
2	Higher spindle power 11/7.5 kw in lieu of standard spindle 7.5/5.5 kw	
3	Flood coolant system around the spindle	
4	Chip flushing system on the base	
5	Coolant gun	
6	24T-BT4O-DIN Automatic tool magazine in lieu of 20 station tool magazine	
7	Automatic Pallet Changer (Pallet Size: 650 x 430) - Linear type to RH side	
8	Pull studs (24 Nos.)	
9	SERVO SPIN - 12 lubricating oil (1 liter)	
10	1 set of glass fuses(inside the cabinet)	
11	Electrical documentation manual & back up CD	
12	Geometrical test report	Yes
13	NASA test report	
14	Laser Calibration test report	
15	Heavy cut & Rigid tapping test report	

In side Tube Record-807

For ACE MANUFACTURING SYSTEMS LTD.

Authorised Signatory

ACE MANUFACTURING SYSTEMS LIMITED

WORK ORDER

Customer : M/s **SNY AUTOTECH PVT. LTD.,** SURVEY NO. 297, 300, 301 & 302, SUGAMTHARUMPEDU VILLAGE ROAD,IRUNGATTUKOTTAI, SRIPERUMBUDUR TALUK, DIST. KANCHEEPURAM 602 105 TAMILNADU	**SO. No:** MO10002171 **PO. No.:** CON/11-12/PUR/112089 **S.O.TYPE:** Standard	**WO. Date :** 14/02/2012 **PO. Date :** 13/02/2012

Sl. No	Item Id	Item Description	Unit	Qty
1	VMC	CNC VERTICAL MACHINING CENTER - M6V-400	Nos	1

MACHINE COLOUR :

Machine Colour : ACE Blue & Light Grey combination.

WARRANTY :

a) One year (i.e. 12 months) from the date of commissioning of the machine or 15 months from the date of supply of the machine whichever is earlier against faulty material and workmanship. Guarantee is not applicable for part, which has normal wear & tear (like springs, balls etc.) and certain other items like Bulbs, Lamps, Indicator Lamps, Revolving Lamps, Seals and Glass Fuses.

b) For Fanuc CNC System, motors and drives, it is Two Years from the date of dispatch. This warranty is applicable only for the first installation of the machine.

Scope of Supply, Technical Specs & CNC Features As Per **Annexure I**

Machine Inspection	Geometrical accuracies and machine acceptable as per test chart and NAS test
Customer Inspection Type	Direct
Customer Inspection Date	14/02/2012
Component Trials Required?	Not Applicable

FOR ACE MANUFACTURING SYSTEMS LTD.,

(P.RAMADAS)
MANAGING DIRECTOR

(L.S.UMESH)
CHIEF EXECUTIVE OFFICER

ACE MANUFACTURING SYSTEMS LTD.

PRODUCT SPECIFICATION SHEET

MODEL: MCV-400

Sl. No.	Description	Unit	MCV-400 (WITH APC)
1	**AUTOMATIC PALLET CHANGER**		
1.1	PALLET SIZE	mm	650 x 430
1.2	T - SLOT DIM.	N x W x P	4 x 18 x 100
1.3	NO. OF PALLETS	Nos.	2
1.4	WEIGHT ON PALLET MAX.	Kgf.	250
1.5	PALLET CHANGE TIME	Sec.	18
1.6	PALLET HEIGHT FROM FLOOR	mm	1070
2	**MACHINE CAPACITY**		
2.1	X - AXIS TRAVEL	mm	600
2.2	Y - AXIS TRAVEL	mm	450
2.3	Z - AXIS TRAVEL	mm	500
2.4	DIST. FROM SPINDLE FACE TO TABLE TOP	mm	160 - 660
2.5	DIST. FROM SPINDLE CENTER TO COLUMN GUIDEWAYS	mm	510
2.6	COOLANT SUPPLY-FLOOD COOLANT AROUND THE SPINDLE	lpm	50
3	**SPINDLE**		
3.1	SPINDLE NOSE TAPER		7 / 24 No.40
3.2	SPINDLE SPEED - INDIRECT DRIVE	rpm	60 - 6000
3.3	SPINDLE BEARING DIAMETER	mm	70
4	**FEED**		
4.1	RAPID TRAVERSE X, Y & Z AXES	m / min.	30 / 30 / 30
4.2	CUTTING FEED	mm / min.	1 - 10000
4.3	GUIDEWAYS FOR X, Y & Z AXES		Linear Motion
5	**AUTOMATIC TOOL CHANGER**		
5.1	NUMBER OF TOOLS	Nos.	24
5.2	MAX. TOOL DIA	mm	80
5.3	MAX. TOOL DIA. WITH ADJACENT POCKETS EMPTY	mm	155
5.4	TOOL SELECTION		Bi - Directional
5.5	TOOL LENGTH MAX.	mm	250
5.6	TOOL WEIGHT MAX.	Kg.	8
5.7	CHIP TO CHIP TIME (WITH Z ONLY / WITH X & Y ALSO)	Sec.	3.9 / 4.8
5.8	TOOL CHANGE SYSTEM		Twin Arm
5.9	TOOL SHANK TYPE		BT - 40
5.10	PULL STUD TYPE		DIN 69872 Form-B
6	**ACCURACY WITH ROTARY ENCODER**		
6.1	POSITIONING FOR X, Y & Z AXES AS PER VDI DGQ 3441	mm	0.015
6.2	POSITIONING FOR X, Y & Z AXES AS PER ISO 230-2	mm	0.010
6.3	REPEATABILITY FOR X,Y&Z AXES AS PER VDI DGQ 3441	mm	± 0.005
6.4	REPEATABILITY FOR X, Y & Z AXES AS PER ISO 230-2	mm	± 0.003
7	**MOTORS**		
7.1	SPINDLE POWER 15 MIN./CONT. (FANUC)	kW	11 / 7.5
7.2	FEED MOTOR X, Y & Z AXES (FANUC)	Nm	11 / 11 / 20

ACE MANUFACTURING SYSTEMS LTD.

PRODUCT SPECIFICATION SHEET

MODEL: MCV-400

Sl. No.	Description	Unit	MCV-400 (WITH APC)
8	**INSTALLATION DATA - STANDARD MACHINE**		
8.1	MACHINE WIDTH	mm	3300
8.2	MACHINE DEPTH	mm	3300
8.3	MACHINE HEIGHT	mm	2875
8.4	MACHINE WEIGHT - NET	Kgf	4900
8.5	POWER SUPPLY FOR BASIC MACHINE	Kva	16
8.6	CNC CONTROL (WITH 8.4" COLOUR LCD)		FANUC 0IMATE-MD
8.7	PNEUMATIC PRESSURE REQUIRED	Kgf / Cm2	5
8.8	PNEUMATIC SUPPLY REQUIRED AT ATMOSPHERIC PR.	Lpm	300
9	**TANK CAPACITY - STANDARD MACHINE**		
9.1	COOLANT TANK	litr	300 400
9.2	LUBRICATION TANK (USEFUL QUANTITY)	litr	5 (4)
9.3	HYDRAULIC POWER PACK	litr	32
10	**MACHINE FEATURES / ACCESSORIES - STANDARD**		
10.1	PANEL COOLER FOR ELECTRICAL CABINET		
10.2	FAIL SAFE BRAKE FOR Z - AXIS		
10.3	STANDARD WORK LIGHT		
10.4	COOLANT TANK & CHIP TRAY		
10.5	THREE TIER INDICATION LAMP TO INDICATE CYCLE RUNNING READY FOR NEXT CYCLE & MACHINE IN ALARM STATUS		
10.6	FLEXIBLE COUPLING FOR X, Y & Z AXES		
10.7	MANUAL PULSE GENERATOR		
10.8	RIGID TAPPING		
10.9	TOTAL MACHINE ENCLOSURE		
10.10	CHIP DISPOSAL TO THE REAR SIDE OF THE MACHINE THROUGH TWO CHUTES ON THE SIDES		
10.11	MANUALS - ONE SET		
10.12	BALL SCREW STRETCHING FOR X & Y AXES		
10.13	TOOL LIFE MANAGEMENT		
10.14	AIR DRIER UNIT (SMALL)		
11	**STANDARD CNC FEATURES OF FANUC-0IMATE-MD**		
11.1	THREE CONTROLLED AXES		
11.2	SIMULTANEOUS CONTROLLABLE AXES - 3 AXES		
11.3	FEED RATE / RAPID OVERRIDE FACILITY		
11.4	CIRCULAR INTERPOLATION - ANY TWO OUT OF THREE AXES		
11.5	JOG MODE		
11.6	BACKLASH COMPENSATION		
11.7	PITCH ERROR COMPENSATION		
11.8	MANUAL DATA INPUT		
11.9	DRY RUN		
11.10	SINGLE BLOCK		
11.11	PROGRAM PROTECT KEY		
11.12	SELF DIAGNOSTIC FUNCTION WITH SYSTEM & MACHINE ALARM STATUS DISPLAY		
11.13	EMERGENCY STOP		
11.14	MIRROR IMAGE (X & Y CO-ORDINATES)		

ACE MANUFACTURING SYSTEMS LTD.

PRODUCT SPECIFICATION SHEET

AMS — In Pursuit of Excellence

MODEL: MCV-400

Sl. No.	Description	Unit	MCV-400 (WITH APC)
11.15	TOOL OFFSET MEMORY - 400 NOS.		
11.16	TOOL LENGTH COMPENSATION		
11.17	PART PROGRAM MEMORY 1280 MTS (512K) BATTERY BACKUP		
11.18	CUTTER DIAMETER COMPENSATION		
11.19	CANNED CYCLES FOR DRILLING, TAPPING AND BORING		
11.20	PROGRAMMABLE DATA INPUT (G10)		
11.21	HELICAL INTERPOLATION		
11.22	RIGID TAPPING		
11.23	CLOCK FUNCTION		
11.24	LOOK AHEAD CONTROL		
11.25	READER PUNCHER INTERFACE (RS 232)		
11.26	WORK CO-ORDINATE SYSTEM (G53 TO G59)		
11.27	ANGLE CHAMFERING / CORNER RADIUS		
11.28	CUSTOM MACRO-B		
11.29	CO-ORDINATE SYSTEM ROTATION		
11.30	NUMBER OF REGISTERED PROGRAMS TOTAL 200		
11.31	BACK GROUND EDITING		
11.32	EXTENDED PART PROGRAM EDITING		
11.33	RUN HOUR PARTS COUNT DISPLAY		
11.34	MANUAL PULSE GENERATOR		
11.35	ALARM HISTORY & OPERATION HISTORY		
11.36	SPINDLE LOAD DISPLAY		
12	**MACHINE FEATURES / ACCESSORIES - ADDITIONALLY ORDERED BY M/S SNY AUTOTECH PVT.LTD., CHENNAI.**		
~~12.1~~	HIGHER SPINDLE POWER 11/7.5 KW IN LIEU OF STANDARD SPINDLE 7.5 / 5.5KW		
~~12.2~~	FLOOD COOLANT SYSTEM AROUND THE SPINDLE		
~~12.3~~	CHIP FLUSHING SYSTEM ON THE BASE		
~~12.4~~	COOLANT GUN		
~~12.5~~	24 TOOL ATC IN LIEU OF 20 TOOL ATC		
~~12.6~~	AUTOMATIC PALLET CHANGER - PALLET SIZE [650 X 430] - LINEAR TYPE TO RH SIDE		

V.G 2-4

ACE MANUFACTURING SYSTEMS LTD

MACHINE SEQUENCE RUNNING TEST REPORT

AMS — In Pursuit of Excellence

MACHINE NUMBER : M11-772

Preconditions for Machine before starting Machine Sequence Running

1. Machine should be fitted with all accessories as per Work Order
2. Machine should be cleared from all Electrical & Mechanical Remarks
3. Machine should be run with Sufficient Coolant.
4. ATC should be full with tool holders
5. No Bypass of any Safety switches and cabinet sealed.

Date	Time	No of Hours Running	Date	Time	No of Hours Running	Remarks / Alarm Observed
09/2/12	8PM	1 ✓			17 ✓	
		2 ✓			18 ✓	–
		3 ✓			19 ✓	
		4 ✓			20 ✓	
		5 ✓			21 ✓	
		6 ✓			22 ✓	
		7 ✓			23 ✓	
		8 ✓			24 ✓	
		9 ✓				
		10 ✓				
		11 ✓				
		12 ✓				
		13 ✓				
		14 ✓				
		15 ✓				
		16 ✓				

Nature of Remarks Related to	1.Hydraulic leakage	2. Coolant leakage	3.Electrical alarm	4. ATC	5. Spindle	6. Axes	7. Others

Note : In case M/c is stopped due to any reason, write the Sl. No. of the above remarks against corresponding number of hours, set right the M/c & continue to run for a total period of 24 hrs.

In charge - Mechanical **In charge - QA**

ACE MANUFACTURING SYSTEMS LTD

SPINDLE TEMPERATURE STABILIZATION REPORT

AMS *In Pursuit of Excellence*

M/c. NUMBER : M11 - 772	MAXIMUM RPM : 6000	DRIVE: ID/DD/IDCTS/DDCTS
SPINDLE SL No. : 10508	PULLEY RATIO : 1:1 / 1:1.25 / 1:1.5 /1:1.75 / 1:2	

Note : Refer TM-QA-07 for test procedure. **Recommended Temp. Rise : 20°C max. above ambient**

% of current drawn at max RPM of the spindle : (Recommended max : DD : β3 = ≤ 15%, β6 / α6 & above = ≤10%. ID : β3 = ≤ 20%, β6 / α6 & above = ≤15%.	Measured = 09 %
Spindle noise at max RPM [Recommended : 80 dB(A) upto 10 kW, 85 dB(A) for > 10 kW]	Measured = 78 dB (A)

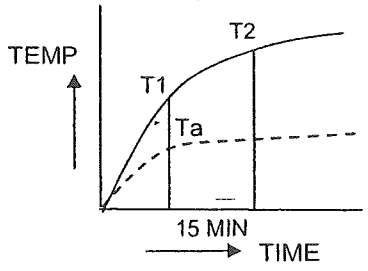
SPINDLE RPM	TIME	AMBIENT TEMPERATURE	SPINDLE TEMPERATURE
500	8:30	26°C	—
	9:00	26 °C	—
—	9:30	26°C	32°C
	10:00	27°C	36 °C
	10:30	27°C	40°C
	11:00	27°C	42 °C
	11:30	28°C	43°C
	12:00	28°C	44°C
	12:30	28°C	45 °C
	01:00	28°C	46°C
	01:30	28°C	47°C

1. The Stabilization point, Calculated as follows. Is the Temperature at which the rate of temperature rise over 15 minutes is less than 5%.

= [(T2 – T1) / (T1 –Ta)] X 100

= < 5%

T1 = Previous Temp., T2 = Temp. after 15 Min & Ta = Ambient Temp. at T1



2. RUN SPINDLE AFTER STABILIZATION FOR 3 HOURS.

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MACHINE HEALTH CHECK POINTS

MACHINE NUMBER : M11-772 **DATE :** 9/2/12

I	**AXIS**	
1	Ensure plugs are fitted for L M Guide ways.	✓
2	Ensure effective wiping of saddle cover wipers.	T.C cover
3	Ensure the lube timer is set 12 min for HG and ~~20~~ min for LM Guide ways.	✓
4	Ensure leveling bolts are fixed with locknut.	✓
5	Check the Table tenon slot 14H8 / ~~18H8~~ (Tolerance 0 to +0.027 mm)	✓
6	Ensure butt reference function.	✓
7	Check all the axes reference arrows are matched. only 2 axes	✓
8	Check Emergency push button function of Machine / APC / Double Hand	✓
9	Ensure marking of all screws with Yellow paint marker	✓
10	Ensure cover fixed to Index table, rotary joint & tool breakage sensor	— N.A —
11	Ensure smooth movement of sliding door without play and unhealthy noise	✓
II	**SPINDLE**	
1	Ensure air jet through all the holes (4/8 holes) at oriented spindle position	✓
2	Ensure supply of air to spindle Air Purge system	— N.A —
3	Check the tightness of push stud.	✓
4	Ensure grub screws are fitted to spindle flange.	✓
5	Check spindle orientation position and note down the value. (Parameter No. : 4031 for Fanuc controller, ... for Siemens controller)	888
6	Ensure spindle balancing is completed	✓
7	Ensure presence of gap between draw bar bush & ejection screw / roller (visually or by feel)	✓
8	Check the "Tool not in spindle" proximity message.	— N.A —
III	**ATC**	
1	Ensure ATC / APC / Counter balance bracket pinning.	✓
2	Ensure oil level is max. in ATC gear box.	✓
3	Check the Tool Number display on panel board is same as in spindle	✓
4	Check tool change position limit switch for proper functioning & band width (Max. 1.5 mm on each side).	✓

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MACHINE HEALTH CHECK POINTS

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IV		COOLANT
1	✓	Ensure no coolant leakage around the machine & presence of coolant in lubrication oil drain pipe
2	✓	Ensure proper working of float switch (by shifting the sensor upwards).
3	✓	Check the function of Flood coolant (M7), Chip push (M182), Base flush (M181), CTS (M52) & Coolant gun (through selector switch).
4	N.A	Ensure drum wash in conveyors (Optional)
5	N.A	Check conveyor overload message & Direction (Optional)
6	N.A	Check oil in conveyor gear box. (Optional)
7	N.A	Ensure functioning of screw conveyor in forward & reverse directions (through selector switch).
8	N.A	Ensure CTS flow switch setting by opening the bag filter air vent ball valve (Optional)
9	N.A	Ensure suction strainer & Check Valve in CTS tank & pr. gauge functioning (Optional)
10	N.A	Ensure CTS pin locator hole is plugged (Optional)
11	N.A	Ensure that twin arm plunger & CTS pin locator screw do not foul (Optional)
12	N.A	Ensure bag filter clog message by closing bag filter outlet ball valve (Optional)
13	N.A	Ensure High pressure air blast after CTS switched off (Optional)
V		APC
1	N.A	Check pallet initialization in APC Machine
2	N.A	Ensure supply of air jet to pallet rest pad cleaning.
3	N.A	Ensure pallet change takes place smoothly with specified load (as per work order) without abnormal / unhealthy noise
VI		HYDRAULIC, PNEUMATIC & LUBRICATION
1	✓	Ensure air pressure switch is working properly.
2	✓	Ensure proper setting of oil (1 drop / tool change) in FRL unit & Oil Level low message
3	✓	Check the FRL drain hose is fixed.
4	✓	Ensure identification name plates are fixed on power pack hoses.
5	N.A	Check the Hyd.oil level low message (by adjusting sensor position) & Pr. gauge function
6	✓	Ensure hydraulic flow control valve grub locking in APC machine.
7	✓	Ensure Ball valve fixed to FRL Unit inlet
8	✓	Ensure no oil leakage in hydraulic valves & fittings
9	✓	Ensure lubrication pressure switch is set to 15 bar.
10	N.A	Ensure pneumatic quick change coupling fixed to 4th axis provision
VII		GENERAL
1		Check all the points of MOM are attended.
2	✓	Ensure coolant tank capacity engraved or marked in name plate
3	✓	Check Hydraulic specification is as per circuit diagram
4	✓	Ensure axis calibration & ball bar parameter as per the report.
5	✓	Ensure Ball screw pitch in the machine and report is same.
6	✓	Ensure spindle Sl. No. and parameters are as per the report.

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MACHINE HEALTH CHECK POINTS

MACHINE No. : M11-772 ASSEMBLY TECHNICIAN : Mr- Hemraju

S.No	REMARKS OBSERVED	STATUS	SIGN
(1)	Column RH Side Lub hose routing to be done	✓	✓
(2)	ATC Pinning	✓	✓
(3)	Coolant float switch setting	✓	✓
(4)	APC door movement is not smooth	✓	✓
(5)	Y axis noise is more to be rectified	✓	✓
(6)	APC base & guide block pinning.	✓	✓
(7)	Lock nut fixing for APC & MC base	✓	✓
-	leveling screws	✓	✓
(8)	All screws tightness to be checked and	✓	✓
	marked with yellow marker.	✓	✓
(9)	ATC gear box, practical test reports to	✓	✓
	be filed		

ASSEMBLY IN CHARGE

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CHECK LIST - MECHANICAL

MACHINE NUMBER : M11-772 **DATE :** 9/2/12

I Z-axis motor current during tool declamp (Turret type ATC) : Recommended Max : BT30-150%, BT40-180%)

A	With Z axis moved at 840 mm/min in Jog mode	N.A
B	In Auto cycle	

II Z - axis drop during power off :

- ve' direction< 0.005 mm		'+ ve' <0.015mm direction	
System Initial Reading :	-250.000	System Initial Reading. :	-250.000
System Final Reading :	-250.009	System Final Reading. :	-250.007
System Diff. in Reading :	0.009	System Diff. in Reading. :	0.007
Dial Reading. :	0.010	Dial Reading. :	0.007

III % of Axis current drawn (Recommended as per Data sheet DS -QA -02)

A	Axis	25 % Rapid Rate	50% Rapid Rate	100% Rapid Rate	600 mm/m Feed rate
	X	15%	18%	48%	12%
	Y	18%	20%	50%	16%
	Z Up	56%	58%	65%	53%
	Z Down	32%	30%	28%	35%
	B - Forward				
	B - Reverse				

B % of Axis holding current (Recommended as per Data sheet DS -QA -02)

Axis	100% Rapid Rate		2000 mm/min. Feed Rate	
	Motor end	Brake end	Motor end	Brake end
X	07%	10%	02%	05%
Y	08%	11%	04%	05%
Z	53%	36%	44%	44%

C Over travel against Axis dead stopper in mm (Recommended as per Drawing)

Axis	Brake end	Motor End
X	07.800	39.050
Y	21.480	17.500
Z	22.300	08.850

IV SPINDLE

A	Push stud stroke length for Tool ejection (mm)	Recommended values	Auto	Manual
	BT – 30/40 /50	Refer DS-DSN-PROD-01	0.75	0.75
	HSK A50/63/100		—	—
B	Tool clamping force-Kgf	Refer DS-DSN-PROD-01	860 kgf	

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C	Axial play in the spindle (Ref. TM-QA-10 for procedure) (Recommended max.except BT50 : 0.005 mm , for BT 50 : 0.008 mm)		
	1. Dial Reading. 'R1' (Ref. Step D)	0 ⟶ +0.040 ⟶ 0	Difference = mm (R1~R2) 0.002
	2. Dial Reading. 'R2' (Ref. Step G)	0 ⟶ -0.035 ⟶ 0.033	
V	Turret type ATC		
A	Check the spring lever compression at :		
	1. After the magazine butts against the dead stopper (mm)		N.A
	2. Z – axis position below zero position (mm)		N.A
	3. Recommended : BT 40 - 2.5 to 3 mm , BT 30 - 1.5 to 2 mm (i.e. 1-2)		N.A
B	Tool picking position for Turret Type ATC Recommended : BT 40 - 60 to 80 mm , BT 30 - 50 to 65 mm		N.A
VI	A.P.C		
A	Pressure setting for the pallet transfer cylinder in kg/cm^2 (As per circuit diagram) - N.A -		
B	Pressure setting for the pallet clamp-de clamp in kg/cm^2 (As per circuit diagram) - N.A -		
VII	Cycle Time in seconds (Refer Work order for recommended value)		
	1. Check the tool to tool time		2.00Sec
	2. Check the chip to chip time with Z axis only		4.10Sec
	3. Check the chip to chip time with all axes		4.80 Sec
	4. Check the pallet changing time		- N.A -

CHECK LIST – MECHANICAL

AMS

1. MACHINE LEVEL READING :

Along X-Axis / Along Y-Axis

3/4 →	0	1 →
3/4 →	0	3/4 →
3/4 →	0	3/4 →

1/2 ↓	0	↑ 1/4
0	1/2	↑ 1/2
1/4 ↓	3/4 ↓	0

2. TABLE TOP READINGS:

Along X-Axis

0	+3	+5
0	+7	+9
0	+3	-1

Along Y-Axis

-4	-3	-2
-12	-10	+1
0	0	0

3. TENON SLOT READING :

0	+6	+2

PALLET – II

0	+16

PALLET – II

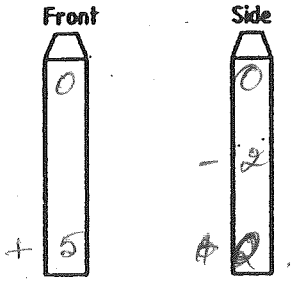
Along X-Axis

0	+2	-3
0	+4	+2
0	+6	+4

Along Y-Axis

0	0	0
+6	+10	+8
-3	+9	+5

5. MANDREL READINGS:

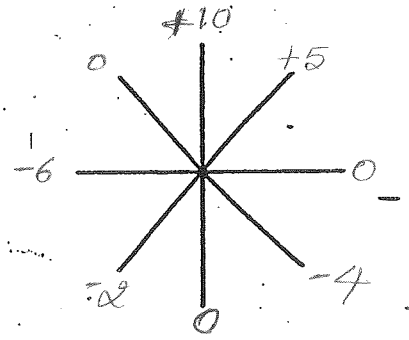


RUNOUT:

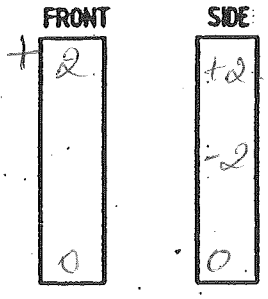
a. TOP : (0.010mm)
1. 0.002

b. BOTTOM :(0.020mm)
1. 0.005
2. 0.007
3. 0.004
4. 0.003

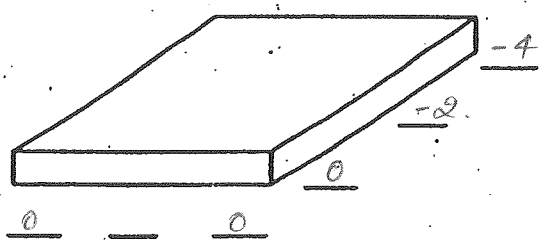
6. TURN AROUND READING:



4.COLUMN READINGS :



7. SQUARENESS READING (X & Y):



8. PALLET HEIGHT VARIATION:

Pallet-1: 0.000

Pallet-2 : 0.007

NOTE: 1. For recommended value refer Design Data sheet DS-DSN-PROD-TC 01-06.

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MACHINE HISTORY - MECHANICAL

AMS — In Pursuit of Excellence

MACHINE NUMBER : M11-772 **DATE :** 9/2/12

1	ATC Details : Model	4020V / 4024V / 4030V / 4020H / 4024H /HSK A63 /HSK A100		
1.1	ATC :Turret Type/Twin Arm	Sl.No	14985	
1.2	Turret type magazine Encoder	Make	Model	Sl. No
		Pragati	ATC12 / ATC16 / ATC20	
1.3	ATC Torque Motor	TM 50 / 63 / 80 / 100 / 125 / Servo ATC		
2	**BELT / COUPLING DETAILS :**			
		GT Belt (For Indirect Drive)	COUPLING (For Direct Drive)	
2.1	Make	Unitta / Power Grip	Mayr	
2.2	Model			
3	**COOLANT SYSTEM DETAILS :**			
	Make : Rajmane Industries		Model	Type
3.1	Flood Coolant Pump - Grown/Flexinox/Disc Nozzle		RKM 206 / RV 160 - 270	Centrifugal Pump
3.2	Chip Pushing Coolant Pump		RKM 206 / RV 160 - 270	
3.3	Base Flushing Coolant Pump		RKM 206 / RV 160 - 270	
3.4	Coolant Gun & Drum Wash Pump		RKM 206 / RV 160 - 270	
3.5	Oil Skimmer - Make / Model	N.A		
3.6	High Pressure Coolant Pump	N.A	Make	GF/ WP / BP
			Model	
3.7	Pressure Relief Valve	N.A	Make	Polyhydron
			Model	DPRH-10T-100-04
3.8	Rotary Joint		Make	Deublin / RIX
			Model	
4	**FLOAT SWITCH :**			
	Make		Filpro Electronics	
5	**CHIP CONVEYOR DETAILS :**			
5.1	Type of Conveyor	Scraper / Slat cum Scraper / Magnetic along X or Y axis.		
5.2	Model			
5.3	Make	Universal Conveyor		
5.4	Sl.No	N.A		
6	**CONVEYOR GEAR BOX DETAILS :**			
6.1	Model	N.A		
6.2	Make	Universal Conveyor		
7	**FILTER SPECIFICATIONS :**			
7.1	Model / Working pressure	8 /15, 50 bar / 100 bar		
7.2	Make	Universal Conveyor		
7.3	Sl.No	N.A		
8	**COIL / SCREW CONVEYOR DETAILS :** Qty : 1 / 2			
8.1	Make	Universal Conveyor		
8.2	Model	N.A		

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MACHINE HISTORY - MECHANICAL

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9	HYDRAULIC SYSTEM :			
9.1	Make		Pascal / Hyd Tech / Rexorth / Yuken /Jai Chamundi /Hymech	
9.2	Product No		QC. No : 6350	
9.3	Tank Capacity (Litres)		32 / 40 / 63 / 70	
9.4	Maximum Working Pressure		45 BAR	
9.5	HYDRAULIC PUMP	Make Rexroth / yuken /Corotech / vickers	Model	
9.6	MOTOR	Make Rotomotive / Siemens	Model 90L-4-1400 / 90S-4-1390 ,	
10	PRESSURE SWITCHES FOR HYDRAULIC SYSTEM :			
10.1	Make		Polyhydron	
10.2	Model		1PS 30-50 / 30-100 / 10 - 30 /	
10.3	Quantity		02	
11	HYDRAULIC ACCUMULATOR :			
11.1	Make	FOX / BOSCH	FOX / BOSCH	
11.2	Set Oil Pressure	Tool de clamp :	Counter balance:	
11.3	Gas Pressure	Tool de clamp :	Counter balance:	
11.4	Volume (Litres)	Tool de clamp : 1.4	Counter balance : 4 or 10	
12	HYDRAULIC CYLINDERS :			
	Make	Model	Function	Qty
12.1	Island Hydraulics	Ø40X22 X [125/152/180/1150]	Pallet Forward / Reverse	02
13	LUBRICATION UNIT DETAILS	Make	Pressure Setting	
		GENLUB / DROPCO	15 Kg/ cm2	
14	FRL UNIT			
	Make	Model	Operating Range	Grade of Oil
	Shavo / Festo / Norgan	MINI / MIDI	0-16 Kg/cm2	Servo Way Spin 12
15	DRYER	Make		Specification
		Trident		VXD - 2
16	PNEUMATIC VALVES			
	Application	Specs	Make	Qty
16.1	Spindle Air Jet	MFH 3-1/8	Festo	01
16.2	Spindle Air Purge	MFH 3-1/8	Festo	
16.3	ATC Pocket up & Down	JMFH 5-1/8	Festo	01
16.4	Manual Tool De Clamp	MFH 5-1/8/JMFH 5-1/2	Festo	01
16.5	APC Auto Door	JMFH 5-1/8	Festo	01
16.6	CTS Air Jet	MFH 3-1/8	Festo	
16.7	Front Auto Door	JMFH 5-1/8 MFH 3-1/8	Festo Festo	
16.8	Pallet Rest Pad Cleaning	MFH 3-1/8	Festo	01
16.9	Fixture		Festo	

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MACHINE HISTORY - MECHANICAL

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17	PRESSURE SWITCH FOR PNEUMATIC SYSTEM			
17.1	Operating Range		2 to 15 Kg /cm2	
17.2	Make		SUCO	
17.3	Quantity		1 + 1	
18	Tool de clamp system:		Mechanical / Pneumatic / Hydraulic / Hydro-Pneumatic	
	Function	Make	Model	Qty
18.1	Manual Tool De Clamp - Pneumatic	Festo	DNU 80-80-PPV DNU 100-80-PPV	1
		Rotex	SCN D -100 / 60 SCN D 80 / 80	1
18.2	Manual Tool De Clamp - Hydraulic	Island Hydraulic / AMS	ϕ40 X 20 X 65	1
18.3	Manual Tool De Clamp - Hydro pneumatic	Preac	PC33-8045-AM02	1
19	PNEUMATIC CYLINDERS USED			
	Function	Make	Model	Qty
19.1	Front Auto Door	Festo	N.A	
19.2	APC Auto Door	Festo	DNC-32-550PPV-A IN0	
20	TURRET / INDEXING TABLE (OPTIONAL)			
	Make	Model	Sl.No	No of Station
	Pragathi / AMS		1	8 / 12 /
21	ROTARY TABLE	Make	Model	Sl.No
		NIKKEN / UCAM		
22	TAIL STOCK N.A	Make	Model	Sl.No
23	PNEUMATIC COUNTER BALANCING			
23.1	Make	SMC / Rotex / Festo		
23.2	Cylinder Diameter :	Ø 80 / 100 / 125		
23.3	Air Reservoir used	N.A Yes / No		
24	HYDRAULIC COUNTER BALANCING			
24.1	Make	N.A		
24.2	Model	Ø 50 X 32 X 670		
25	ADDITIONAL INFORMATION / REMARKS			
1				
2				
3				

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CHECK LIST - ELECTRICAL

M/C NUMBER:- M11-772 | DATE:- 03/02/2012

I.AXES

SL.No	PARAMETERS	STATUS	REMARKS
1	Specifications of servo motors are as per work order	✓	
2	Axes strokes are as per work order. (P1320 onwards)	✓	
3	All servomotors are taking normal current while running	✓	
4	Check Motor ID & other parameters of axis motor (P2020)	✓	
5	Axis cables are routed properly with suitable clamps & ties	✓	
6	Cables are connected properly both at motor & drive end	✓	
7	Lubrication timers & signals are as required to the machine	✓	
8	Rapid rate & feed rate of axes are as per workorder (P1420/1428)	✓	

II.SPINDLE

SL.No	PARAMETERS	STATUS	REMARKS
1	Specifications of spindle motor /speed /power is as per work order	✓	
2	Speed locking parameters If any are perfectly set (P3735,3736,)	✓	
3	Power locking parameters If any are perfectly set (P4028,4029)	✓	
4	Rigid tap parameters are set (P4016#4,4044,4052,4085,5241,5261)	✓	
5	Spi. running is checked for direction, noise if any, low & high speed	✓	
6	Spindle motor fan rotation /mpcb rating is checked	✓	
7	Spindle orientation is perfectly set (Proximity signal, parameter)	✓	
8	Manual tool declamp in jog is checked in twin arm models	✓	
9	Switching of Contactors for required speed incase of star/delta motor	N/A	
10	Check cable connections both at motor & drive end	✓	

III.ATC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for latest tool change program(**O9001**) of the machine	✓	
2	Check for macro of twin arm machines	✓	
3	Check for Magazine indexing in both directions	✓	
4	Auto tool change is checked both in jog & auto mode	✓	
5	Check for position sw parameters,maco& reference parametres	✓	
6	Proximity sw cables are properly connected with tags & routed neatly	✓	
7	Tool display is matching with atc position	✓	
8	Power OFF conditions - Message and retrievals are checked	✓	

IV.APC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for pallet change program(**O9003&100**) of the machine.	✓	
2	APC is operating manually through mop & in auto through **M80**	✓	
3	Proximity sw & limit sw signals are checked for condition during	✓	
	different operation	✓	
4	Working of pallet setup sequence	✓	
5	Interrupt conditions checked & initialization procedure is displayed	✓	
6	Working of fixture interlock if any for pallet change is checked	N/A	
7	Cables are routed properly with suitable clamps & ties	✓	
8	Check for position sw parameters,maco& reference parametres	✓	

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V.PNUEMATIC

SL.No	PARAMETERS	STATUS	REMARKS
1	All solenoids / pressure switches are working for specified condition	✓	
2	Messages related to pressure sw settingis appearing when disturbed	✓	
3	Suitable element tags & indicators are put	✓	
4	Cables / plugs are terminated & routed properly	✓	
5	FRL unit signals are functioning properly (check FRL counter)	✓	

VI.HYDRAULIC

SL.No	PARAMETERS	STATUS	REMARKS
1	All solenoids /pressure switches are working for specified conditions	✓	
2	Messages related to pressure sw settings appearing when disturbed	✓	
3	Hydraulic Motor / cooling fan direction is checked	✓	
4	Oil level ,clog, temperature sensors if any, are working	✓	
5	Suitable element tags & indicators are put	✓	
6	Cables / plugs are terminated & routed properly	✓	

VII.COOLANT

SL.No	PARAMETERS	STATUS	REMARKS
1	All coolant motors are running in cw direction	✓	
2	Timers for running all coolant / conveyor motors is checked	✓	
3	Float sw signals are functioning properly	✓	
4	Flow sw,clog sw & air blast is functioning properly in case of cts m/c	NA	
5	Door close condition for coolant 'on' is checked if it is a option	NA	
6	Chip/ screw conveyors are working properly in auto & manual mode	NA	
7	Limit sw / proximity sw signals are functioning properly	NA	
8	MPCB used and settings done is as per the requirement	✓	

VIII.ROTARY/ INDEX TABLES NA

SL.No	PARAMETERS	STATUS	REMARKS
1	Table is checked for commanded positions & shortest path		
2	Related logic is checked for proper running at safe position		
3	Feedback signals are functioning properly		
4	Related displays are marching with table position		
5	Cable/ hoses are routed properly at safe position		

IX.PANEL

SL.No	PARAMETERS	STATUS	REMARKS
1	MCB's are proper rated & connected as per circuit diagram	✓	
2	Rated MPCB's are set & connected as per circuit diagram	✓	
3	Contactors' of required rating & connected as per circuit diagram	✓	
4	Transformers of required rating & connected as per circuit diagram	✓	
5	Relay boards of proper rating & connected as per circuit diagram	✓	
6	Cables of proper rating & properly crimped at connection end	✓	
7	Drive cooling fans are working normally	✓	
8	Panel cooler-setting & working is checked	✓	
9	Proper identification stickers / tags are fixed for all elements	✓	

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CHECK LIST - ELECTRICAL

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10	Cabinet is fully checked for loose connections if any	✓	
11	PCB , drive, I/O module connectors are firmly fixed	✓	
12	Machine operating panel connections are checked and routed	✓	
13	Moisture free beadings for cabinet are properly fixed	✓	
14	Moisture free beadings for operating panel are properly fixed	✓	

X.GENERAL

SL.No	PARAMETERS	STATUS	REMARKS
1	Illuminations of work lamps & indication lamps are satisfactory	✓	
2	Emergency circuit & monitor is functioning properly	✓	
3	Connections inside pendent are checked	✓	
4	Protective covers for the elements are given if required	✓	
5	All channel covers are fixed with proper fasteners	✓	
6	Protective earth is connected for elements which requires earthing	✓	
7	Safety instructions & other standard stickers are fixed on cabinet	✓	
8	PMC timer values are checked as per data sheet	✓	
9	New design changes are incorporated in the machine	✓	

INTERNAL CHECKLIST POINTS

→ ~~MT27, XRSTO1, X18, X12, X13~~ tags (All API related tags).
→ ~~X14~~ Sticker, ~~X25~~, X33
→ ~~TB1~~ Jn box name plate & sticker.
→ Servo guide beeeep pending
→ ~~Fuse~~ panel, ~~IBT~~ & ~~TBT~~ fuse to be put.
→ ~~MT~~ no/c plate to be fixed.
→ TB1 ferrule has to be put.

XI. PMC DATA

Instruction: Please note all D-bits which is enabled at the time of shipping .
Also give the details if it is used for special function

D0	7	6	5	(4)	3	(2)	(1)	(0)	D13	7	6	5	4	3	2	1	0
D1	7	6	5	4	3	2	1	0	D14	7	6	5	4	3	2	1	0
D2	7	6	5	4	3	2	1	0	D15	7	6	5	4	3	2	1	(0)
D3	7	6	5	4	(3)	2	1	0	D16	7	6	5	4	3	2	1	0
D4	7	6	(5)	4	3	2	1	0	D17	7	6	5	4	3	2	1	0
D5	7	6	5	4	3	2	1	0	D18	7	6	5	4	3	2	1	0
D6	(7)	6	5	4	3	2	1	0	D19	7	6	5	4	3	2	1	0
D7	7	6	5	4	(3)	2	1	0	D20	7	6	5	4	3	2	1	0
D8	7	6	(5)	4	3	(2)	1	(0)	D21	7	6	5	4	3	2	1	0
D9	7	6	5	4	3	2	1	(0)	D22	7	6	5	4	3	2	1	0
D10	7	6	5	4	3	2	1	0	D23	7	6	5	4	3	2	1	0
D11	7	6	5	4	3	2	1	0	D24	7	6	5	4	3	2	1	0
D12	7	6	5	4	3	2	1	0	D25	7	6	5	4	3	2	1	0
D																	
D																	

CHECKED BY:- VERIFIED BY:-

D110	00000000	1	1	
D100	00000000	0	10	
D400	00000010	0	21**	** NO. OF ATC TOOLS +1 (NOT APLICABLE FOR DISC TYPE ATC
D500	00000010	0	21**	
D600	00000010	0	21**	
D700	00000010	0	21**	
D800	00000010	0	21**	
D000	00000000	3	30	

XII.CNC PARAMETERS

Instruction: Please Set the following parameters at the time of shipping .
Also give the details if any special parameter has set

1301#7 = 1 Over travel before move	
3103#2 = 1 System alarm history display.	
3106#4 = 1.Operation history enable.	
3108#6 = 1 Spindle load meter	
3111#7 = 0 (NPA) - Message display	
3112#0 = 0 Graphics display enable	
3112#2 = 1 Operator message history display	
3112#3 = 1 Alarm description display in alarm history screen for external alarms	
3112#5 = 0 Operator history enable	
3117#1 = 1. Spindle position feedback display in dgn 442	
3205#4 = 1 In the offset screen offset value CLEAR by soft key to be disabled.	
3402#6 =1 (CLR)Set the parameter	
1601#4 = 1 (RTO)Set the parameter	
1722 = 10/10/10 Set for RTO percentage	
1826 =(500,500,500) ,1827=(200,200,200)-check inposition width	
6910 - 6965 -check position switches values for ATC and APC	
8132.0=1, - TOOL LIFE Parameters	
6800#0 , 6801#3 , 6801#7 ,6804#2,6805#4 =1 6810=100, 6811=6, 6844=10, 6845=0	
1620 & 1621 Rapid time constant	

Check the following with system:		"1"		"0"	
No.of Ladder steps**	Dgn:1141 # 6	8000		5000	**N/A for 0 iMD
4th axis option**	1279 # 4 OR 1148 # 7	Yes		No	**N/A for 0 iMD.
Macro executer option	Dgn:1236 # 1	Yes		No	

XIII.SHIPPING

Instruction: Please perform following activities before shipping .
Also give the details if any special activity is done

SL.No	PARAMETERS	REMARKS
1	Take the servo guide back up	
2	Check the various communication like RS232C,PCMCIA	
3	Edit title,& set time, date other original factory set values	
4	Take all programs back up &Lock 8000 & 9000 series programs (P3202)	
6	Take the back up of following :	
	a)CNC parameters b) Pitch error compensation c)PMC parameters	
	d) ID- Info f)M- info e) Ladder g) macro i)S-RAM data	
7	Make I/O channel '0' for RS232C communication	
8	Keep spares inside the cabinet if any (eg.;Fuse)	
9	Check relevant circuit diagram of latest edition	
10	Ensure all the elements are in good condition before shipping	
11	Compare CNC parametrers with standard data & store in backup file	

CHECKED BY:-

VERIFIED BY:-

M11-772.

43193

FANUC Series 0i Mate-M Model D DATA SHEET

得意先名:	FANUC INDIA PRIVATE LIMITED	DATE OF MANUFACTURE:	2011-10
CUSTOMER:	YD73672- 001	DATE OF TEST	:2011/10/05
CONTRACT NO.:	A02B-0321-B500	CHECKED BY	:A. Koike
MODEL NO.:	E11XB0035	APPROVED BY	:MIYASAKA
SERIAL NO.:			
SETTING OF INPUT:			

CNC IDENTIFICATION NUMBER

3C7B5D01
F9916215
5E34C389
4B92C4E2

EDITION OF PRINTED BOARD & ROM

NAME	EDIT	NAME	EDIT	NAME	EDIT
A20B-8200-0849	01A	A02B-0319-H500#60W3	04		
		A02B-0319-H580#40B0	05		
		A02B-0319-J541#60VM	06		
		A02B-0319-J571#658E	04		
		A02B-0319-J572#658F	04		
		A02B-0319-J579#BW50	02		
		A02B-0320-J578#DX50	02		
		A02B-0320-J578#DX51	01		
		A02B-0322-H501#D5F1	20		
		A06B-6057-H507#90C5	06		

LIST OF CARD , DIMM & BACK BOARD

A20B-3300-0638	1		
A20B-3900-0286	1		
A20B-8002-0703	1		
A20B-8200-0560	1		

本データシートはファナック工場出荷時のCNC装置の装置情報です
Information contained in this Data Sheet is system configuration
at the time of FANUC factory shipment.

NAME	NAME	SPECIFICATION	ORDR
BASIC CONTROL UNIT	ベーシック ユニット	A02B-0321-B500	1
AXIS CONTROL CARD A2 FROM/SRAM MEMORY A1 MAIN BOARD POWER SUPPLAY UNIT DISPLAY UNIT NAME PLATE	ジク セイギョ カード A2 FROM/SRAMメモリ A1 メインボード デンゲン ユニット ヒョウジ ユニット メイバン	A02B-0319-H032 A02B-0319-H051 A02B-0319-H105 A02B-0319-H111 A02B-0319-H144#M A02B-0322-H190	1 1 1 1 1 1
PACKAGE 3 CUSTOM SOFTWARE SIZE 512K BYTE MACRO EXECUTOR	パッケージ 3 カスタムソフト ヨウリョウ 512Kbyte マクロ エグゼキュータ	A02B-0322-H532 A02B-0322-J738#512K A02B-0322-J888	1 1 1
I/O UNIT FOR Oi *MANUAL PULSE GENERATOR	I/O UNIT *シュドウ パルス ハッセイ	A02B-0309-C001 A860-0203-T001	1 1
	SPINDLE AMP.	A06B-6164-H343#H580	1
	BIS12/3000 BIS12/3000 BIS22B/3000 SPB6/10000I	A06B-0078-B103 A06B-0078-B103#0100 A06B-0082-B403 A06B-1445-B103	1 1 1 1

2021

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

M/C NO.: M11-772	DATE: 03/01/12
MODEL: MCV-400	CUSTOMER: Unity Engineering. SNY Autotech

M/C BACKUP (PATH) : \\arya2\E-Backup-II\ELECTRICAL MACHINE DATA\AMS DISPATCH MACHINE BACKUPS (20011-12)\FANUC 7.11

CNC SYSTEM

0i-MD ☐ 0i-MATE-MD ☑ ☐

PART No A02B-0321-B500

Sl.No.: ~~E11XB0035~~ E11X52468

SYSTEM OPTIONS:

β SERVO DRIVE SVPM

7.5i (20/20/40) ☐ 15i (40/40/80) ☑

PART No A06B-6164-H343 #H580

Sl.No.: V11740281

α SERVO DRIVE PSM

11i ☐ 15i ☐ 26i ☐ ☐

PART No A06B-

Sl.No.:

α SERVO DRIVE SPM

11i ☐ 15i ☐ 22i ☐ ☐

PART No A06B-

Sl.No.:

α SERVO DRIVE SVM2

40/40i ☐ 80/80i ☐ 160/160i ☐ N/A

PART No A06B-

Sl.No.:

α SERVO DRIVE SVM1(1'st)

40i ☐ 80i ☐ 160i ☐

PART No A06B-

Sl.No.:

α SERVO DRIVE SVM1(2'nd)

40i ☐ 80i ☐ 160i ☐

PART No A06B-

Sl.No.:

FANUC I/O MODULE / I/O CARD

I/O MODULE WITH MPG ☑	I/O CARD WITH MPG ☐	I/O CARD WITHOUT MPG ☑
PARTNo.: A02B-0309-C001	PART No.: A20B-2002-0520	PART No.: A20B-2002-0521
Sl No. E11BD6268	Sl No.	Sl No.

AC REACTOR1

0.14 0.11mH/62A ☑ 0.07mH/110A ☐ ☐

PART No A81L-0001-0156

Sl.No.: RT1162330

CHECKED BY:- [signature]

VERIFIED BY:- [signature]

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS *In Pursuit of Excellence*

SERVO MOTOR - X AXIS

β8/3K	β12/3K	β22/3K	
☐	☑	☐	☐

PART No.: A06B-0078-B103 #100

Sl.No.: C119B4339

SERVO MOTOR - Y AXIS

β8/3K	β12/3K	β22/3K	
☐	☑	☐	☐

PART No.: A06B-0078-B103

Sl.No.: C119BUESA

SERVO MOTOR - Z AXIS

β12/3K	β22/3K	β22/2K	
☐	☑	☐	☐

PART No.: A06B-0082-B403

Sl.No.: C11XADDA1

B AXIS DETAILS

N/A

ROTARY TABLE ☐ INDEX TABLE ☐ ☐

DRIVE:	βiSV-20	βiSV-40
	☐	☐

PART No.: A06B-____-H002

Sl.No.:

MOTOR:	β4/4K	β8/3K
	☐	☐

PART No.: A06B-0___-____

Sl.No.:

REACTOR: A81L-0001-0083#3C (0.15mH/24A) ☐

Sl No.:

DISCHARGE RESISTOR: 16Ohms ☐ 30Ohms ☐

PMM AXIS DETAILS

N/A

SERVO ARM ☐ SERVO MAG ☐ TILTING TABLE ☐

DRIVE:	βiSV-20	βiSV-40
	☐	☐

PART No.: A06B-____-H002

Sl.No.:

MOTOR:	β4/4K	β8/3K
	☐	☐

PART No.: A06B-0___-____

Sl.No.:

REACTOR: A81L-0001-0083#3C (0.15mH/24A) ☐

Sl No.:

DISCHARGE RESISTOR: 16Ohms ☐ 30Ohms ☐

SPECIAL INFORMATIONS:

CHECKED BY:- 10/5/18

VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

ISOLATTION SWITCH (SW1) — AC 3 Ph 415V ☑

40A ☐ | 63A ☑ | 100A ☐

MAKE: SALZER ☑ ABB ☐ ☐

TYPE:

ISOLATTION TRANSFORMER (TFR1) PRIMARY- 415V/3Ph SECONDARY - 220V/3Ph

10KVA ☐ | 12KVA ☐ | 15.5KVA ☑ | ☐

MAKE: SALZER ☐ T.S INTERNATIONAL ☑

SL.No: 3910 [illegible] - 11-11

PRIMARY MCB (F15) — AC 3 Ph 415V / 3POLE ☑

25A ☐ | 32A ☑ | 40A ☐ | ☐

MAKE: SCHNEIDER ☑ MERLIN GERIN ☐

TYPE:

SECONDARY MCB (F19) — AC 3 Ph 220V / 3POLE ☑

40A ☐ | 50A ☐ | 63A ☑ | ☐

MAKE: SCHNEIDER ☑ ABB ☐

TYPE:

DRIVE CONTROL MCB (F20) N/A — AC 3 Ph 220V / 2POLE ☐

1A ☐ | 3A ☐ | ☐ | ☐

MAKE: SCHNEIDER ☐ MERLIN GERIN ☐

TYPE:

SPINDLE MOTOR FAN MPCB (F56) — AC 3 Ph 220V / 3POLE ☑

0.1-0.16A ☑ | 0.25-0.4A ☐ | ☐

MAKE: ABB ☐ SPRECHER &SCHU ☑

MPCB SETTING: 0.13A

PSM CONTACTOR (K19) — 2 30V AC ☑

CA3-30-10 ☐ | A -30-10 ☐ | CA3- 37 -N-11 ☑ | ☐

MAKE: ABB ☐ SPRECHER &SCHU ☑

TYPE:

REGULATED POWER SUPPLY DETAILS

PS1 ☑ PS2 ☑ PS3 ☐ — OUTPUT: 24V DC 10A ☑ 24V DC 20A ☐

MAKE: TELEMECANIQUE(ABL 1RPM24100) ☐ OMRON(S8VE 24024) ☑ KB ELECTORNICS ☐

Sl. No.: PS1: PS2: PS3:

ISOLATTION TRANSFORMER (TFR4) PRIMARY- 415V/1Ph SECONDARY - 200V/1Ph

900VA ☑ | 1.5KVA ☐ | ☐

MAKE: SALZER ☐ T.S INTERNATIONAL ☑

SL.No: 2522-12-11

TFR4 PRIMARY MCB (F16) : 3A/2P ☑ ☐ — MAKE:SCHNEIDER ☑ MERLIN GERIN ☐

TFR4 SEC. MCB (F58) : 6A/1P ☑ ☐ — MAKE:SCHNEIDER ☑ MERLIN GERIN ☐

TFR4 SEC. MCB (F59) : 2A/1P ☑ ☐ — MAKE:SCHNEIDER ☑ MERLIN GERIN ☐

SPECIAL INFORMATIONS:

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VERIFIED BY:- [signature]

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

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CONTROL CIRCUIT

SEALED KEY BOARD/MCP: Sl.No.: 2021

TYPE: (YSL-AMS-SKB-2-REV1-10/09) ☑ YSL-SPARK-MCP2-REV2-05/05 ☐

ATC BRAKE RELAY (PCB 1) :

TYPE: 1 SSR RELAY -YSL-SSRP-BR-3-REV1-06/10 ☐ 2 SSR RELAY - YSL-S26-009-77-REV3-08/10 ☑

I/O DISTRIBUTON PCB (PCB 2) : Sl.No.:

TYPE: YSL-S26-009-56-REV0-04/08 ☑ YSL IFB- 50-FCC TB - REV3 01/06 ☐

CNC ON/OFF RELAY (PCB 3) :

TYPE: YSL -SSR-CNC ON / OFF-REV8-08/10 ☑

Z AXIS BRAKE RELAY (PCB 5) :

TYPE: YSL-SSRP-BR3-REV1-06/10 ☑

Z AXIS BRAKE CURRENT SENSINGRELAY (CSR1) :

TYPE: YSL ELECTRONICS (CSR1-2) ☑

AC RELAYS: Z BRAKE/EMG (ACR1) ☑ ATC/CAB. THERMAL(ACR2) ☑ I. TABLE THERMAL(ACR3) ☐

MAKE: OMRON ☑ ELESTA ☐

RELAY BOARD(STD-X20) 1RELAY(ELECTRO MAGNETIC-DC) ☑

TYPE: YSL-UMK-1RM-24V- DC-CN-FI-2-REV1-12/09 ☑

RELAY BOARD(STD-X31) 1RELAY(ELECTRO MAGNETIC-DC/AC) ☑

TYPE: YSL-S26-009-082-REV1-06/10 ☑

RELAY BOARD(APC-X25) 8 RELAY (SSR TYPE-DC) ☑

TYPE: YSL-S26-009-064-REV3-08/08 ☑

RELAY BOARD(FIXTURE-X26) 2 / 4 / 8 RELAY (SSR TYPE-DC) ☑

TYPE: YSL-S26-009-062-REV3-08/08 ☐ S26-009-063-REV3-08/08 ☑ S26-009-064-REV3-08/08 ☐

RELAY BOARD(TOOL DECLAMP - X33) 1 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-061-REV3-08/08 ☐

RELAY BOARD(- X) 1 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-061-REV3-08/08 ☐

EMERGENCY CONTACTOR (K22) 24V DC ☑

CS8C-40E	KC6-40E	MAKE: ABB ☐ SPRECHER &SCHU ☑
☑	☐	TYPE:

CHECKED BY:- VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

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POWER FAILURE DETECTION MODULE (PFDM) :		24V DC ☐
	MAKE:	YSL-3PFMR-2 ☐
CAPACITOR BANK (DP1) :		24V DC ☑
	MAKE:	YSL-PF-CHU-1 ☑

SINGLE PHASE POWER SUPPLY CIRCUIT

MCB (F18)		AC 1 Ph 230V / 1POLE ☑
10A ☑ ☐ ☐	MAKE: SCHNEIDER ☑	MERLIN GERIN ☐
	TYPE:	

PANEL COOLER:	POWER: 500W ☐		1000W ☐
MAKE: ADVANCE (TURBO1000R33) ☑	ADVANCE (TURBO500R50) ☐		ACE (IGLOO - 750) ☐
Sl.No.: 40849-11-11			

UTILITIES (CABINET):		
CABINET LAMP: PEARL(CFL-11W) ☑	PHILIPS(CFL-11W) ☐	☐
GANG BOX:	2 SWITCHES + 1SOCKET + I INDICATOR ☐	
DOOR INTERLOCK:	LIMIT SW (HONEYWELL-~~GLD-B01D~~ ZCDA01D) ☑	
MCB FOR CAB. LAMP, 5 PIN SOCKET	(F66-3A/1PMAKE: SCHNEIDER) ☑	

UTILITIES (SWITCHGEARS):		
ADD ON BLOCK(PCB5/PSM(K19) INTERLOCK):	SPC&SCH(CS7-PV) ☐	SPC&SCH(CA3-P) ☐
ADD ON BLOCK(STAR DELTA INTERLOCK):	SPC&SCH(CS7-PV) ☐	SPC&SCH(CA3-P) ☐
ADD ON BLOCK(HYDRAULIC FAN POWER): (K9)	SPC&SCH(CS8-P22Z) ☑	ABB(CAF6-20-M) ☐
ADD ON BLOCK(AUTO DRAIN SOL): (K12)	SPC&SCH(CS8-P22Z) ☑	ABB(CAF6-20-M) ☐

SPINDLE STAR / DELTA CIRCUIT

CONTACTORS (K20 / K21) N/A		230V AC ☐
CA3-72N-L11 ☐ CA7 - 85 ☐ ☐	MAKE: ABB ☐	SPRECHER &SCHU ☐
	TYPE:	

RELAY BOARD(X32)		2 RELAYS (SSR TYPE) ☐
TYPE: S26-009-80-CN--F-24DC-230AC ☐		

SPECIAL INFORMATIONS:

CHECKED BY:-	VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

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MAGAZINE MOTOR POWER

MPCB (F1)	1-1.6A ☑	☐		CONTACTOR (K1/K2)	CA8-9C ☑	VBC6-30-01 ☐	
MPCB SETTING:	1A ☑	☐		MOTOR POWER:	1.1KVA/1.5A ☑	1.6KVA/2.3A ☐	
SOLID STATE RELAY (SSR1):				MAKE: ERI ELECTRONICS- SJDA - 482524(DSPST) ☑			

ARM MOTOR POWER

MPCB (F2)	1-1.6A ☑	☐		CONTACTOR (K3) K16	CA8-9C ☑	BC6-30-01 ☐	
MPCB SETTING:	1A ☑	☐		MOTOR POWER:	1.1KVA/1.5A ☑	2.3KVA/3.2A ☐	
SOLID STATE RELAY (SSR3):				MAKE: ERI ELECTRONICS- SJDA - 482524(DSPST) ☑			

LUBRICATION MOTOR POWER

MCB (F3)	1A/3P ☑	MPCB(0.63-1A) ☐		CONTACTOR (K4)	CA8-9C ☑	BC6-30-01 ☐	
				MOTOR POWER:	0.09KW/0.5A ☐	0.12KW/0.6A ☑	

FLOOD COOLANT MOTOR POWER

MPCB (F4)	0.63-1A ☑	2.5-4A ☐		CONTACTOR (K5)	CA8-9C ☑	BC6-30-01 ☐	
MPCB SETTING:	0.9A ☑	3.4A ☐		MOTOR POWER:	0.37KW/0.9A ☑	1.5KW/3.4A ☐	
SURGE SUPRESSOR(RC4):				YSL-NIR-RC3-1/022-575VAC ☑			

CHIP PUSH COOLANT MOTOR POWER

MPCB (F5)	0.63-1A ☐	2.5-4A ☑		CONTACTOR (K6)	CA8-9C ☑	BC6-30-01 ☐	
MPCB SETTING:	0.9A ☐	3.4A ☑		MOTOR POWER:	0.37KW/0.9A ☐	1.5KW/3.4A ☑	
SURGE SUPRESSOR(RC5): YSL-NIR-RC3-1/022-575VAC ☑				COMMON CONTACOR KIT - PUSH/FLUSH(ABB-VBC6 ☐			

CHIP FLUSH COOLANT MOTOR POWER

MPCB (F6)	0.63-1A ☑	2.5-4A ☐		CONTACTOR (K7)	CA8-9C ☑	BC6-30-01 ☐	
MPCB SETTING:	0.9A ☐	3.4A ☐		MOTOR POWER:	0.37KW/0.9A ☐	1.5KW/3.4A ☑	
SURGE SUPRESSOR(RC6): YSL-NIR-RC3-1/022-575VAC ☑				COMMON MPCB WITH CHIP PUSH ☑			

COOLANT GUN MOTOR POWER

MPCB (F27)	0.63-1A ☑	2.5-4A ☐		CONTACTOR (K25)	CA8-9C ☑	BC6-30-01 ☐	
MPCB SETTING:	0.9A ☑	3.4A ☐		MOTOR POWER:	0.37KW/0.9A ☑	1.5KW/3.4A ☐	
SURGE SUPRESSOR(RC27):				YSL-NIR-RC3-1/022-575VAC ☑			

HIGH PRESSURE COOLANT MOTOR POWER

MPCB (F9)	4-6.3A ☐	2.5-4A ☐		CONTACTOR (K10)	CA8-12C ☐	BC7-30-01 ☐	
MPCB SETTING:	4.9A ☐	5.2A ☐		MOTOR POWER:	2.2KW/4.9A ☐	2.2KW/5.2A ☐	
SURGE SUPRESSOR(RC9):				YSL-NIR-RC3-1/022-575VAC ☐			

HYDRAULIC 1 MOTOR POWER

MPCB (F8)	4-6.3A ☐	2.5-4A ☑		CONTACTOR (K9)	CA8-9C ☑	BC6-30-01 ☐	
MPCB SETTING:	2.7A ☑	3.5A ☑		MOTOR POWER:	1.1KW/2.7A ☑	1.5KW/3.5A ☑	
SURGE SUPRESSOR(RC8):				YSL-NIR-RC3-1/022-575VAC ☑			

CHECKED BY:- VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS *In Pursuit of Excellence*

INDEX TABLE MOTOR POWER NA

MPCB (F12)	2.5-4A ☐	☐		CONTACTOR(K15/K16)	CA8-9C ☐	VBC6-30-01 ☐	
MPCB SETTING:	☐	☐		MOTOR POWER:	☐	1.6KVA/2.3A ☐	
SOLID STATE RELAY (SSR5):				MAKE: ERI ELECTRONICS- SJDA - 482524(DSPST) ☐			

CHIP CONVEYOR 1 MOTOR POWER NA

MPCB (F10)	0.63-1A ☐	1-1.6A ☐		CONTACTOR(K11/K12)	CA8-9C ☐	VBC6-30-01 ☐	
MPCB SETTING:	0.74A ☐	☐		MOTORPOWER:	0.18KW/0.65A ☐	☐	
SURGE SUPRESSOR(RC10):				YSL-NIR-RC3-1/022-575VAC ☐			

CHIP CONVEYOR 2 MOTOR POWER

MPCB (F11)	0.63-1A ☐	1-1.6A ☐		CONTACTOR(K13/K14)	CA8-9C ☐	VBC6-30-01 ☐	
MPCB SETTING:	0.74A ☐	☐		MOTORPOWER:	0.18KW/0.65A ☐	☐	
SURGE SUPRESSOR(RC11):				YSL-NIR-RC3-1/022-575VAC ☐			

CONVEYOR DRUM WASH MOTOR POWER NA

MPCB (F7)	0.63-1A ☐	2.5-4A ☐		CONTACTOR (K8)	CA8-9C ☐	BC6-30-01 ☐	
MPCB SETTING:	0.9A ☐	3.4A ☐		MOTOR POWER:	0.37KW/0.9A ☐	1.5KW/3.4A ☐	
SURGE SUPRESSOR(RC7):				YSL-NIR-RC3-1/022-575VAC ☐			

SCREW CONVEYOR 1 MOTOR POWER NA

MPCB (F13)	0.63-1A ☐	1-1.6A ☐		CONTACTOR(K17/K18)	CA8-9C ☐	VBC6-30-01 ☐	
MPCB SETTING:	0.74A ☐	1.4A ☐		MOTORPOWER:	0.18KW/0.65A ☐	0.37KW/1.1A ☐	
SURGE SUPRESSOR(RC13):				YSL-NIR-RC3-1/022-575VAC ☐			

SCREW CONVEYOR 2 MOTOR POWER

MPCB (F30)	0.63-1A ☐	1-1.6A ☐		CONTACTOR(K28/K29)	CA8-9C ☐	VBC6-30-01 ☐	
MPCB SETTING:	0.74A ☐	1.4A ☐		MOTORPOWER:	0.18KW/0.65A ☐	0.37KW/1.1A ☐	
SURGE SUPRESSOR(RC30):				YSL-NIR-RC3-1/022-575VAC ☐			

MOTOR POWER (OPTIONAL-1) NA

MPCB (F)	☐	☐		CONTACTOR (K)	CA8-9C ☐	BC6-30-01 ☐	
MPCB SETTING:	☐	☐		MOTOR POWER:	☐	☐	
SURGE SUPRESSOR(RC):				YSL-NIR-RC3-1/022-575VAC ☐			

MOTOR POWER (OPTIONAL - 2)

MPCB (F)	☐	☐		CONTACTOR (K)	CA8-9C ☐	BC6-30-01 ☐	
MPCB SETTING:	☐	☐		MOTOR POWER:	☐	☐	
SURGE SUPRESSOR(RC):				YSL-NIR-RC3-1/022-575VAC ☐			

SPECIAL INFORMATIONS: MPCB'S- SPRECHER &SCHU(KTA7-25S) ☑ ABB (MS 116) ☐

CHECKED BY:-	VERIFIED BY:-

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MACHINE HISTORY - ELECTRICAL

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ATC & SPINDLE DETAILS

SPINDLE MOTOR:	MODEL:	β3 ☐	β6 ☐	β12 ☑	☐
	MOTOR RPM:	6000 ☐	8000 ☑	10000 ☐	☐
	MOTOR POWER:	15/11 KW			
	PART No:	A06B-14A1-B10?			
	Sl. No:	CUXJ1082 #6402			
	GEAR RATIO:	1:1 ☐	1:1.25 ☑	1:1.5 ☐	DIRECT DRIVE ☐
	POWER LIMIT:	IN KW: 11/7.5 KW			
	SPEED LIMIT:	IN RPM: 60-6000 rpm			

PROXIMITY SENSORS-SPINDLE

MAKE; TEKNIC ☑ TYPE: (24V DC - PNP) ☑

SPINDLE ORIENTATION (PXS1) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
TOOOL CLAMP(PXS3) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
TOOOL DECLAMP(PXS4) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
NO TOOL IN SPINDLE(PXS32) ☐	M08X50	M08X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐
ONE ROTATION(PXS39) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑

PROXIMITY SENSORS-ATC(DISC TYPE)

MAKE; TEKNIC ☐ TYPE: (24V DC - PNP) N/A ☐

MAGAZINE IN POSITION (PXS2) ☐	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐
MAGAZINE REFERENCE (PXS46) ☐	M12X73	FG2X05APO24-5000LFLAT		
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐		☐

PROXIMITY SENSORS-ATC(TWIN ARM TYPE)

MAKE; TEKNIC ☑	TYPE:EGT 12 X 02 AP - 024 SEM4L(24V DC - PNP) ☑
MAGAZINE IN POSITION PROXIMITY (PXS2) ☑	MAG REFERENCE PROXIMITY (PXS46) ☑
POCKET UP PROXIMITY (PXS5) ☑	POCKET DOWN PROXIMITY (PXS6) ☑
ARM AT HOME PROXIMITY (PXS7) ☑	☐
TOOL CLAMP START PROXIMITY (PXS58) ☑	TOOL DECLAMP START PROXIMITY (PXS59) ☑

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MACHINE HISTORY - ELECTRICAL

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PNUEMATIC ELEMENTS (SPINDLE + ATC)

MAKE; SUCO ☑	TYPE:(24V DC - NO CONTACT) ☑
MAIN AIR PRESSURE SWITCH (PRS 2) ☑	SPINDLE AIR PRESSURE SWITCH (PRS 101) ☑
AIR PURGE PRESSURE SWITCH (PRS 102) ☐	☐
MAKE; FESTO ☑	TYPE:(24V DC -COIL) ☑
SPINDLE AIR JET SOLENOID (Y1) ☑	SPI AIR PURGE SOLENOID (Y115) ☑
POCKET UP SOLENOID (Y2) ☑	POCKET DOWN SOLENOID (Y3) ☑
TOOL CLAMP SOLENOID (Y51) ☑	TOOL DECLAMP SOLENOID (Y4) ☑
AIR THROUGH NOZZLE SOLENOID (Y99) ☐	☐

HYDRAULIC ELEMENTS (SPINDLE + ATC) NA

MAKE; YUKEN ☐ REXROTH ☐	TYPE:(24V DC -COIL) ☐
☐	TOOL DECLAMP SOLENOID (Y4) ☐

UTILITIES(SPINDLE + ATC)

TOOL CHANGE POSITION LIMIT SWITCH- (LS3) :	TEKNIC SG-01-R-610 ☑	☐
TOOL DECLAMP PUSH BUTTON :	ILLUMINATED GREEN ☑	☐
ATC / I.TABLE DISPLAY SEL SWITCH :	2 POSITION ☐	☐
MAG TEST MODE KEY SWITCH (KSW7) :	2 POSITION ☐	☐
	☐	☐

SPECIAL INFORMATIONS:

AUTO DOOR & CYCLE START DETAILS NA

MAKE; FESTO ☐	TYPE:(24V DC -COIL) ☐
AUTO DOOR OPEN SOLENOID (Y6) ☐	AUTO DOOR CLOSE SOLENOID (Y7) ☐
AUTO DOOR SAFETYSOLENOID (Y8) ☐	☐
DOOR SWITCHES	TYPE: LS5 - NG1 RS 510 (24V DC -NO CONTACT) ☐
MAKE; TECNIK ☐ FESTO ☐	READ SWITCHES (RSW30& RSW31) ☐
CYCLE START SWITCH	TYPE: LS45 - XCE-154 (24V DC -NO CONTACT) ☐
MAKE; ☐ TELEMECHANIC ☐	☐
DOUBLE HAND CYCLE START BOX	CYCLE START PB (MUSHROOM HEAD GREEN) (02 Nos) ☐
MAKE; TECNIK ☐ ☐	EMERGENCY PB (MUSHROOM HEAD SPRING RETURN) ☐
AUTO DOOR OPERATION	AUTO DOOR OPEN/CLOSE PB (GREEN) ☐
MAKE; TECNIK ☐ ☐	AUTO DOOR INTERLOCK KEY SW (2 POSITION) ☐

SPECIAL INFORMATIONS:

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COOLANT TANK / CONVEYORS DETAIL

UTILITIES (STD COOLANT)				
COOLANT LEVEL FLOAT SWITCH (PXS 8) :	FILPRO(LIM 18PF SC)	☑		☐
COOLANT GUN SEL SW :	2 POSITION	☑		☐
FLUSH COOLANT VALVE (Y128) :	AVCON	☐		☐
COOLANT PLUG (HC1) :	16 PIN	☑	24 PIN	☐
LOCATION(FROM M/C REAR) :	LHS	☐	RHS	☑

UTILITIES (HP COOLANT):				
FLOW SWITCH (FLW1) :	IFM (SH0 500)	☐ N/A		☐
CLOG SWITCH (FCLG1) :	HIRLEKAR	☐	SALVIN	☐
AIR BLAST SOLENOID (Y5) :	FESTO	☐		☐
M/C DOOR CLOSE LIMIT SW (LS5) :	TEKNIC(NG1 RS 510)	☐		☐

UTILITIES (CHIP CONVEYOR):				
CHIP CONVEYOR 1 O/L LIMIT SW (LS12) :	TEKNIC(NG1 RS 510)	☐ N/A		☐
CHIP CONVEYOR 2 O/L LIMIT SW (LS13) :	TEKNIC(NG1 RS 510)	☐		☐
CHIP CONVEYOR FWD/REV SEL SW :	3 POSITION(SPRING RETURN)	☐		☐
CHIP CONVEYOR PLUG (HC3) :	10 PIN	☐	24 PIN	☐
LOCATION(FROM M/C REAR) :	LHS	☐	RHS	☐

UTILITIES (SCREW CONVEYOR):				
SCREW CONVEYOR 1 O/L PROXIMITY (PXS20) :	M12 X 52MM (CONN. TYPE)	☐ N/A	M12 X 72MM (CONN. TYPE)	☐
SCREW CONVEYOR 2 O/L PROXIMITY(PXS104) :	M12 X 52MM (CONN. TYPE)	☐	M12 X 72MM (CONN. TYPE)	☐
SCREW CONVEYOR FWD/REV SEL SW :	3 POSITION(SPRING RETURN)	☐	COMMON WITH CHIP.CNYR	☐

SPECIAL INFORMATIONS:

HYDRAULIC TANK DETAIL

OIL LEVEL FLOAT SWITCH (FL4) :	SHRIDHAN	☐		☐
OIL CLOG SWITCH (CLG2) :	HYDAC	☐		☐
HYDRAULIC THERMAL SWITCH (TH4) :	CARE SYSTEM	☐		☐
HYDRAULIC PLUG (HC2) :	16 PIN	☑	24 PIN	☐
MAKE;INDO ELECTRIC ☑ LOCATION(FROM M/C REAR) :	LHS	☐	RHS	☑

SPECIAL INFORMATIONS:

CHECKED BY:- R.V. Gaikwad

VERIFIED BY:-

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MACHINE HISTORY - ELECTRICAL

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LUBRICATION TANK / FRL DETAIL

OIL LEVEL FLOAT SWITCH (FL1) :	CENLUB	☑	☐
LUB PRESSURE SWITCH (PRS1) :	SUCO	☑	☐
AUTO DRAIN SOLENOID (Y44) :	ROTEX	☑	☐

MACHINE LAMPS

MAKE:			PATANKAR ☐	AISHWARYA	☑
MACHINE LAMP1 (ML1) :	11W	☐	9W+9W ☑	36W	☐
MACHINE LAMP2 (ML2) :	11W	☐	9W+9W ☑	36W	☐
MACHINE LAMP3 (ML3) :	11W	☐	9W+9W ☐	36W	☐
MAHINE INDICATION LAMP(PL5) :	1 TIER	☐	3 TIER ☑	3 TIER WITH HOOTER	☐
MAKE:			PATANKAR ☑	LAXMI(VANTECH	☐

UTILITIES(MAHINE)

EMERGENCY MONITOR (PCB 12)			YSL-EMG-PCB-4-REV2-06/10	☑
HAND HELD MPG:			FANUC ☐	FUTURE(TAIWAN) ☐
		Sl. No.:		
TPM TRACK :			PIONEER CT ☐	☐
		Sl. No.:		
SERVO VOLTAGE STABILIZER:			15KVA ☐	25KVA ☐
INPUT:	OUTPUT:		MAKE: SERVOMAX ☐	☐
		Sl. No.:		
COOLANT / SPINDLE CHILLER UNIT:			MAKE: KAUKAN ☐	☐
POWER:	MCB(F31): 4A / 3P		Sl. No.:	
CONNECTOR :	INDO ELECTRIC (10 PIN) ☐		LOCATION: LHS ☐	RHS ☐

FIXTURE DETAIL

MAKE;	☐	PNEUMATIC (NA) ☐	HYDRAULIC ☐
STATION 1 CLAMP SOLENOID (Y70)	☐	STATION 1 DECLAMP SOLENOID (Y71)	☐
STATION 2 CLAMP SOLENOID (Y72)	☐	STATION 2 DECLAMP SOLENOID (Y73)	☐
STATION 3 CLAMP SOLENOID (Y84)	☐	STATION 3 DECLAMP SOLENOID (Y85)	☐
STATION 4 CLAMP SOLENOID (Y86)	☐	STATION 4 DECLAMP SOLENOID (Y87)	☐
MAKE;	☐	PNEUMATIC ☐	HYDRAULIC ☐
STATION 1 CLAMP PRESSURE SW (PRS20)	☐	STATION 2 CLAMP PRESSURE SW (PRS22)	☐
STATION 3 CLAMP PRESSURE SW (PRS31)	☐	STATION 4 CLAMP PRESSURE SW (PRS32)	☐

SPECIAL INFORMATIONS: CLAMP PB'S (GREEN) -

DECLAMP PB'S (RED) -

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MACHINE HISTORY - ELECTRICAL

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APC (LINEAR TYPE) DETAIL

MANUAL OPERATION PANEL :	YSL-AMS-APC-PCB-REV0-07/06	☑
PALLET SET UP PB :	LUMINOUS GREEN	☑
EMERGENCY PB :	MUSHROOM HEAD RED	☑

PROXIMITY SENSORS:

MAKE; TEKNIC ☑	TYPE: (24V DC - PNP)	☑

PALLET RECEIVED PROXIMITY(PXS11) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
ARM 1 IN PROXIMITY (PXS13) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
ARM 1 OUT PROXIMITY (PXS14) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
ARM 2 IN PROXIMITY (PXS15) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
ARM 2 OUT PROXIMITY (PXS16) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑

PNEUMATIC SOLENOIDS:

MAKE; FESTO ☑	TYPE: 24V DC -COIL ☑
APC DOOR OPEN SOLENOID (Y12) ☑	APC DOOR CLOSE SOLENOID (Y13) ☑
ALLET PAD CLEAN SOLENOID (Y18) ☑	☐

HYDRAULIC SOLENOIDS:

MAKE; YUKEN ☑ REXROTH ☐	TYPE: 24V DC -COIL ☑
PALLET CLAMP SOLENOID (Y10) ☑	PALLET DECLAMP SOLENOID (Y11) ☑
ARM 1 IN SOLENOID (Y14) ☑	ARM 1 OUT SOLENOID (Y15) ☑
ARM 2 IN SOLENOID (Y16) ☑	ARM 2 OUT SOLENOID (Y17) ☑

HYDRAULIC PRESSURE SWITCHES:

MAKE; POLYHYDRON ☑	TYPE: ☐
PALLET CLAMP PRSW (PRS3) ☑	PALLET DECLAMP PRSW (PRS60) ☑

LIMIT SWITCHES:

MAKE; TEKNIC(NG1 VS SQ 510) ☐	JAI BALAJI (JLSSPR) ☐
PALLET 1 ON STATION LIMIT SW (LS10) ☑	PALLET 2 ON STATION LIMIT SW (LS11) ☑

READ SWITCHES:

MAKE; FESTO ☑	APC DOOR INTERMEDIATE READ SW (RSW 65) ☑
APC DOOR OPEN READ SW (RSW 58) ☑	APC DOOR CLOSE READ SW (RSW 59) ☑

SPECIAL INFORMATIONS:

CHECKED BY:- K.V. Pandey

VERIFIED BY:- Rao

ACE MANUFACTURING SYSTEMS LTD

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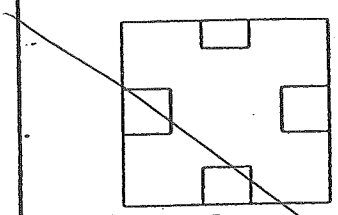
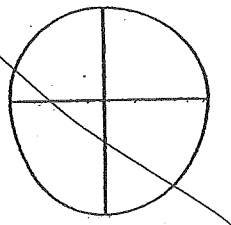
FINAL ASSEMBLY - INSPECTION REPORT (VMC / ~~CMV~~) - MECHANICAL

MACHINE NUMBER: M11 - 772	CUSTOMER : UNITY ENGINEERS

I. MOTOR MOUNTING BRACKET FACE OUT AND CONCENTRICITY FOR DIRECT DRIVE

	FACE OUT (Recommended≤ 0.015 mm)	CONCENTRICITY (Recommended≤ 0.010 mm)
		

S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
1	SPINDLE CONNECTOR RADIAL RUN OUT	≤ 0.010 MM	
2	MOTOR CURRENT AT MAX. RPM	β3 < 15%, β6 / α6 & ABOVE <10%	
3	CTS ROD GUIDE RADIAL RUN OUT (DDCTS)	≤ 0.010 MM	
4	SPINDLE CLAMPING FORCE	AS PER DS-DSN-PROD-01	
5	SPINDLE MOTOR SPECIFICATION	AS PER WORK ORDER	
II	INDIRECT DRIVE		
1	MOTOR PULLEY RADIAL RUN OUT	≤ 0.050MM	0.05
2	SPINDLE PULLEY RADIAL RUN OUT	≤ 0.050MM	0.05
3 a)	BELT TENSION	AS PER DS-DSN-PROD-02	as per data sheet
b)	BELT SPECIFICATION	Make : Size : - 820 - 5 9T	Quantity : 01
4	MOTOR CURRENT AT MAX. RPM	β3 < 20%, β6 / α6 & ABOVE <15%	13 %
5	ROTARY JOINT RADIAL RUN OUT (IDCTS)	≤ 0.040MM	— NA —
6	SPINDLE CLAMPING FORCE	AS PER DS-DSN-PROD-01	~~840~~ K.g
7	SPINDLE MOTOR SPECIFICATION	AS PER WORK ORDER	B12 80001
8	PULLEY RATIO	AS PER WORK ORDER	1:1.25
III	AXIS DEAD STOPPER VALUE / BUTT REFERENCING VALUE (AS PER DRAWING)		
	AXIS	MOTOR END	BRAKE END
1	X	39.05	7.8
2	Y	17.50	21.48
3	Z	8.850	22.3
IV	% of axis current drawn (Recommended as per Data sheet DS -QA -02)		
	AXIS	100 % rapid rate	
	X	48%	
	Y -	52%	
	Z Down	22%	
	Z Up	67%	
V	ACCUMULATOR OIL PRESSURE	AS PER CIRCUIT DIAGRAM	
	ACCUMULATOR GAS PRESSURE	AS PER CIRCUIT DIAGRAM	

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FINAL ASSEMBLY - INSPECTION REPORT (VMC / CMV) - MECHANICAL

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S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
VI	ATC ALIGNMENT (TURRET TYPE)		
1	Parallelism-of gripper Arm face wrto. X - Axis	0.050 MM MAX.	
2	Tool Ejection Stroke	AS PER DS - DSN-PROD-01	
3	Swing Lever Gap	AS PER DRAWING	
4	Side Cam Alignment	0.020 MM MAX.	
5	Gap between Draw Bar bush & Roller/Screw	1 ± 0.1 MM	
VII	ATC ALIGNMENT (TWIN ARM)		
1	X - Axis	0.2 MM MAX.	0.1
2	Y - Axis	0.2 MM MAX.	0.1
3	Z - Axis	0.050 MM MAX.	0.03
4	Twin Arm bottom surface parallelism in Y-Axis	0.1MM MAX.	0.08
5	Gap between Draw Bar Bush & Roller	1 ± 0.1 MM	1 mm
6	Tool Ejection stroke	AS PER DS - DSN-PROD-01	0.65
7	Gap between deflection Pads	0.02 TO 0.050 MM	
VIII	APC ALIGNMENT - LINEAR TYPE		
1	Height difference of rest pads	0.010 MM MAX.	0.006
2	Pallet change Time	AS PER WORK ORDER	
IX	APC ALIGNMENT- ROTARY TYPE		
1	APC Base parallelism w.r.to. X - Axis	≤ 0.1 MM	
	APC Base parallelism w.r.to. Y - Axis	≤ 0.1 MM	
2	Arm Top parallelism w.r.to. X - Axis	≤ 0.05 MM	
	Arm Top parallelism w.r.to. Y - Axis	≤ 0.05 MM	
3	Arm 1 Alignment ' FACE ' w.r.to X- Axis	≤ 0.020 MM	
	Arm 1 Alignment ' Taper ' w.r.to X - Axis	≤ 0.020 MM	
	Arm 2 Alignment ' FACE ' w.r.to X - Axis	≤ 0.020 MM	
	Arm 2 Alignment ' Taper ' w.r.to X - Axis	≤ 0.020 MM	
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FINAL ASSEMBLY - INSPECTION REPORT (VMC / CMV) - MECHANICAL

S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
4	Offset between Arm 1 & Arm 2 - Face	≤ 0.030 MM	
	Offset between Arm 1 & Arm 2 - Taper	≤ 0.030 MM	
5	Pallet - 1.Wedge Alignment along X - Axis	≤ 0.020 MM	
	Pallet - 2 Wedge Alignment along X - Axis	≤ 0.020 MM	
6	Gap between Pallet -1 side & Arm along X - Axis	Min 0.3 MM	
	Gap between Pallet -1 side & Arm along Z - Axis	0.1 ± 0.02 MM	
	Gap between Pallet -2 side & Arm along X - Axis	Min 0.3 MM	
	Gap between Pallet -2 side & Arm along Z - Axis	0.1 ± 0.02 MM	
7	Pallet Sagging - Pallet -1 'W.r.to. X & Y - Axis	≤ 1 MM	
	Pallet Sagging - Pallet -2 'W.r.to. X & Y - Axis	≤ 1 MM	
8	Pallet Height Variation	AS PER TEST CHART	
9	Backlash between Rack & Pinion at distance of 500 MM	≤ 1 MM	
10	Pallet Change Time	AS PER WORK ORDER	

S.No.	CHECK POINTS	REMARKS
A	**AXIS**	
1	Ensure geometrical accuracies as per test chart.	✓
2	Ensure plugs are fitted for LM guide ways.	✓
3	Ensure butt reference function.	✓
4	Ensure effective wiping of saddle cover wipers.	NA
5	Ensure lube timer is set 12 min for HG and 20 min for LM Guide ways.	✓
6	Ensure leveling bolt locknuts are tighten after machine leveling.	✓
7	Check the all axes reference arrows are matched & Direction name plate fixed.	✓
8	Ensure tightness & marking of all screws with Yellow paint marker	✓
9	Ensure proper setting of Limit switches.	✓
B	**ATC**	
1	Check oil level is maximum in ATC gear box.	✓
2	Check tool change position limit switch for proper functioning & band width (Max. 1 mm on each side).	✓
3	Ensure smooth function of Tool change without abnormal/unhealthy noise.	✓
4	Ensure ATC bracket & Magazine Dowelling.	✓

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FINAL ASSEMBLY-INSPECTION REPORT (VMC / CMV) - MECHANICAL		In Pursuit of Excellence
S.No.	CHECK POINTS	REMARKS
C	SPINDLE	
1	Check the tightness of push stud.	✓
2	Ensure air jet through all the (4 / 8) holes at oriented spindle position	✓
3	Ensure supply of air to spindle Air Purge system (If provided)	—
4	Ensure proper setting of proximities.	✓
5	Ensure smooth running of spindle without abnormal/unhealthy noise	✓
6	Ensure dummy plate fixing on CTS slot in Non CTS machine.	NA
7	Check the Tool not in spindle proximity message.	NA
D	COOLANT	
1	Ensure no coolant leakage around the machine & presence of coolant in lubrication oil drain pipe	✓
2	Ensure proper working of float switch (by shifting the sensor upwards).	✓
3	Ensure all the hoses, fittings tightened and routed properly.	✓
4	Check the function of Flood coolant (M7), Chip push (M181), Base flush (M182), CTS (M52) & Coolant gun (through selector switch).	✓
5	Ensure CTS flow sw.setting by opening the bag filter air vent ball valve.(Optional)	NA
6	Ensure suction strainer & Check Valve in CTS tank & pressure gauge working (If provided)	—
7	Ensure CTS pin locator hole is plugged (If provided)	
8	Ensure that twin arm plunger & CTS pin locator screw do not foul (if provided)	
9	Ensure bag filter clog message by closing bag filter outlet ball valve (Optional)	
10	Ensure High pressure air blast after CTS switched off (If provided)	
11	Ensure drum wash in conveyors (If provided)	
12	Check conveyor overload message & Direction (If provided)	
13	Check oil in conveyor gear box. (if provided)	
14	Ensure functioning of screw conveyor in forward & reverse directions (through selector switch).	
15	Ensure splash guard fixed to conveyor.	
E	APC	
1	Ensure proper setting of proximities.	✓
2	Ensure supply of air jet to pallet rest pad cleaning.	✓
3	Ensure pallet change takes place smoothly with specified load (as per work order) without abnormal / unhealthy noise	✓
4	Ensure APC base Dowelling.	✓
5	Check pallet initialization in APC Machine	✓
NAME & SIGN	CHECKED BY — APPROVED BY	4...6 Page No

ACE MANUFACTURING SYSTEMS LTD		AMS In Pursuit of Excellence
FINAL ASSEMBLY -INSPECTION REPORT (VMC / CMV)- MECHANICAL		
S.No.	CHECK POINTS	
F	HYDRAULIC, PNEUMATIC & LUBRICATION	
1	Ensure pneumatic main line pressure switch setting (4 bar) & error message are satisfactory	✓
2	Ensure Spindle air jet pressure switch setting (3 bar) & error message are satisfactory	✓
3	Ensure Spindle air purge pressure switch setting (1.2 bar) & error message are satisfactory	— NA —
4	Ensure proper setting of oil flow (1 drop/tool change) in FRL unit. & Oil Level low message	✓
5	Ensure proper routing of FRL drain hose.	✓
6	Ensure identification name plates are fixed on power pack hoses.	✓
7	Ensure Ball valve fixed to FRL Unit inlet	✓
8	Ensure pneumatic quick change coupling fixed to 4th axis provision	—
9	Ensure no oil leakage in hydraulic valves & fittings	✓
10	Ensure hydraulic flow control valve grub locking in APC machine.	✓
11	Ensure pneumatic counter balance bracket Dowelling.	—
G	GUARDING ASSEMBLY	
1	Ensure all internal guards are assembled.	✓
2	Ensure smooth movement of sliding door without play and unhealthy noise	✓
3	Ensure Nylock nuts are fixed to door rollers.	✓
4	Ensure ATC door LM rail plugs fixed.	—
5	Ensure Proper tuning of Front auto door / ATC auto door cylinder.	—
H	ANY OTHER POINTS	
*	Head Stock greased	✓
*	Chip tray.	✓
*	Roof guard	✓
*	coolant Gun	✓
NAME & SIGN	CHECKED BY APPROVED BY	5...6 Page No.

ACE MANUFACTURING SYSTEMS LTD

GEOMETRICAL TEST CHART(VMC / CMV)

AMS — The Pursuit of Excellence

PERMISSIBLE TOLERANCE AS PER TEST CHART (DS-DSN-PROD-TC-01)

1. SPIRIT LEVEL READING:

Along X-Axis			Along Y-Axis		
3/4 →	0	1 →	1/2 ↓	0	↑ 1/4
3/4 →	0	3/4 →	0	1/2 ↑	↑ 1/2
3/4 →	0	3/4 →	1/4 ↓	3/4 ↓	0

2. TABLE TOP READINGS:

Along X-Axis		
0	+5	+3
0	+5	+3
0	+4	+6

Along Y-Axis		
-1	-1	-5
-4	-3	-5
0	0	0

PALLET - II

Along X-Axis		
0	+12	+3
0	+4	+2
0	+6	+4

Along Y-Axis		
0	0	0
+6	+10	+8
-3	+9	+5

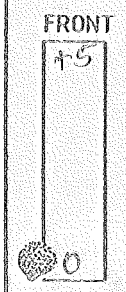
3. T SLOT READING:

0	+8

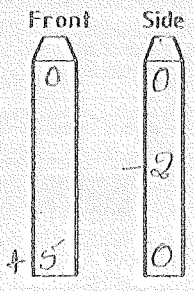
PALLET - II

0	+12

4. COLUMN READINGS :



5. MANDREL READINGS:

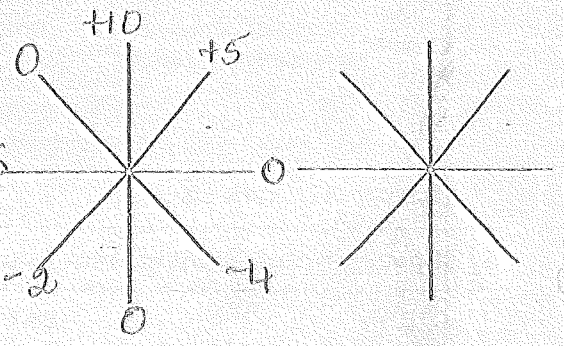


6. TURN AROUND READING:

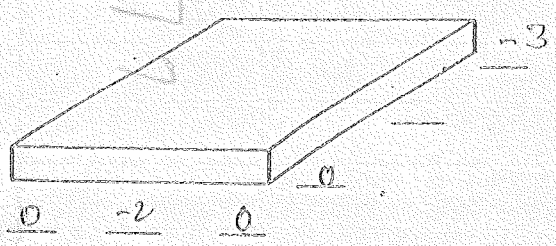
RUNOUT:

a. TOP :
1. 0.002

b. BOTTOM : -6
1. 0.005
2. 0.007
3. 0.004
4. 0.003



7. SQUARENESS READING (X&Y):



8. PALLET HEIGHT VARIATION:

Pallet-1: 0.000

Pallet-2 : 0.007

SIGN & DATE				
	CHECKED BY	APPROVED BY	PAGE NUMBER	DOCUMENT NUMBER
	Hemanth		6....	F-QA-FA-05-R1

ACE MANUFACTURING SYSTEMS LTD

AMS
In Pursuit of Excellence

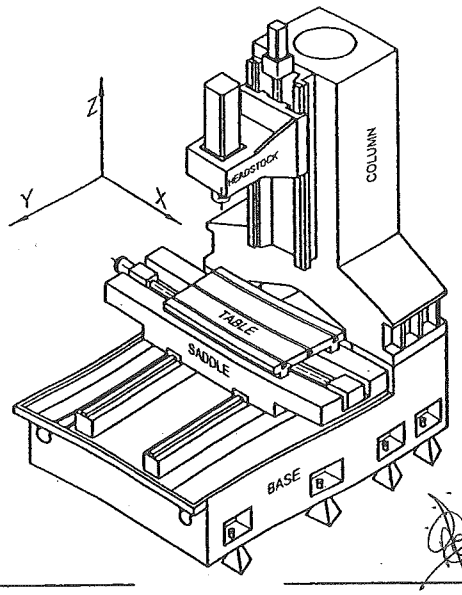
DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

PRODUCT : VERTICAL MACHINING CENTER (FIXED COLUMN)

MODEL NAME : MCV-400

MACHINE NUMBER : M11-772



Pradeep.K.B

CHECKED BY
QUALITY ASSURANCE

CHECKED BY
ASSEMBLY - MECHANICAL

ACE MANUFACTURING SYSTEMS LTD

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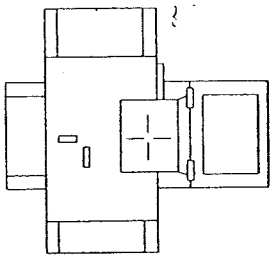
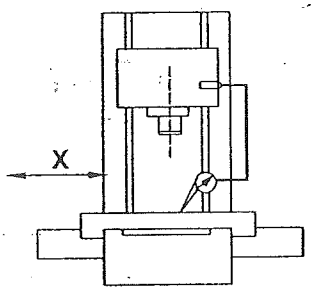
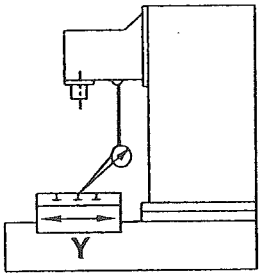
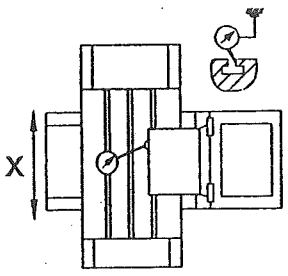
DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

All measurements in mm

SL. NO.	Description of the test	Figure	Tolerance	
			Permissible	Actual
G1	LEVELLING OF MACHINE X & Y AXES AT THEIR MIDDLE OF STROKE		0.03 / m ALONG X & Y AXES	LEVEL ALONG X-AXIS : 0·009 ⟷ LEVEL ALONG Y-AXIS : 0·006 ↕
G2	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO X-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : PALLET 1 : 0·010 PALLET 2 : 0·009 (FOR APC MACHINE ONLY)
G3	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO Y-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : – PALLET 1 : 0·013 PALLET 2 : 0·013 (FOR APC MACHINE ONLY)
G4	PARALLELISM OF REFERENCE T-SLOT TO X-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : PALLET 1 : 0·006 PALLET 2 : 0·016 (FOR APC MACHINE ONLY)

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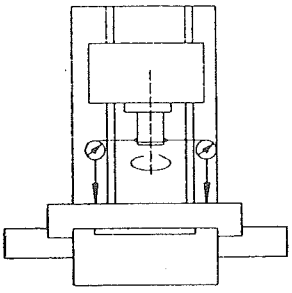
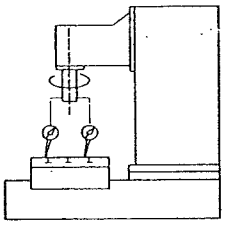
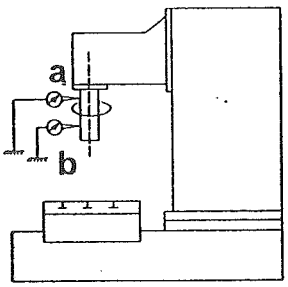
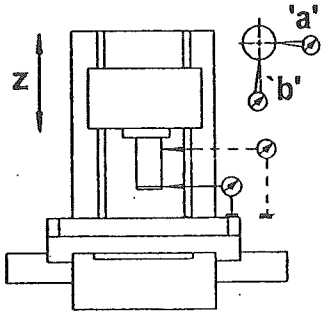
DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

All measurements in mm

SL. NO.	Description of the test	Figure	Tolerance	
			Permissible	Actual
G5	SQUARENESS OF SPINDLE AXIS TO TABLE SURFACE IN XZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : PALLET 1 : 0.006 PALLET 2 : 0.008 (FOR APC MACHINE ONLY)
G6	SQUARENESS OF SPINDLE AXIS TO TABLE SURFACE IN YZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : PALLET 1 : 0.010 PALLET 2 : 0.008 (FOR APC MACHINE ONLY)
G7	RUN OUT OF SPINDLE TAPER a) NEAR SPINDLE NOSE b) AT A DISTANCE OF 300 FROM SPINDLE NOSE		a. 0.01 NEAR SPINDLE NOSE b. 0.02 AT A DISTANCE OF 300mm	a. 0.002 b. 0.007
G8	PARALLELISM OF SPINDLE AXIS & Z-AXIS IN a) XZ PLANE b) YZ PLANE TO BE CHECKED AT MEAN POINT OF RUNOUT		a. 0.02 / 300 b. 0.02 / 300	a. 0.002 b. 0.005

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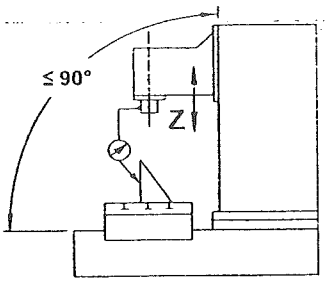
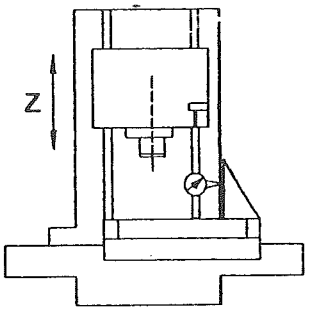
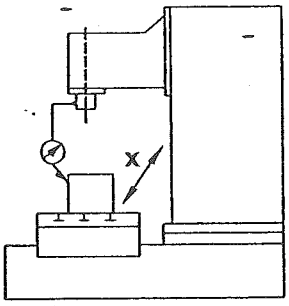
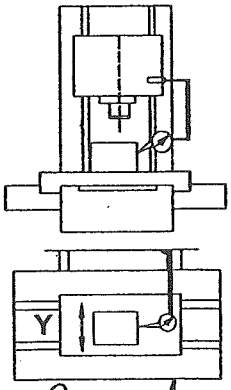
DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

All measurements in mm

SL. NO.	Description of the test	Figure	Tolerance	
			Permissible	Actual
G9	SQUARENESS BETWEEN TABLE SURFACE & Z-AXIS MOVEMENT IN YZ PLANE		0.015 / 300	TABLE : PALLET 1 : 0.002 PALLET 2 : 0.004 (FOR APC MACHINE ONLY)
G10	SQUARENESS BETWEEN TABLE SURFACE & Z-AXIS MOVEMENT IN XZ PLANE		0.015 / 300	TABLE : PALLET 1 : 0.004 PALLET 2 : 0.005 (FOR APC MACHINE ONLY)
G11	STRAIGHTNESS OF X-AXIS MOVEMENT IN XY - PLANE		0.010 / 300	0.002
G12	STRAIGHTNESS OF Y-AXIS MOVEMENT IN XY - PLANE		0.010 / 300	0.001

ACE MANUFACTURING SYSTEMS LTD

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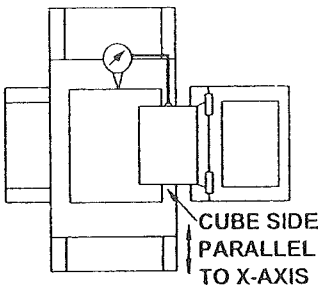
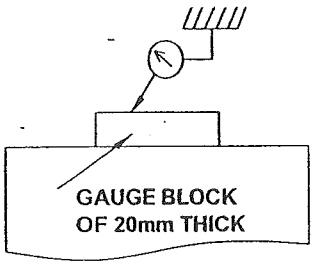
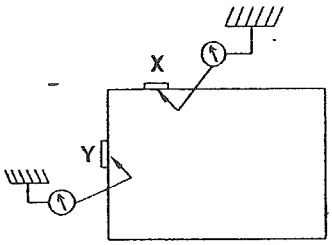
DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance: Permissible	Actual
G13	SQUARENESS OF Y - AXIS MOVEMENT WITH RESPECT TO X - AXIS MOVEMENT	MEASURE READING ALONG 'Y' 	0.015 / 300	0.004
G14	VARIATION BETWEEN THE PALLET HEIGHT (FOR APC MACHINE ONLY)		0.030	0.007
G15	PALLET REPEATABILITY IN X & Y AXES WHEN LOADED ON THE MACHINE FROM THE PALLET STATION FOR SAME PALLET (FOR APC MACHINE ONLY)		0.020 IN X & Y AXES DIRECTION	PALLET 1 : 0.002 PALLET 2 : 0.003 (FOR APC MACHINE ONLY)

SIGN & DATE	BSR	IG	RP	31.08.11	5 OF 5	DS-DSN-TC-PROD-01-R2
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

ACE MANUFACTURING SYSTEMS LTD.

STANDARD TEST REPORT

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1) HEAVY CUT REPORT :

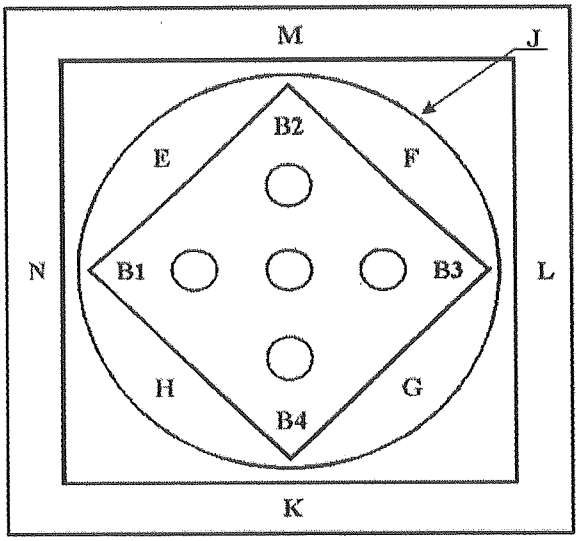
MODEL	: MCV-400	BASE SPEED OF MOTOR	:1100 rpm	MATERIAL	: C.I.(Gr 25)
M/C NO.	: M11-772	SPINDLE SPEED RANGE	:60-6000 rpm	CUTTER	: Ø80 Sq.Sh FMC
TYPE OF MOTOR	: βi 12/8000	MOTOR SPEED RANGE	:75-7500 rpm	WIDTH OF CUT	: 55 mm
PULLEY RATIO	: 1:1.25	MOTOR CAPACITY	:11/7.5 kW	C.S. (V)	: 250 mtr/min.

Spindle speed in rpm	Feed rate mm/ins.	Feed rate mm/rev.	Feed rate mm/min.	Depth Of Cut in mm	M R R cc/min.	'X' Axis % as.per CRT	Spl.load % as per CRT	Remarks
1000	0.15	1.05	1050	2.8	162	28	56	SMOOTH CUTTING
1000	0.15	1.05	1050	5.7	330	45	115	SMOOTH CUTTING

RIGID TAPPING : M8×1.25 RIGID TAPPING TEST CONDUCTED, CHECKED WITH GAUGE & FOUND OK.

2) NAS TEST REPORT :

No.	Description of Test	Test Applied to	Permissible deviation	Actual Readings in mm
1	Flatness of Surface	A	0.015	0.008
2	Parallelism of Surfaces	N-L	0.020	0.010
		M-K	0.020	0.003
		H-F	0.030	0.005
		G-E	0.030	0.006
3	Squareness of side faces	K-L	0.020	0.004
		L-M	0.020	0.004
		M-N	0.020	0.013
		N-K	0.020	0.007
		E-F	0.030	0.005
		F-G	0.030	0.006
		G-H	0.030	0.009
		H-E	0.030	0.004
4	Circularity of Machining	J	0.040	0.010
5	Linear pitch in ' X ' axis	B1-B3 (100)	0.025	0.003
	Linear pitch in ' Y ' axis	B2-B4 (100)	0.025	0.000
	Diagonal pitch	C(70.71)	0.035	0.001
6	Bore size variation		0.020	
	Hole No.	Max.	Min.	Ovality
	B1=	27.452	27.450	0.002
	B2=	27.453	27.449	0.004
	B3=	27.450	27.448	0.002
	B4=	27.454	27.451	0.003
7	Straightness	K	0.010	0.006
		L	0.010	0.004
		M	0.010	0.003
		N	0.010	0.008
		E	0.015	0.005
		F	0.015	0.006
		G	0.015	0.005
		H	0.015	0.003
8	Step difference in axis change over		0.010	0.004



SINGLE AXIS CONTROL K=L=M=N = 240 mm

DOUBLE AXIS CONTROLE=F=G=H = 160 mm

CIRCLE " J " = Ø 243.30 mm

Heavy cut Done By	NAS TEST DONE BY	Department incharge	QC / CMM
ASHWATH A S	ASHWATH A S	14/2/12	14/2/12
Date: 2/2/2012	DATE:04/02/2012	Date:	Date:

F-TSG-34-R1

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-772-X.RTL

Machine:MCV-400.
Date:09:10 Feb 04 2012
Axis:X.

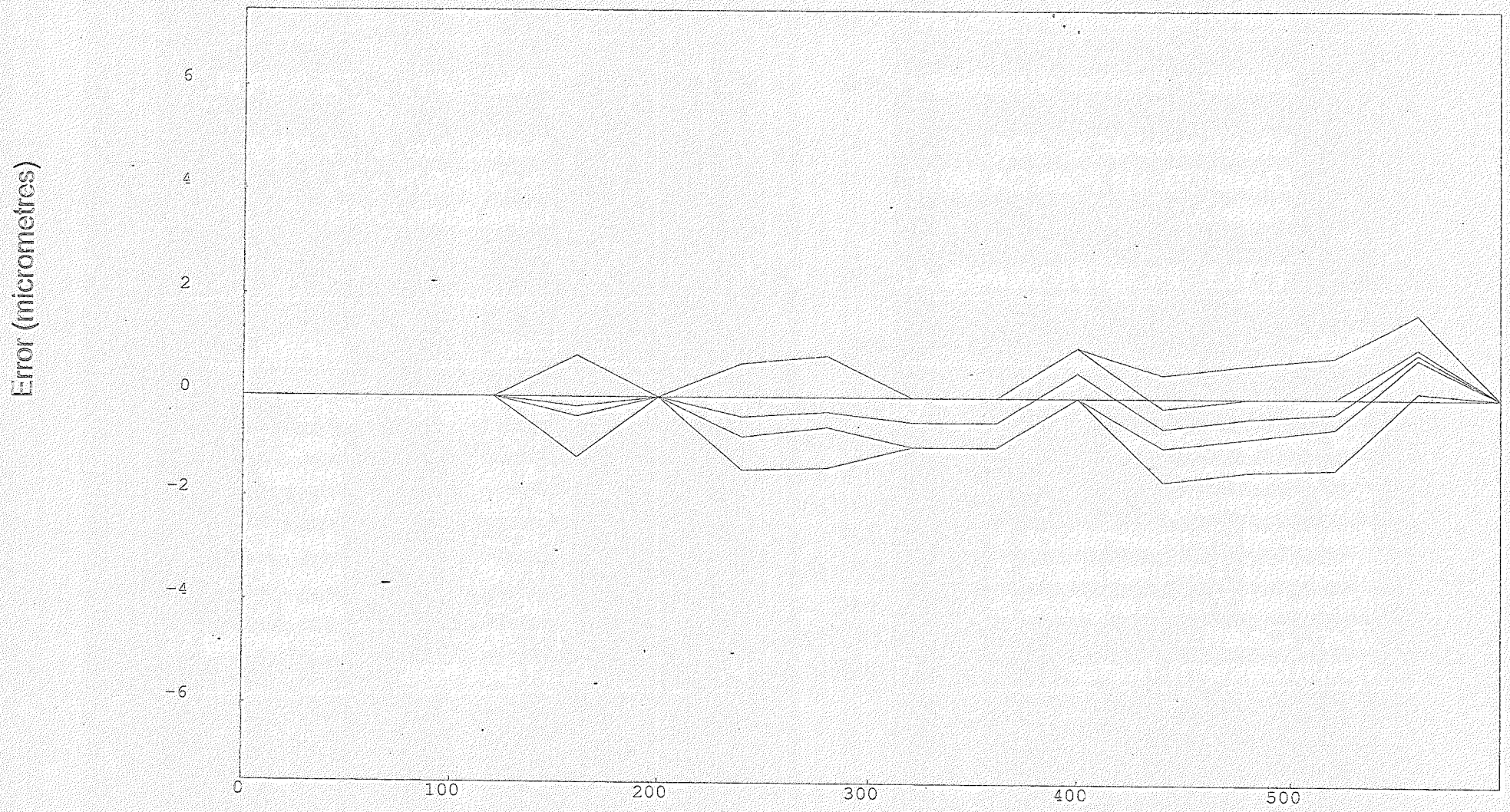
Serial No:M11-772.
By:QAS/RKU.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	09:10 Feb 04 2012	09:20 Feb 04 2012	
Air temperature :	21.14	21.27	deg C
Air pressure :	913.80	914.00	mbar
Rel. humidity :	69.08	68.39	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	20.05	20.06	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31641806	0.31641805	

VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:MCV-400.	Axis:X.	U max : 1.000
Serial No:M11-772.	Location:AMS.	P : 3.342
Date:09:10 Feb 04 2012	U mean : 0.450	Ps max : 1.643
By:QAS/RKU.	Ps mean : 0.644	Pa : 1.500

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-772.
Date:09:10 Feb 04 2012 By:QAS/RKU.
Axis:X. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M11-772-X.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 2 (+) (39.9990 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 2 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 3 (+) (79.9980 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 3 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 4 (+) (119.9970 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 4 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 5 (+) (159.9960 millimetres)
1) 0.0 2) -1.0 3) 0.0 4) -1.0
5) 0.0
Target 5 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 6 (+) (199.9950 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 6 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 7 (+) (239.9940 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 7 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (+) (279.9930 millimetres)
1) 0.0 2) -1.0 3) 0.0 4) -1.0
5) -1.0

Target 8 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 9 (+) (319.9920 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 9 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 10 (+) (359.9910 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 10 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 11 (+) (399.9900 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 11 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 12 (+) (439.9890 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 12 (-)
1) 0.0 2) 0.0 3) 0.0 4) -1.0
5) 0.0

Target 13 (+) (479.9880 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) 0.0
5) -1.0
Target 13 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 14 (+) (519.9870 millimetres)
1) -1.0 2) -1.0 3) 0.0 4) 0.0
5) -1.0
Target 14 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 15 (+) (559.9860 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 0.0
Target 15 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 16 (+) (599.9850 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 16 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-772-Y.RTL

Machine:MCV-400.
Date:10:05 Feb 04 2012
Axis:Y.

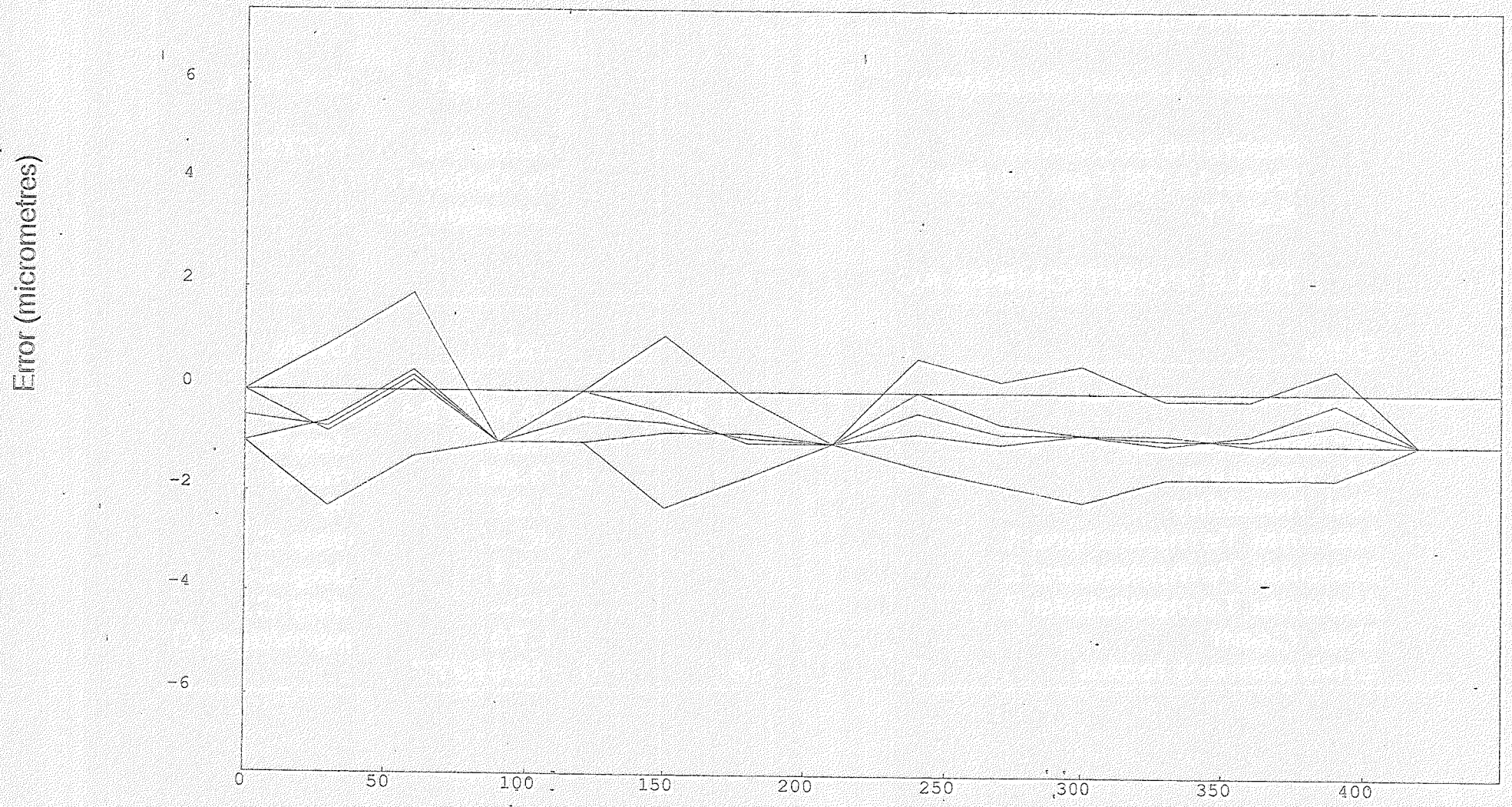
Serial No:M11-772.
By:QAS/RKU.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	10:05 Feb 04 2012	10:15 Feb 04 2012	
Air temperature :	22.12	22.23	deg C
Air pressure :	913.80	913.80	mbar
Rel. humidity :	63.83	63.37	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	20.17	20.21	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31641788	0.31641777	

VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:MCV-400.	Axis:Y.	U max : 1.000
Serial No:M11-772.	Location:AMS.	P : 4.185
Date:10:05 Feb 04 2012	U mean : 0.337	Ps max : 2.985
By:QAS/RKU.	Ps mean : 1.249	Pa : 1.300

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-772.
Date:10:05 Feb 04 2012 By:QAS/RKU.
Axis:Y. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M11-772-Y.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 2 (+) (29.9990 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 2 (-)
1) 0.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0

Target 3 (+) (59.9980 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0
Target 3 (-)
1) 1.0 2) 1.0 3) 0.0 4) 0.0
5) 0.0

Target 4 (+) (89.9970 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 4 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 5 (+) (119.9960 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 5 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 6 (+) (149.9950 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 6 (-)
1) 0.0 2) 0.0 3) -1.0 4) 0.0
5) -1.0

Target 7 (+) (179.9940 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) 0.0
Target 7 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 8 (+) (209.9930 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 8 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 9 (+) (239.9920 millimetres)
1) -1.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0
Target 9 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 10 (+) (269.9910 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 10 (-)
1) 0.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0

Target 11 (+) (299.9900 millimetres)
1) -1.0 2) -1.0 3) 0.0 4) -1.0
5) -1.0
Target 11 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 12 (+) (329.9890 millimetres)
1) -1.0 2) -1.0 3) 0.0 4) -1.0
5) -1.0
Target 12 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 13 (+) (359.9880 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 13 (-)
1) -1.0 2) -1.0 3) 0.0 4) -1.0
5) -1.0

Target 14 (+) (389.9870 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 14 (-)
1) -1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 15 (+) (419.9860 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 15 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 16 (+) (449.9850 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 16 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-772-Z.RTL

Machine:MCV-400.
Date:11:28 Feb 04 2012
Axis:Z.

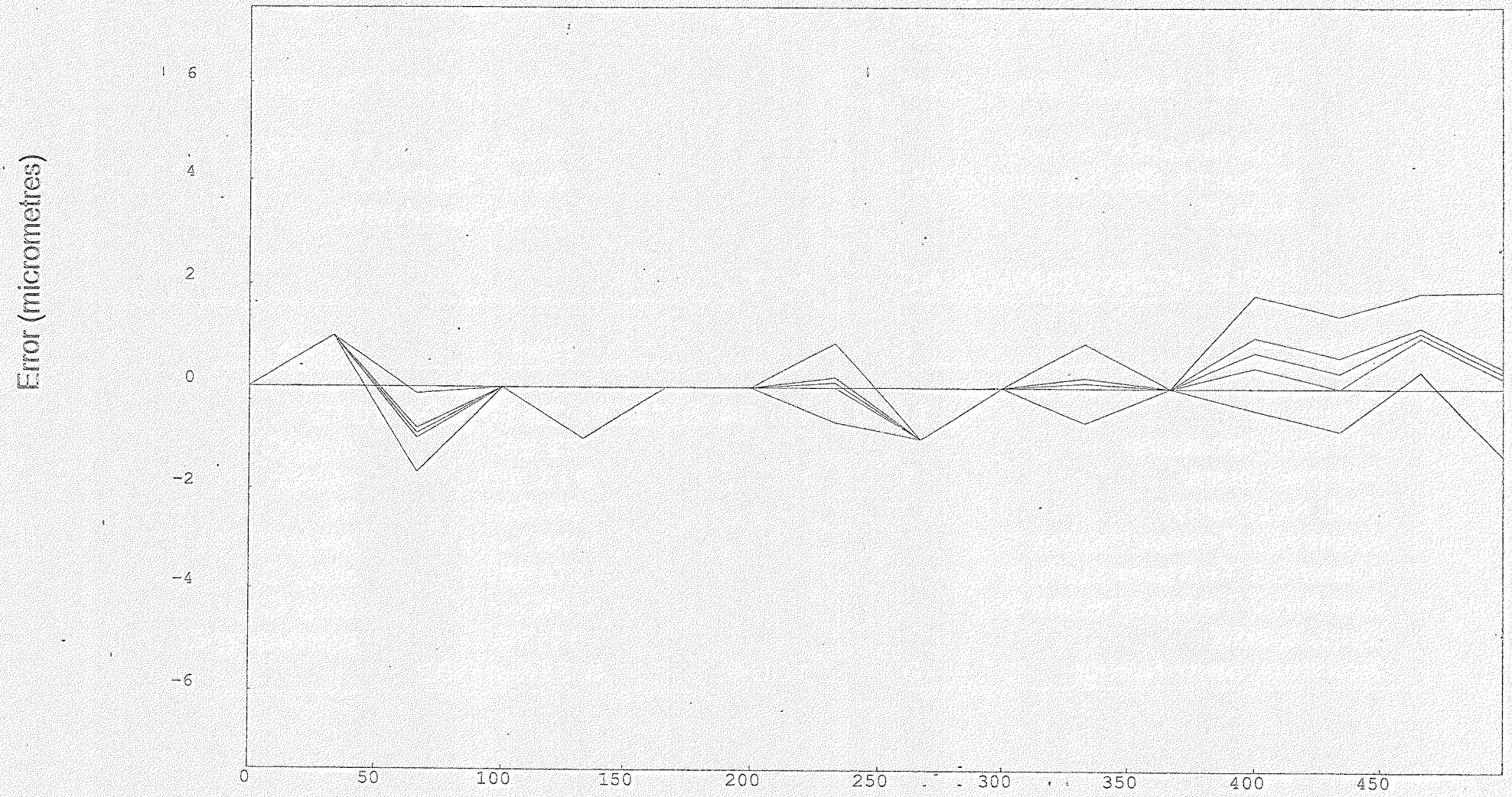
Serial No:M11-772.
By:QAS/RKU.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	11:28 Feb 04 2012	11:38 Feb 04 2012	
Air temperature :	23.41	23.65	deg C
Air pressure :	913.20	913.00	mbar
Rel. humidity :	58.35	56.80	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	20.61	20.66	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31641664	0.31641652	

VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:MCV-400.	Axis:Z.	U max : 0.600
Serial No:M11-772.	Location:AMS.	P : 3.563
Date:11:28 Feb 04 2012	U mean : 0.138	Ps max : 2.985
By:QAS/RKU.	Ps mean : 0.727	Pa : 2.100

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-772.
Date:11:28 Feb 04 2012 By:QAS/RKU.
Axis:Z. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M11-772-Z.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 2 (+) (33.3330 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 2 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 3 (+) (66.6660 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 3 (-)
1) -1.0 2) -1.0 3) -1.0 4) 0.0
5) -1.0

Target 4 (+) (99.9990 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 4 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 5 (+) (133.3320 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 5 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 6 (+) (166.6650 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 6 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 7 (+) (199.9980 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 7 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (+) (233.3310 millimetres)
1) 0.0 2) 1.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 9 (+) (266.6640 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 9 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 10 (+) (299.9970 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 10 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 11 (+) (333.3300 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 11 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0

Target 12 (+) (366.6630 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 12 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 13 (+) (399.9960 millimetres)
1) 1.0 2) 1.0 3) 0.0 4) 0.0
5) 0.0
Target 13 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 14 (+) (433.3290 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 14 (-)
1) 1.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0

Target 15 (+) (466.6620 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 15 (-)
1) 1.0 2) 1.0 3) 1.0 4) 2.0
5) 1.0

Target 16 (+) (499.9950 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0
Target 16 (-)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 1.0

ACE MANUFACTURING SYSTEM LTD.

AMS
In Pursuit of Excellence

Machine No :-	M11-772.	Date :- 07:11 Feb 04 2012
Machine Modle :-	MCV-400.	Cond by:- QAS/RKU.

AXIS CALIBRATION PITCH ERROR REPORT

X				Y				Z			
SL NO 100 (X)	0			SLNO 200 (Y)	0			SL NO 300 (Z)	0		
101	1	141	0	201	0	241	0	301	1	341	0
102	0	142	0	202	1	242	1	302	1	342	-1
103	-1	143	0	203	0	243	0	303	1	343	-1
104	0	144	0	204	-1	244	0	304	0	344	-1
105	1	145	0	205	0	245	2	305	2	345	-1
106	1	146	0	206	-1	246		306	1	346	-1
107	0	147	1	207	0	247		307	0	347	-1
108	-1	148	0	208	-1	248		308	0	348	0
109	0	149	0	209	0	249		309	0	349	-1
110	0	150	0	210	-1	250		310	-1	350	-1
111	0	151	0	211	0	251		311	0	351	
112	0	152	0	212	0	252		312	0	352	
113	1	153	1	213	0	253		313	0	353	
114	0	154	0	214	-1	254		314	-1	354	
115	0	155	0	215	-1	255		315	0	355	
116	1	156	-1	216	0	256		316	-1	356	
117	0	157	-1	217	0	257		317	-1	357	
118	2	158	0	218	0	258		318	-1	358	
119	0	159	0	219	0	259		319	0	359	
120	0	160	0	220	0	260		320	-1	360	
121	1	161		221	0	261		321	0	361	
122	0	162		222	0	262		322	0	362	
123	0	163		223	0	263		323	0	363	
124	0	164		224	0	264		324	0	364	
125	0	165		225	0	265		325	0	365	
126	0	166		226	0	266		326	-1	366	
127	1	167		227	1	267		327	0	367	
128	0	168		228	0	268		328	0	368	
129	1	169		229	1	269		329	0	369	
130	0	170		230	0	270		330	1	370	
131	-1	171		231	1	271		331	0	371	
132	0	172		232	0	272		332	0	372	
133	0	173		233	0	273		333	0	373	
134	1	174		234	1	274		334	0	374	
135	0	175		235	-1	275		335	0	375	
136	-1	176		236	0	276		336	1	376	
137	0	177		237	-1	277		337	1	377	
138	0	178		238	-1	278		338	1	378	
139	-1	179		239	-1	279		339	0	379	
140	0	180		240	0	280		340	-1	380	

SL NO	PARAMETER NO	DISCRIPTION	X	Y	Z
1	3620	P COMP REF NO	160	245	350
2	3621	P COMP MIN NO	100	200	300
3	3622	P COMP MAX NO	165	250	355
4	3623	PCOMP MULTIPLICATION	10	10	10
5	3624	P COMP INTERVAL	10	10	10

ACE MANUFACTURING SYSTEM LTD.

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M/c NO :-	M11-772.	COND BY:- QAS/RKU.
Machine Modle :-	MCV-400.	DATE:- 04-Feb-12

AXIS CALIBRATION BALL BAR PARAMETERS

SERVO PARAMETERS RELATED TO CIRCULARITY FOR FANUC SYSTEM

SL No.	PARAMETER No.	TYPICAL VALUE	DESCRIPTION	X	Y	Z	REMARKS
1	1851	###	BACKLASH IN FEED	35	65	35	
2	1852	###	BACKLASH IN RAPID	15	30	10	
3	1825	8000	SERVO LOOP GAIN	5000	5000	5000	
4	1826	500	IN POSITION WIDTH	500	500	500	
5	2021	256	LOAD INERTIA	192	128	161	*****
6	2048	600 (###)	BACK ACCEL COMPENSATION VALUE	35	35	35	
7	2094	600 (###)	BACK ACCEL COMPENSATION VALUE	35	35	35	
8	2066	10 TO -20	250 MICRO SEC ACC FEEDBACK	-10	-10	-10	*****
9	2068	70 TO700	FEED FWD CO- EFFICIENT	100	100	100	
10	2069	50 TO 200	VELOCITY FEED FWD CO- EFFICIENT FOR RAPID	100	100	100	
11	2071	50 TO 200	TIME DURING WHICH BACK ACCELERATION IS EFFECTIVE	100	100	100	
12	2145	100	VELOCITY FEED FWD CO- EFFICIENT FOR CUTTING	100	100	100	SHOULD NOT BE CHANGED
13	2144	9900	LOOK AHEAD FFW COEFFICIENT FOR CUTTING	9900	9900	9900	
14	2092	9200	LOOK AHEAD FFW COEFFICIENT FOR RAPID	9200	9200	9200	
15	2082	100	STAGE 2 START / END PARAMETER	100	100	100	
16	2039	100	STAGE 2 ACCELERATION AMOUNT	300	300	300	
17	1622	NA	CUT TIME CONSTANT	64	64	64	Same value for 3 axis
NOTE:-		*****	DON'T CHANGE THE VALUE				
		###	VARIES FOR ALL MACHINES				

Ballbar diagnostics table
M11-772-XY-B

Operator: Shivu.
Date: 2012-Feb-04 12:31:08

Machine: QuickCheck
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash X	▸-0.1	◂0.1 µm	0.1 µm	(1%)	(12)
Backlash Y	▴0.2	▾0.1 µm	0.2 µm	(1%)	(9)
Reversal spikes X	▸1.5	◂-1.8 µm	1.8 µm	(10%)	(4)
Reversal spikes Y	▴0.5	▾-5.0 µm	5.0 µm	(29%)	(1)
Lateral play X	▸0.0	◂0.0 µm	0.0 µm	(0%)	(13)
Lateral play Y	▴0.4	▾-0.4 µm	0.2 µm	(1%)	(10)
Cyclic error X	↑1.1	↓1.1 µm	0.9 µm	(5%)	(7)
Cyclic error Y	↑0.9	↓0.6 µm	0.8 µm	(5%)	(8)
Servo mismatch	0.01 ms		0.2 µm	(1%)	(11)
Squareness	24.6 µm/m		3.7 µm	(21%)	(2)
Straightness X	2.7 µm		1.4 µm	(8%)	(5)
Straightness Y	-3.7 µm		1.9 µm	(11%)	(3)
Scaling mismatch	-2.3 µm		1.1 µm	(7%)	(6)
Cyclic pitch X	50.8000 mm				
Cyclic pitch Y	18.0000 mm				
Scaling error X	-66.8 ppm				
Scaling error Y	-59.2 ppm				
Calculated feedrate	1000.8 mm/min				
Centre offset X	-4.6 µm				
Centre offset Y	-6.8 µm				
Positional tolerance	56.4 µm				
Best fit radius	149.9905 mm				
Circularity	10.1 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/360°/180°
Run sequence	CW CCW

ISO 230-4:1996(E) Circular Deviation
M11-772-XY-B

Operator: Shivu.
Date: 2012-Feb-04 12:31:08

RENISHAW

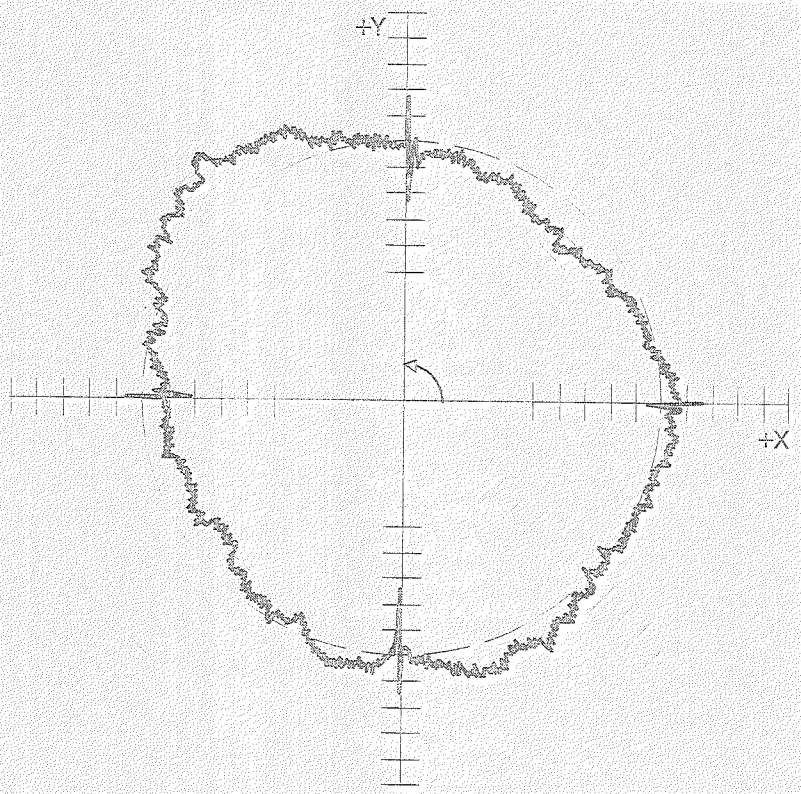
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	10.0µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	XY
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 1

2.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M11-772-XY-B

RENISHAW

Operator: Shivu.
Date: 2012-Feb-04 12:31:08

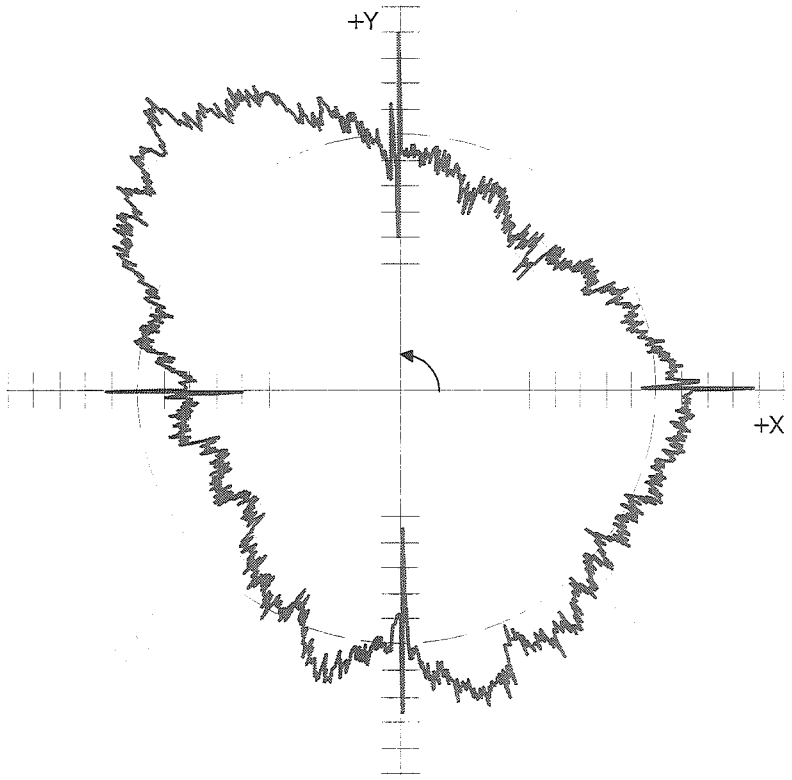
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	9.5µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	XY
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 2

1.0 µm/div

Ballbar diagnostics table
M11-772-XZ-B

Operator: Shivu.
Date: 2012-Feb-04 12:48:02

RENISHAW

Machine: QuickCheck
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash Z	▵-0.6	▿0.5 µm	0.6 µm	(3%)	(9)
Backlash X	‹ -0.4	▹0.1 µm	0.4 µm	(2%)	(10)
Reversal spikes Z	▵-2.1	▿-2.0 µm	2.1 µm	(11%)	(3)
Reversal spikes X	‹ -0.2	▹-1.6 µm	1.6 µm	(8%)	(4)
Lateral play Z	▵0.2	▿-0.4 µm	0.2 µm	(1%)	(12)
Lateral play X	‹ -0.6	▹0.4 µm	0.3 µm	(2%)	(11)
Cyclic error Z	↑1.6	↓1.5 µm	1.5 µm	(8%)	(6)
Cyclic error X	↑1.6	↓1.6 µm	1.4 µm	(7%)	(7)
Servo mismatch	-0.07 ms		1.1 µm	(6%)	(8)
Squareness	1.0 µm/m		0.1 µm	(1%)	(13)
Straightness Z	-7.8 µm		3.9 µm	(20%)	(2)
Straightness X	8.8 µm		4.4 µm	(23%)	(1)
Scaling mismatch	-2.9 µm		1.5 µm	(8%)	(5)
Cyclic pitch Z	50.8000 mm				
Cyclic pitch X	45.0000 mm				
Scaling error Z	-52.6 ppm				
Scaling error X	-42.7 ppm				
Calculated feedrate	1000.4 mm/min				
Centre offset Z	-2.1 µm				
Centre offset X	0.9 µm				
Positional tolerance	47.2 µm				
Best fit radius	149.9929 mm				
Circularity	13.7 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	90°/90°/180°
Run sequence	CW CCW

ISO 230-4:1996(E) Circular Deviation
M11-772-XZ-B

Operator: Shivu.
Date: 2012-Feb-04 12:48:02

RENISHAW

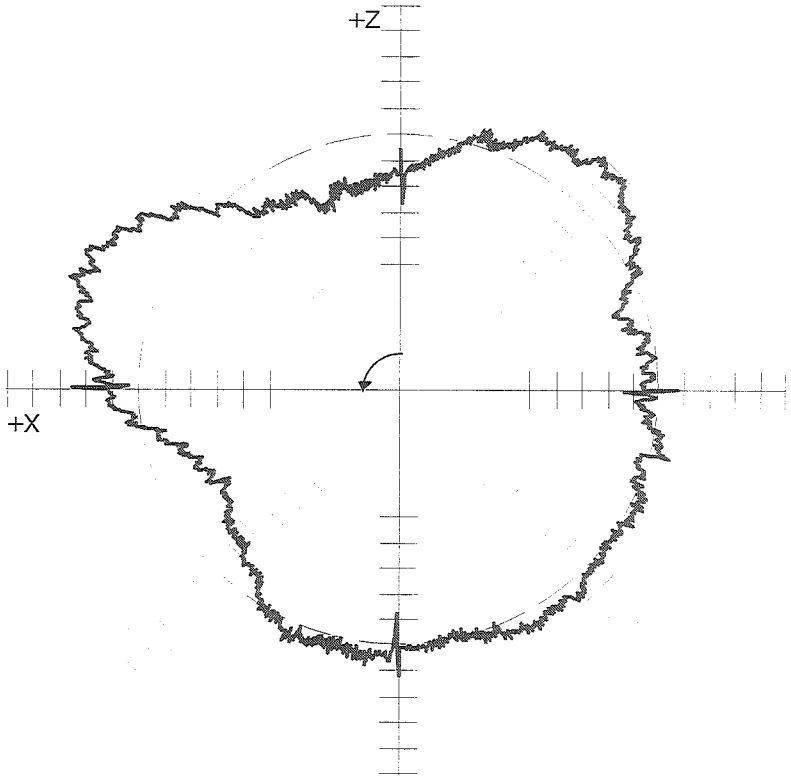
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	12.2µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 1

2.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M11-772-XZ-B

Operator: Shivu.
Date: 2012-Feb-04 12:48:02

RENISHAW

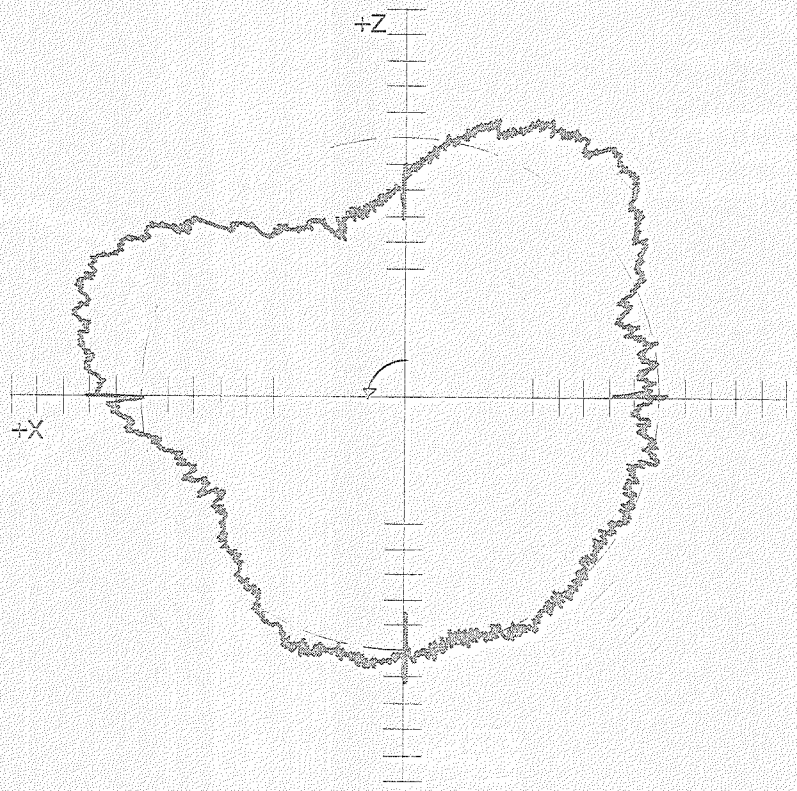
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	13.4µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 2

2.0 µm/div

Ballbar diagnostics table
M11-772-YZ-A

RENISHAW

Operator: Shivu.
Date: 2012-Feb-04 13:02:45

Machine: QuickCheck
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash Y	▸0.2	◂0.3 µm	0.3 µm	(1%)	(12)
Backlash Z	▴-0.9	▾0.1 µm	0.9 µm	(5%)	(8)
Reversal spikes Y	▸-5.0	◂-3.4 µm	5.0 µm	(26%)	(1)
Reversal spikes Z	▴-1.6	▾-2.3 µm	2.3 µm	(12%)	(3)
Lateral play Y	▸-0.4	◂0.1 µm	0.3 µm	(1%)	(13)
Lateral play Z	▴0.7	▾0.1 µm	0.5 µm	(2%)	(10)
Cyclic error Y	↑0.8	↓0.8 µm	0.8 µm	(4%)	(9)
Cyclic error Z	↑1.7	↓1.6 µm	1.5 µm	(8%)	(5)
Servo mismatch	0.07 ms		1.1 µm	(6%)	(7)
Squareness	2.0 µm/m		0.3 µm	(2%)	(11)
Straightness Y	6.2 µm		3.1 µm	(16%)	(2)
Straightness Z	3.6 µm		1.8 µm	(9%)	(4)
Scaling mismatch	2.4 µm		1.2 µm	(6%)	(6)
Cyclic pitch Y	30.4800 mm				
Cyclic pitch Z	50.8000 mm				
Scaling error Y	-44.1 ppm				
Scaling error Z	-52.2 ppm				
Calculated feedrate	1000.8 mm/min				
Centre offset Y	0.4 µm				
Centre offset Z	-0.5 µm				
Positional tolerance	43.7 µm				
Best fit radius	149.9928 mm				
Circularity	11.4 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/360°/180°
Run sequence	CW CCW

ISO 230-4:1996(E) Circular Deviation
M11-772-YZ-A

Operator: Shivu.
Date: 2012-Feb-04 13:02:45

RENISHAW

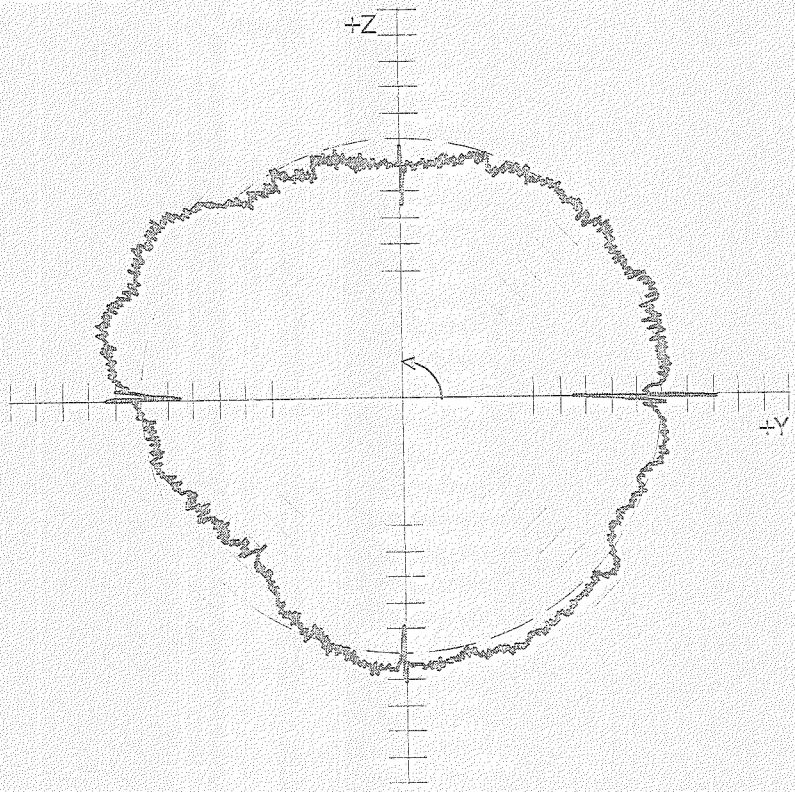
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	11.1µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 1

2.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M11-772-YZ-A

RENISHAW

Operator: Shivu.
Date: 2012-Feb-04 13:02:45

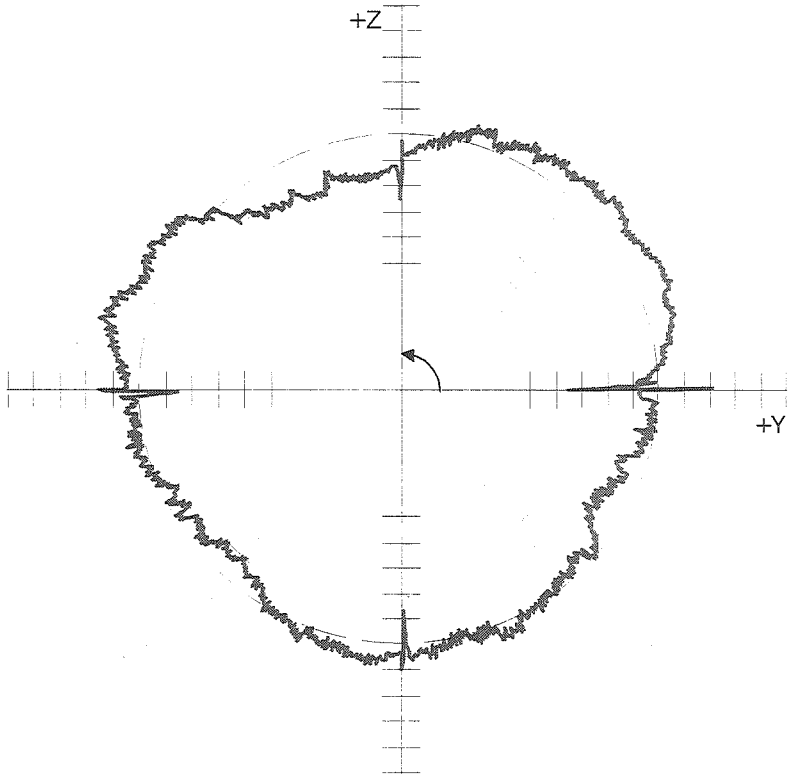
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	11.2µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 2

2.0 µm/div

ACE MANUFACTURING SYSTEMS LTD

SPINDLE BALANCING REPORT

AMS — In Pursuit of Excellence

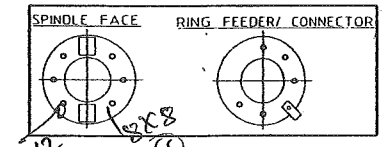
Machine Number : M11-772

Spindle Serial Number : 10508

Spindle Type : BD 8hp

Spindle Speed (Max.) : 6000 rpm

Note : Refer TM-QA-09 for measurement procedure.



Sl No.	Speed in RPM	Velocity of vibration (mm/sec) before Balancing Along X-axis	After Balancing along X-axis (8000 rpm : 0.8 mm/sec max. > 8000 rpm : 1 mm/sec max.)	After Balancing along Y-axis (8000 rpm : 0.8 mm/sec max. > 8000 rpm : 1 mm/sec max.)
1	500			
2	1000	0.015	0.013	0.033
3	1500	0.055	0.038	
4	2000	0.060	0.034	0.010
5	2500	0.033	0.017	
6	3000	0.095	0.056	0.019
7	3500	0.257	0.076	
8	4000	0.405	0.235	0.116
9	4500	0.509 - 125°	0.385	
10	5000	0.707 - 156°	0.365	0.021
11	5500	0.757 - 85°	0.672	
12	6000	0.603 - 89°	0.256	0.043
13	6500			
14	7000			
15	7500			
16	8000			
17	8500			
18	9000			
19	9500			
20	10000			
21	10500			
22	11000			
23	11500			
24	12000			
25	12500			
26	13000			
27	13500			
28	14000			
29	14500			
30	15000			

Spindle maximum vibration mm / sec :

Correction in gms :

REMARK IF ANY:

nil

Date: 02/02/12

Checked By

APPROVED BY

ACE MANUFACTURING SYSTEMS LTD

SPINDLE ASSEMBLY - INSPECTION REPORT

AMS — In Pursuit of Excellence

Machine Number : MH-772	Spindle Unit Number: 10508
Spindle Drive ID / ID -CTS / DD / DD -CTS	Max. Spindle Speed : 6000 RPM
Pull stud type DIN / MAS	Draw bar Type ID / ID-CTS / DD / DD-CTS
Bearing Make & Specification : SKF / B7004-E-T-P4S-UL	Draw bar SL No : 327/1
Spindle Number: ACU-165	Cartridge Number: 780

Spindle Nose Taper ISO-30 / 40 / 50 / HSK50 / 63 / 100

SL No.	Description	Recommended	Actual
1	Locknut Tightening Torque	Ref.DS-PDN-MEC-SA-08	115 Nm
2	Spindle Bearing Preload	Ref.DS-PDN-MEC-SA-07	-0.009 mm
3	Spindle Runout on taper	0.002 mm (*)	0.002 mm
4	Mandrel Runout near the spindle nose	0.005 mm (*)	0.002 mm
5	Mandrel Runout at 4 positions @ 90° at a distance of 300 mm (ISO 40/50, HSK 63), 250 mm (ISO 30, HSK 50) from spindle nose	10 micron (*)	1. 5 2. 3 3. 5 4. 4
6	Spindle stabilization temperature on Test rig	Upto 15° C above ambient	21°C - 31°C = 10°C
7	Check for abnormal / unhealthy noise	Abnormal / unhealthy noise not acceptable	OK
8	Pulley ratio (For ID machines)	Refer Work order	1:1.08 Ratio
9	Number of Teeth on spindle pulley (For ID machines)	Ref.DS-DSN-PROD-02	
10	Spindle pulley Runout (For ID machines)	0.050 mm (*)	0.050 mm
11	Spindle connector Runout (For DD machines)	0.010 mm (*)	—
12	Disc spring specification (Single / double stack)	Ref.DS-DSN-PROD-01	Groove Dia 8x - 3mm
13	Number of disc springs used	Ref.DS-DSN-PROD-01	6 Only
14	Disc spring Compression, Free length & Compressed length	Ref.DS-DSN-PROD-01	97h - 951 = 03mm
15	Tool clamping force	Ref.DS-DSN-PROD-01	Satisfy
16	ID-CTS Deublin Runout in Assembly condition (For CTS Spindle)	0.030 mm	Deublin Sl. No. : ..... Run out : .....

(*) To be measured after temperature stabilisation of spindle at max. speed.

REMARK IF ANY:

Date: 05/1/17	Checked By	Approved By

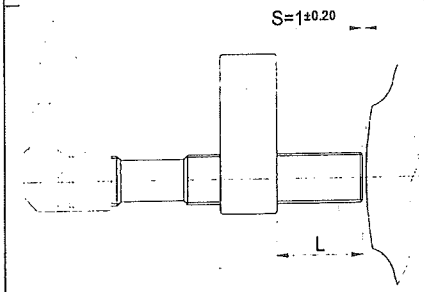
PRAGATI	AUTOMATIC TOOL CHANGER TEST REPORT	
SL NO: 14985	CUSTOMER I.D: S01-011-79	CUSTOMER: AMS
PRODUCT I.D.: ATC1024V-032	PRODUCT DESCRIPTION: ATC1024V-R225-BT.DIN	
DESPATCH DATE: 24-01-2012		

TEST RESULTS:

TESTING HOURS	SPECIFIED: 6 Hrs.	ACTUAL: 6 Hours

PARAMETERS	SPECIFIED	ACTUAL		
		DURING ASSLY	AFTER 6 Hrs	DURING DESPATCH
ARM STROKE	114 -115mm	114.30	114.60	114.60
ARM AXIAL PLAY(IN PARKING)	0.20-1.50mm	1.20	1.20	1.20
ARM RADIAL PLAY @R250 IN PARKING	0.20-1.0mm	0.20	0.40	0.40
ARM RADIAL PLAY @R250 WHILE MOVING OUT	0.20-1.0mm	0.20	0.40	0.40
ARM RADIAL PLAY @R250 WHILE MOVING IN	0.20-1.0mm	0.20	0.40	0.40
ARM PARLLALITY	0-0.20mm	0.05	0.10	0.10
ARM PARK SIGNAL	Visual Check	OK	OK	OK
TOOL DECLAMP SIGNAL	Visual Check	OK	OK	OK
TOOL CLAMP SIGNAL	Visual Check	OK	OK	OK
BACKLASH OF WHEEL @R300	0.05-0.40mm	0.10	0.20	0.20
AXIAL PLAY OF WHEEL	0.01-0.10mm	0.06		

SENSING DISTANCE



PROXIMITY	SENSING GAP(S)	SENSOR LENGTH(L)
PS1	1.00	23.00
PS2	1.00	23.10
PS3	1.00	24.90
PS4	1.00	30.00
PS5	1.00	—
PS6	0.80	—
PS7	1.00	—

DESCRIPTION	NAME	SIGNATURE	DATE
ASSEMBLED BY	Manjunatha.P.C	Manu	22/01/12
CHECKED BY AFTER 6Hrs	Hanumappa	HSK	23-01-12
INSPECTED BY	Channalasavanna	Chal	24.01.12
APPROVED BY	HEBBAR		24.01.12

ACE MANUFACTURING SYSTEMS LTD

HYDRAULIC ASSEMBLY - INSPECTION REPORT

AMS — In Pursuit of Excellence

MACHINE NUMBER : M11- 778 — 2STN APL

I. Power Pack Details			II. Pump Details	
1	Make :	Pascal / Hyd Tech / Rexorth / Yuken /Jai Chamundi	Pump Make :	Rexroth / Yuken /Corotech / vickers
2	Sl No. :	Q.C. Code no:- 6350	Pump Sl. No. :	1150
3	Tank capacity :	32 / 40 / 63 / 70 Liters	Motor Make :	Rotomotive / Siemens
4	Cooler make :	Pascal System Technik Pvt Ltd	Motor Sl. No :	No M1111250S
5	Cooler fan make :	EBM NADI	—	—
6	No.of stations :	1 / 2 / 3 / 4 / 5 / 6	—	—

III	CHECK POINTS	Remark
1	Ensure Power pack is assembled as per circuit diagram & same is attached to inspection report.	✓
2	Ensure Accessories like Minnimess coupling, Guages are assembled.	✓
3	Ensure instruction name plates, power pack details, name plate are fitted	✓
4	All ports are tied with identification plate.	✓
5	Ensure there is no visual damage to the elements.	✓
6	Ensure oil level (at level indicator) in the hyd tank is at satisfactory	✓
7	Ensure tightness of all screws & marking neatly with yellow marker.	✓

IV	RUNNING CHECK POINTS	Remark
1	Check Motor direction is as per arrow marking.	✓
2	Ensure direction of rotation of cooler fan is as per arrow marking	✓
3	Ensure hydraulic pressure switch set as per circuit diagram.	✓
4	Ensure Accumulator oil & gas pressure as per circuit diagram (if applicable).	NA
5	Ensure no oil leakage from hydraulic system including at the oil level indicator & pipe joints	✓
6	Ensure continuous running of power pack for 4 hrs with solenoid actuation once per minute	✓
7	Power pack oil temp rise is within 30°C above ambient temp or 65 °C maximum.	✓
8	Check for any abnormal / unhealthy noise	✓

REMARK IF ANY:

10/1/12 Date:	Veerabhadraiah Checked by	Approved by	

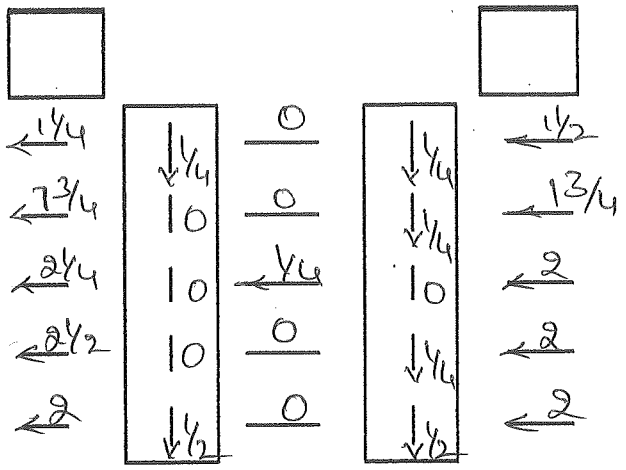
ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS — In Pursuit of Excellence

MACHINE NUMBER: M11-772 **DATE:** 26/12/11

I. MACHINE BASE LEVELING: Least count of the Instrument: 1 div = 0.020mm / meter



SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	1 div / meter	1/2 div		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	1/2 div		
3	Leveling - Individual (Transverse)	3 div / meter	3 div		
4	Straightness of locating guide way	± 0.005	0.003 MM		
5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.004 MM		
6	Ball screw Run out at Motor end	±0.010	0.010 MM		
7	Ball screw Run out at Brake end	±0.010	0.010 MM		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010 MM		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010 MM		
10	Ball screw Stretching	0.025 (±0.002 /meter)	0.025 MM		
			M.E	CEN	B.E
11	Ball screw Torque	0.6 to 1.8 N-m	1.2	1.2	1.3
12	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
13	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0		NA	
14	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
15	Final slide Torque – box guide way	2.5 to 3 N-m			
16	Final slide Torque – L M rail	1.6 to 2.5 N-m	1.6	1.6	1.5
17	Variation in torque for final slide	.10%	—	—	—
18	Remarks if any :		—	—	—

SIGN & DATE	Hemanth Basetha		13/01	1...10	F-QA-IPR-BBA-03-R0
	PREPARED	CHECKED	APPROVED	PAGE NUMBER	FORMAT NUMBER

ACE MANUFACTURING SYSTEMS LTD

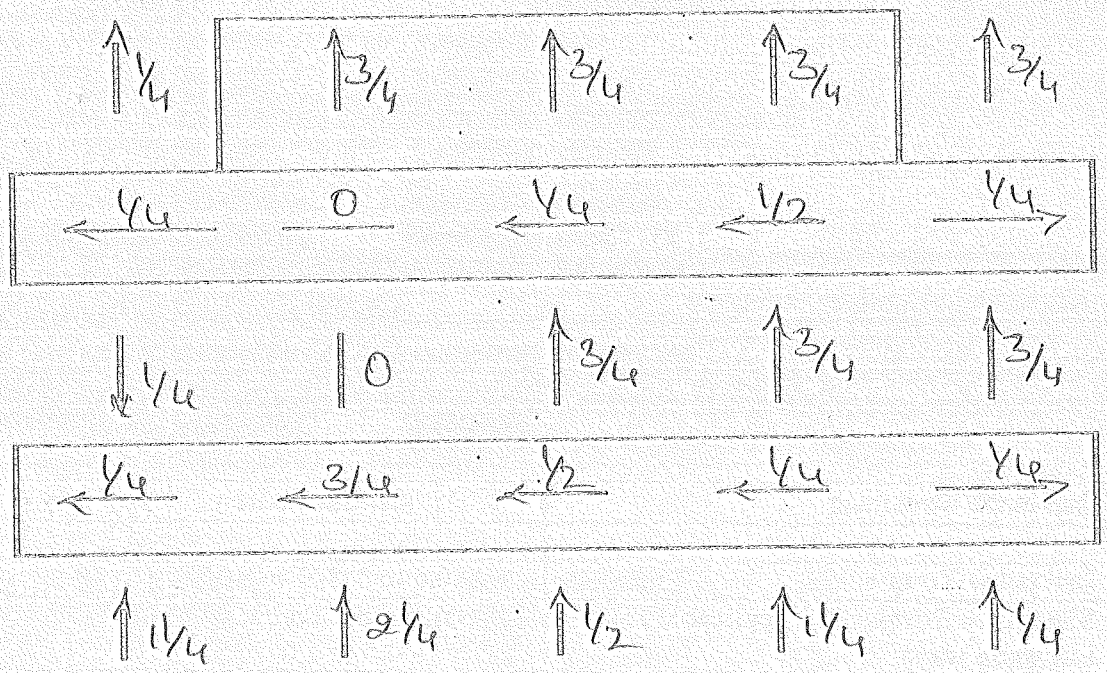
INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

MACHINE NUMBER: M11-772 DATE: 26/12/14

SL NO	Check Points	Remark
1	Ensure all leveling screws & locknuts are fixed to the base.	✓
2	Ensure guide way butting face blue matching is minimum 70%	✓
3	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	✓
4	Ensure position of LM block ground surface butted to slide butting face.	✓
5	Ensure required height blocks are fitted to the base as per Work Order / Planning chart.	✓
6	Ensure plugs are fitted for L M guide way	✓

DATE: 26/12/14

II. SADDLE LEVEL READING: Least count of the Instrument: 1 div = 0.020mm / meter



Remarks if any :

SIGN & DATE	Hemanth Bwesh		13/01	2...10	F-QA-IPR-BBA-03-R0
	PREPARED	CHECKED	APPROVED	PAGE NUMBER	FORMAT NUMBER

ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS
In Pursuit of Excellence

MACHINE NUMBER: M11-772 **DATE:** 26/12/19

SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	2 div / meter	1 division		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	1 div		
3	Leveling - Individual (Transverse)	3 div / meter	1½ div		
4	Straightness of locating guide way	± 0.005	0.003 MM		
5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.005 MM		
6	Ball screw Run out at Motor end	±0.010	0.010 MM		
7	Ball screw Run out at Brake end	±0.010	0.010 MM		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010 MM		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010 MM		
10	Ball screw Stretching	0.025 (±0.002 /meter)	0.025 MM		
			M.E	CEN	B.E
11	Ball screw Torque	0.6 to 1.8 N-m	1.1	1.1	1.2 N/M
12	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
13	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
14	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0		NA	
15	Final slide Torque – box guide way	2.5 to 3 N-m			
16	Final slide Torque – L M rail	1.6 to 2.5 N-m	1.3	1.3	1.6 N/M
17	Variation in torque for final slide	10%	—	—	—

Remarks if any :

SIGN & DATE	Parameshwaran Nagendra		13/01	3...10	F-QA-IPR-BBA-03-R0
	PREPARED	CHECKED	APPROVED	PAGE NUMBER	FORMAT NUMBER

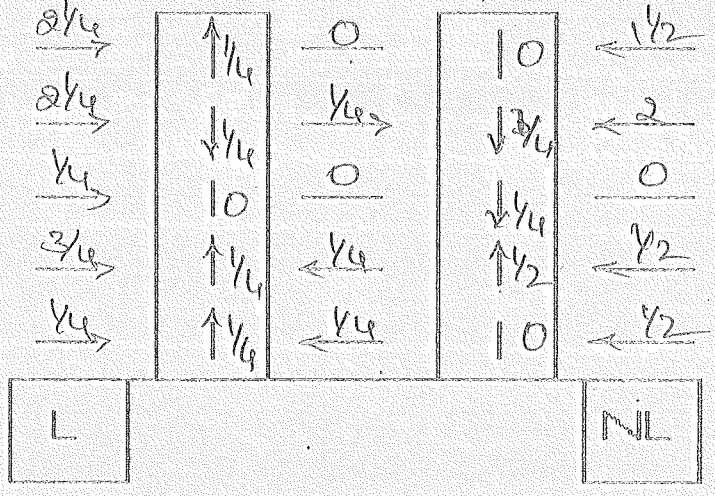
ACE MANUFACTURING SYSTEMS LTD
INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

MACHINE NUMBER: 11-772 DATE: 26/12/11

SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air & kerosene	L
2	Ensure guide way butting face & saddle bottom blue matching is minimum 80%	L
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	—
4	Ensure proper taper matching of jib. ((H G m/c only)	—
5	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	L
6	Ensure position of LM block ground surface butted to slide butting face. (L M m/c only)	L
7	Ensure Equal gap between ball nut & nut housing bore.	L
8	Ensure Nut housing alignment is with in 0.010 mm/meter	L
9	Ensure White Prussian matching between nut housing & ball nut butting face is80%	L
10	Ensure bearing housing Dowelling and grub locking.	L
11	Ensure Ball screw lock nut grub locking with lock tite.	L
12	Ensure guide way wipers are fixed. (H G m/c only)	—
13	Ensure Nylon stoppers are fixed as per specification.	L
14	ensure plugs are fitted for lm guide way & grub screws are fixed to saddle mounting holes	L

DATE: 29/12/11

III. COLUMN READING: Least count of the Instrument: 1 div = 0.020mm / meter



Remarks if any :

SIGN & DATE	Parameshwar Nagendra		13/01	4...10	F-QA-IPR-BBA-03-R0
	PREPARED	CHECKED	APPROVED	PAGE NUMBER	FORMAT NUMBER

ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS — In Pursuit of Excellence

MACHINE NUMBER: MU-772 **DATE:** 29/12/11

SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	1 div / meter	3/4 div		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	1/2 div		
3	Leveling - Individual (Transverse)	3 div / meter	2 1/4 div		
4	Straightness of locating guide way	± 0.005	0.006 MM		
5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.003 MM		
6	Ball screw Run out at Motor end	±0.010	0.010 MM		
7	Ball screw Run out at Brake end	±0.010	0.010 MM		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010 MM		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010 MM		
10	Ball screw Stretching	0.025 (±0.002 /meter)	— NA —		
11	Mandrel radial run out	±0.010	0.003 MM		
12	Mandrel reading at Top (parallelism wrt.head stock movement)	±0.010	0.010 MM		
13	Mandrel reading at side (parallelism wrt.head stock movement)	±0.010	0.010 MM		
			M.E	CEN	B.E
14	Ball screw Torque	0.6 to 1.8 N-m	1.3	1.3	1.4
15	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
16	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
17	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
18	Final slide Torque – box guide way	2.5 to 3 N-m	—	—	—
19	Final slide Torque – L M rail	1.6 to 2.5 N-m	0.6	0.6	1.6
20	Variation in torque for final slide	10%	—	—	—

Remarks if any :

SIGN & DATE	Parameshakar Nagendra		13/01	5...10	F-QA-IPR-BBA-03-R0
	PREPARED	CHECKED	APPROVED	PAGE NUMBER	FORMAT NUMBER

ACE MANUFACTURING SYSTEMS LTD
INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS
In Pursuit of Excellence

MACHINE NUMBER: M11-772 DATE: 29/12/11

SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air & kerosene.	✓
2	Ensure guide way butting face, head stock bottom face,& column bottom face blue matching is minimum 80%	✓
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	—
4	Ensure proper matching of jib taper. ((H G m/c only)	—
5	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	✓
6	Ensure position of LM block ground surface butted to slide butting face. (L M m/c only)	✓
7	Ensure Equal gap between ball nut & nut housing bore.	✓
8	Ensure nut housing alignment is with in 0.010 mm	✓
9	Ensure White Prussian matching between nut housing & ball nut butting face is 80 %	✓
10	Ensure spindle cartridge butting face blue matching is 80%	✓
11	Ensure head stock is fixed as per requirement (ID, DD, and DD CTS).	✓
12	Ensure bearing housing Dowelling and grub locking.	✓
13	Ensure Ball screw lock nut grub locking with locktite.	✓
14	Ensure guide way wipers are fixed. (H G m/c only)	—
15	Ensure Nylon stoppers are fixed as per specification.	✓
16	Ensure machine lifting tackles are fitted.	✓
17	Ensure plugs are fitted for lm guide way & grub screws are fixed to head stock mounting holes	✓

MACHINE NUMBER: M11-772 (APC) DATE:

IV. SIDE ARM ALINGMENT READING: (Turret Type ATC only)

SL NO	DESCRIPTION	Recommended (mm)	ACTUAL (mm)
1	Shaft reading at Top (longitudinal)	±0.050	
2	Shaft reading at side (transverse)	±0.030	
3	Square ness of locating side	±0.010	N/A
4	Distance between Z axis guide way top surface & side arm bore centre	REFER AS PER DRAWING NO	
5	Distance between sidearm front butting face on column & side arm bore centre	REFER AS PER DRAWING NO	

Remarks if any :

SIGN & DATE	Parameshwar Nagendra	2/13/01		6...10	F-QA-IPR-BBA-03-R0
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ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS *In Pursuit of Excellence*

MACHINE NUMBER: M11-772 APC **DATE:** 02-01-12

RECOMMENDED TOLERANCE AS PER DS – DSN – TC – PROD – 01 – R0

V. LEVELLING OF MACHINE: **L.C of Instrument: 1 div = 0.020mm / m**

Along X-axis			Along Y-axis		
Motor End	Middle	Brake End	Motor End	Middle	Brake End
←¼	0	0	↑½	10	↑½
←½	0	←½	10	10	↑¼
0	0	←½	10	↓¼	↓¼

VI. TABLE TOP READING ALONG X- AXIS:

−3	−6		0
+4	+3	−1	0
—	—	—	—
0	−3	−2	0

VII. TABLE TOP READING ALONG Y- AXIS:

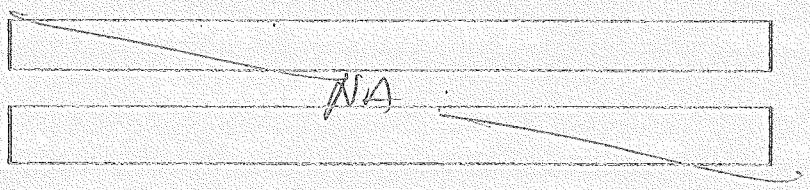
0	−3	+2	+4
+1	−2	+2	0
+2	−2	+1	−2
0	0	0	0

SIGN & DATE	Parameshalch Nagendra		13/01	7...10	F-QA-IPR-BBA-03-R0
	PREPARED	CHECKED	APPROVED	PAGE NUMBER	FORMAT NUMBER

ACE MANUFACTURING SYSTEMS LTD INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS	AMS In Pursuit of Excellence

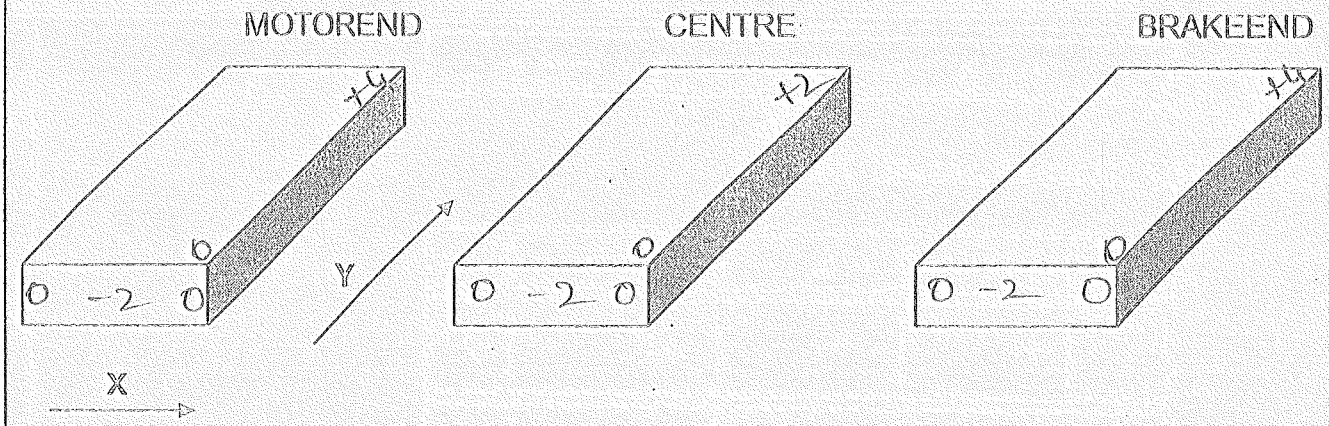
MACHINE NUMBER: M11-772 APC DATE: 02-01-12

VIII. PARALLELISM OF REFERENCE T-SLOT TO X-AXIS MOVEMENT AT FRONT & REAR SURFACE:



SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air	
2	Ensure table bottom face blue matching IS minimum 70%	
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	
4	Ensure proper matching of jib taper. ((H G m/c only)	
5	Ensure position of LM block ground surface butted to slide butting face.	
6	Ensure Equal gap between ball nut & nut housing bore.	
7	Ensure nut housing parallelism is with in 0.010 mm	
8	Ensure Blue matching between nut housing & ball nut butting face is 70%	
9	Ensure Table Tenon slot size as per specification	

IX. SQUARENESS OF Y –AXIS MOVEMENT WITH RESPECT TO X – AXIS MOVEMENT:



Remarks if any :

SIGN & DATE	Parameshwaran Nagendra		13/01	8...10	F-QA-IPR-BBA-03-R0
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per MCM

ACE MANUFACTURING SYSTEMS LTD
INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS
In Pursuit of Excellence

MACHINE NUMBER: M11-772 **DATE:** 02-01-12

X. COLUMN READING (INITIAL):

XZ PLANE(SIDE)

500	
450	+65
400	
350	
300	
250	+35
200	
150	
100	
50	
0	0

YZ PLANE (FRONT)

500	
450	-30
400	
350	
300	
250	-20
200	
150	
100	
50	
0	0

COLUMN READING - FINAL:

XZ PLANE(SIDE)

500	
450	
400	+30
350	
300	
250	
200	+15
150	
100	
50	
0	0

YZ PLANE (FRONT)

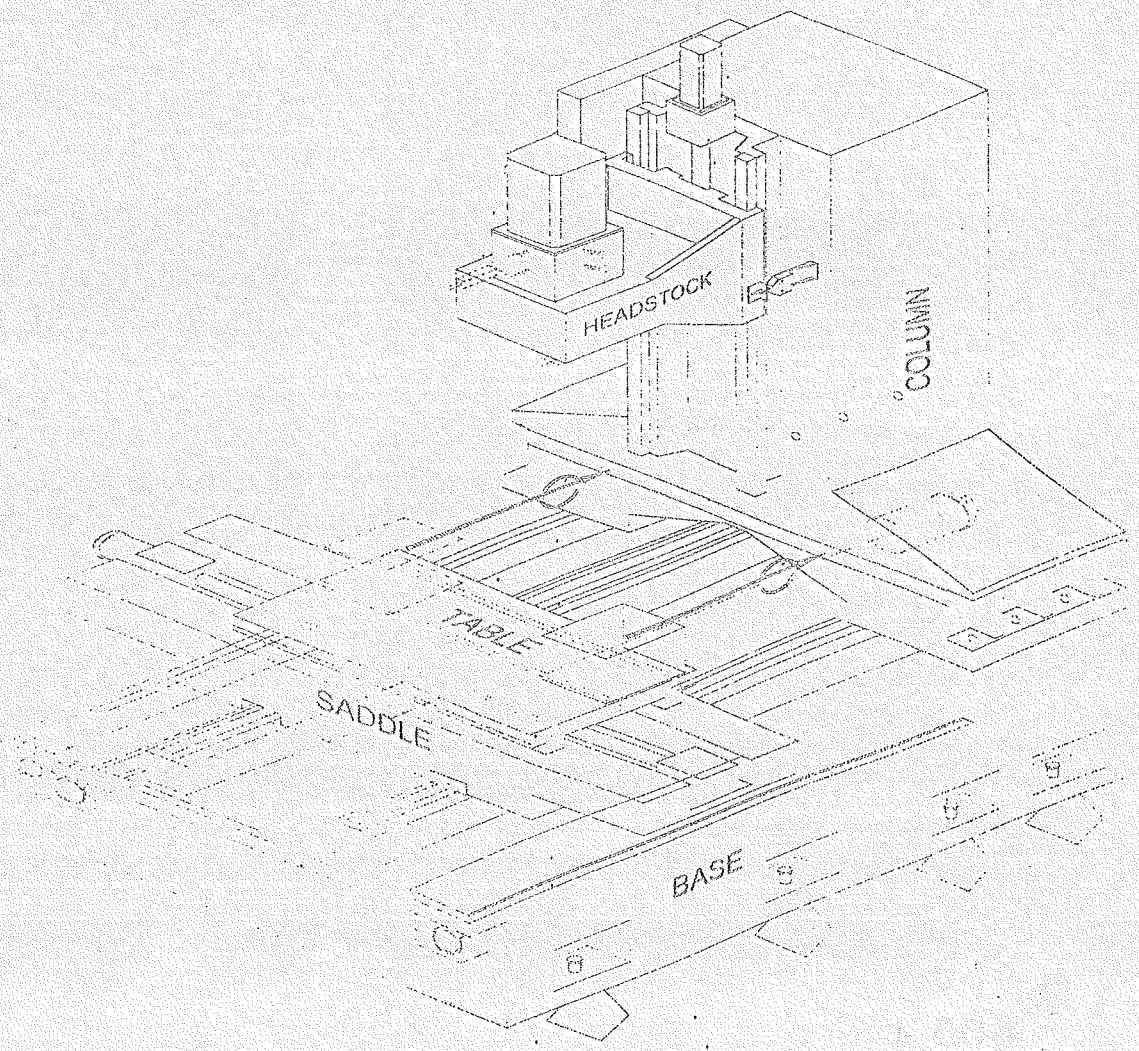
500	
450	
400	+5
350	
300	
250	
200	
150	
100	
50	
0	0

SIGN & DATE	Paramellakan Nagendra 13/01			9...10	F-QA-IPR-BBA-03-R0
	PREPARED	CHECKED	APPROVED	PAGE NUMBER	FORMAT NUMBER

ACE MANUFACTURING SYSTEMS LTD INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS	AMS In Pursuit of Excellence

MACHINE NUMBER: M11-772 DATE: 02-01-12

XI. COLUMN ALINGMENT READING WITH RESPECT TO X-AXIS MOVEMENT:
PERMISSIBLE VALUE = 0.30MM/METRE



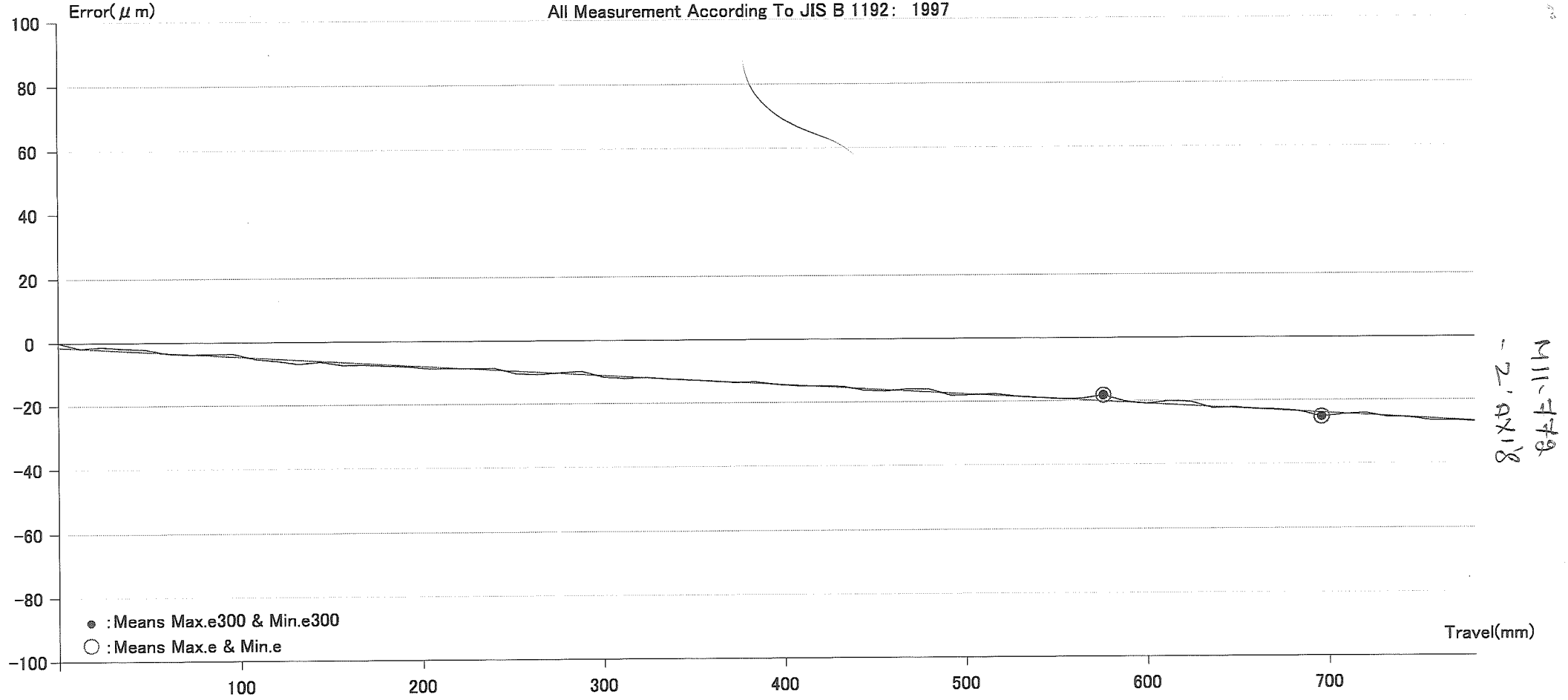
0 +0.070 MM

Remarks if any :

SIGN & DATE	Parameshakki Nagendra	13/01	10...10	F-QA-IPR-BBA-03-R0
	PREPARED	CHECKED	APPROVED	PAGE NUMBER	FORMAT NUMBER

PMI BALLSCREWS — BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



Specifications:	
·Customer	:AMS
·Order No.	:S3-110020142010
·Drawing No.	:BA300510004-0
·Part No.	: *: S12-016-96
·Specification	:R40-12E1-FDWE-841-1105-0.008
·Nominal Diameter (mm)	:40
·Nominal Pitch (mm)	:12
·Travel Length (mm)	:780
·E(μm)	:±18
·T(μm)	:-25

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μm)	-43~-7	-25.40
·e(μm)	13	3.12
·e300(μm)	8	3.12
·Tq(kgf - cm)	5.72~9.88	6.76-7.8
·Accuracy Grade	C3	C0

Manufactured By
PRECISION MOTION IND., INC.
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Measuring Environment:
·Refernce Temperatrue : 20℃ ± 0.5℃
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

QC
2011/07/27
PASS

Inspector: ARJAY

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

BALL SCREW ASSEMBLY REPORT

Machine Number :	M11-779	Ball screw Sl No:	11002014 2010	Axis :	2

I . Ball screw Make & Specification : PMI (RM0-12E1-POWE- 8U1-1105-0.008)

II. Ball Screw Bearing Make & Specification : NACHI (25TAB06DB L4N)

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS - PDN - MEC - SA - 02	1.9 N/M	1.4N/M
2	Bearing housing Bore diameter	As per drawing	170 3U	3705U
3	Bearing housing centre height difference	0.01 mm	—	—
4	Bearing housing bore parallelism with respect to pads	0.01 mm	—	0.006MM

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	(2U-291-8123	T1	13.754
2	Bearing width B1	-	T2	
3	Bearing width B2	NACHI	T3	14.945
4	Bearing width B3	—11—	T4	14.950
5	Spacer cylindrical width B4	—	T5	-
6	Flange cover (2) spigot depth	(2U-291-8123	T6	-
7	Housing Bearing depth	CB-701-2123	T7	57.002

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 701/22

T6 = 13.435 mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	-	T1	-
2	Bearing width B1	-	T2	-
3	Bearing width B2	6805 CB	T3	-
4	Bearing width B3	-	T4	-
5	Spacer cylindrical width B4	-	T5	-
6	Flange cover (2) spigot depth	(2U-280.8123	T6	-
7	Housing Bearing depth	CB-151-2123	T7	48.470

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 154/05

T6 = mm

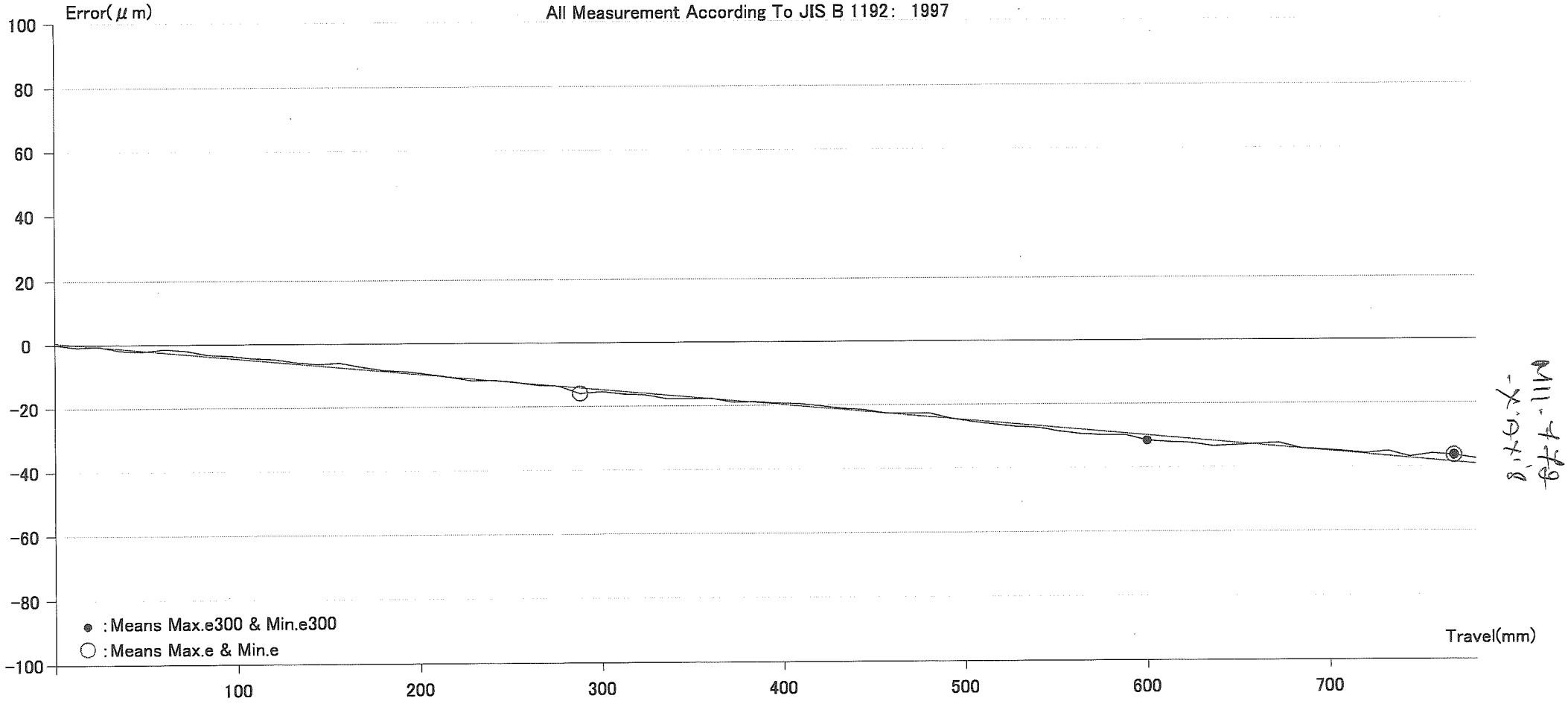
VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	—11—
3	Ensure smooth rotation of ball screw assembly.	—11—
4	Ensure ball screw run out within 0.010 mm	—11—
5	Ensure tightness of all screws & marking with yellow marker	—11—
6	Ensure ball Screw report attached to Inspection Report.	—11—

Date:	Checked by	Approved by

PMI BALLSCREWS — BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



Specifications:	
·Customer	:AMS
·Order No.	:S3-110032521002
·Drawing No.	:BA300510002-0
·Part No.	: *:S12-016-95
·Specification	:R40-12C1-FDWE-846-1110-0.008
·Nominal Diameter (mm)	:40
·Nominal Pitch (mm)	:12
·Travel Length (mm)	:780
·E(μm)	:±18
·T(μm)	:-25

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μm)	-43~-7	-39.80
·e(μm)	13	3.93
·e300(μm)	8	3.79
·Tq(kgf - cm)	4.16~7.28	5.2-6.24
·Accuracy Grade	C3	C3

Manufactured By
PRECISION MOTION IND., INC.
www.pmi-amt.com

Measuring Environment:
·Refernce Temperatrue : 20℃ ± 0.5℃
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

QC
2011/10/05
PASS

Inspector: CAST

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

BALL SCREW ASSEMBLY REPORT

Machine Number : M11-779	Ball screw Sl-No: 110032521002	Axis : X

I . Ball screw Make & Specification : PMI | R40-10H-FOWE-846-1110-0.008)

II. Ball Screw Bearing Make & Specification : NACHI | 25TAB06DB(U+U)

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS - PDN - MEC - SA - 02	1.1N|M	1.2N|M
2	Bearing housing Bore diameter	As per drawing	90+0.014	58+0.014
3	Bearing housing centre height difference	0.01 mm	0.01MM	—
4	Bearing housing bore parallelism with respect to pads	0.01 mm	0.006MM	0.008MM

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	NACHI	T3	14.945
4	Bearing width B3	—11—	T4	14.945
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	CW-275-8123	T6	—
7	Housing Bearing depth	CBT-103-2123	T7	43.510

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 108/22

T6 = 13.680 mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	NACHI	T3	14.960
4	Bearing width B3	—11—	T4	14.960
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	CW-860-8123	T6	—
7	Housing Bearing depth	CBT-153-2123	T7	43.510

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 158/22

T6 = 13.680 mm

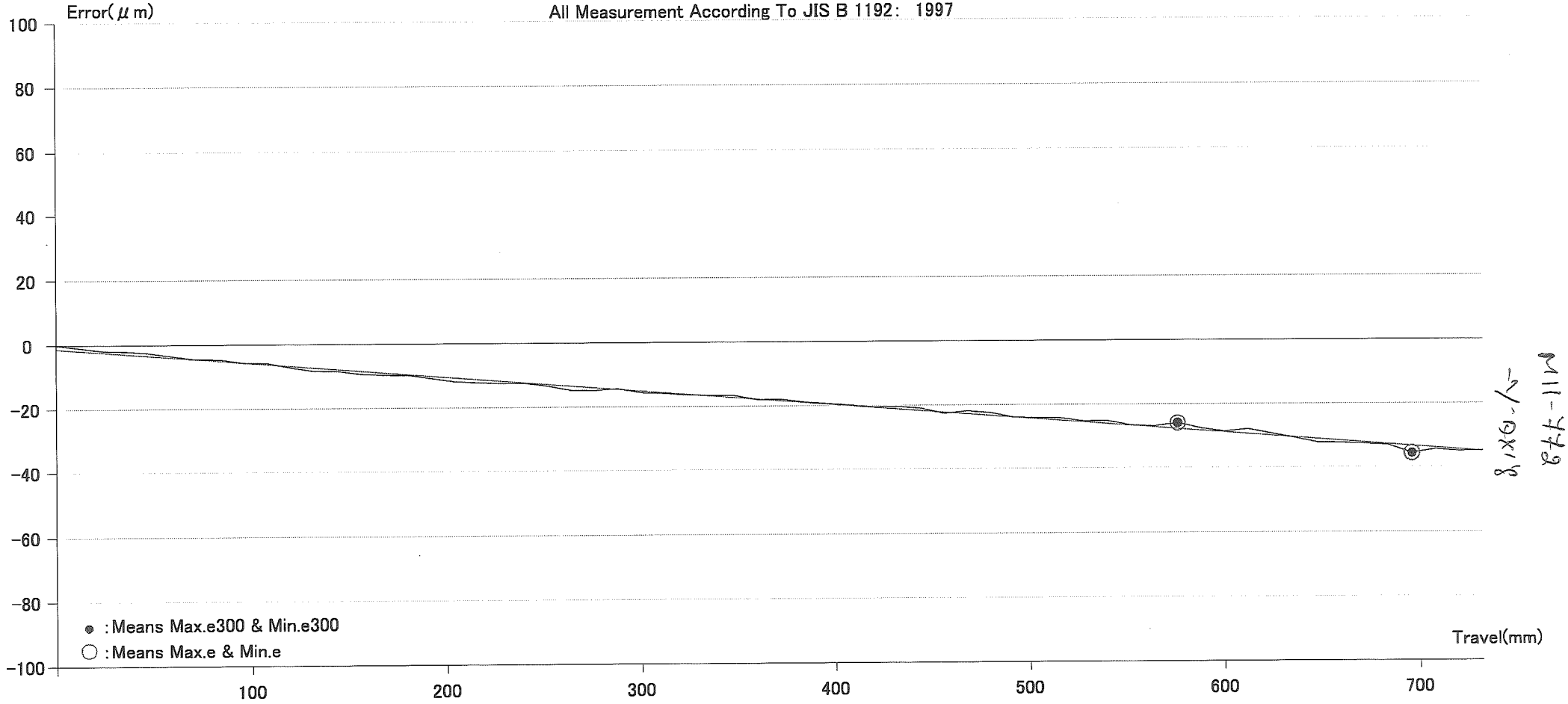
VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	O/C
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	—11—
3	Ensure smooth rotation of ball screw assembly.	—11—
4	Ensure ball screw run out within 0.010 mm	—11—
5	Ensure tightness of all screws & marking with yellow marker	—11—
6	Ensure ball Screw report attached to Inspection Report.	—11—

Date:	Checked by	Approved by

PMI BALLSCREWS BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



Specifications:	
·Customer	:AMS
·Order No.	:S3-110020143008
·Drawing No.	:BA300510003-0
·Part No.	: *:S12-016-97
·Specification	:R40-12C1-FDWE-796-1060-0.008
·Nominal Diameter (mm)	:40
·Nominal Pitch (mm)	:12
·Travel Length (mm)	:732
·E(μm)	:±18
·T(μm)	:-24

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μm)	-42～-6	-33.94
·e(μm)	13	4.06
·e300(μm)	8	4.06
·Tq(kgf - cm)	4.16～7.28	4.68-5.72
·Accuracy Grade	C3	C1

Manufactured By
PRECISION MOTION IND., INC.
www.pmi-amt.com

Measuring Environment:
·Refernce Temperatrue : 20℃ ± 0.5℃
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

QC
2011/07/28
PASS

Inspector: ARJAY

ACE MANUFACTURING SYSTEMS LTD

BALL SCREW ASSEMBLY REPORT

AMS — In Pursuit of Excellence

Machine Number : M11-772	Ball screw Sl-No: 11002014 3008	Axis : Z, Y

I. Ball screw Make & Specification : PMI (RU0-12C1-FOWE-796-1060-0.008)

II. Ball Screw Bearing Make & Specification : NACHI (25TAB06DB(UM))

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS-PDN-MEC-SA-02	1.2N/M	1.8N/M
2	Bearing housing Bore diameter	As per drawing	5to4	5to4
3	Bearing housing centre height difference	0.01 mm	0.01MM	—
4	Bearing housing bore parallelism with respect to pads	0.01 mm	0.006MM	0.008MM

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	NACHI	T3	14.935
4	Bearing width B3	—11—	T4	14.935
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	CW-275-8128	T6	—
7	Housing Bearing depth	(37-183-2123	T7	43.490

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 103/01

T6 = 13.680 mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	NACHI	T3	14.930
4	Bearing width B3	—11—	T4	14.930
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	(2W-360-8123	T6	—
7	Housing Bearing depth	(37-153-2123	T7	43.480

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 153/01

T6 = 13.680 mm

VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	O/a
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	—11—
3	Ensure smooth rotation of ball screw assembly.	—11—
4	Ensure ball screw run out within 0.010 mm	—11—
5	Ensure tightness of all screws & marking with yellow marker	—11—
6	Ensure ball Screw report attached to Inspection Report	—11—

Date:	Checked by	Approved by